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Ito et al.

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(54) **HETEROCYCLIC COMPOUND**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

This patent is subject to a terminal disclaimer.

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(51) **Int. Cl.**

C07D 413/14 (2006.01)

A61P 25/28 (2006.01)

C07D 413/10 (2006.01)

(52) **U.S. Cl.**

CPC **C07D 413/14** (2013.01); **A61P 25/28** (2018.01); **C07D 413/10** (2013.01)

(58) **Field of Classification Search**

CPC C07D 413/14; C07D 413/10; A61P 25/28

USPC 514/364

See application file for complete search history.

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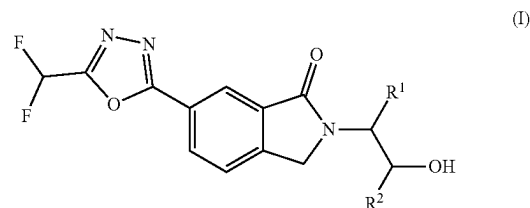
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(57)

ABSTRACT

The present invention provides a heterocyclic compound having a HDAC6 inhibitory action, which is useful for the treatment of central nervous system diseases including neurodegenerative diseases, and the like, and a medicament comprising the compound. The present invention relates to a compound represented by the formula (I):



wherein each symbol is as defined in the description, or a salt thereof.

5 Claims, 5 Drawing Sheets

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Fig. 1

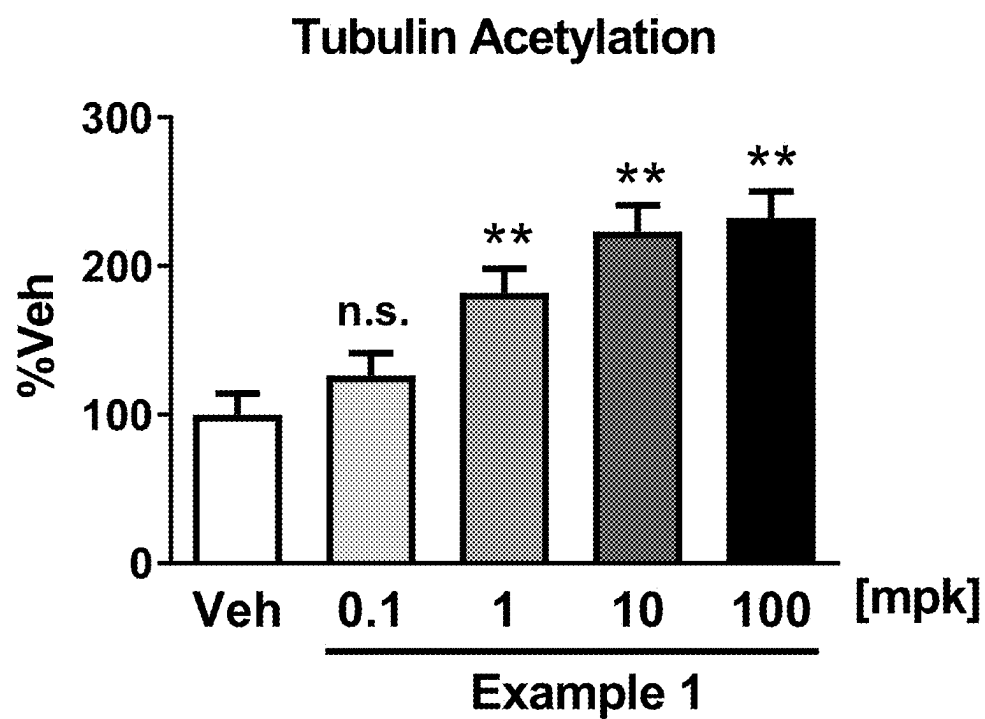


Fig. 2

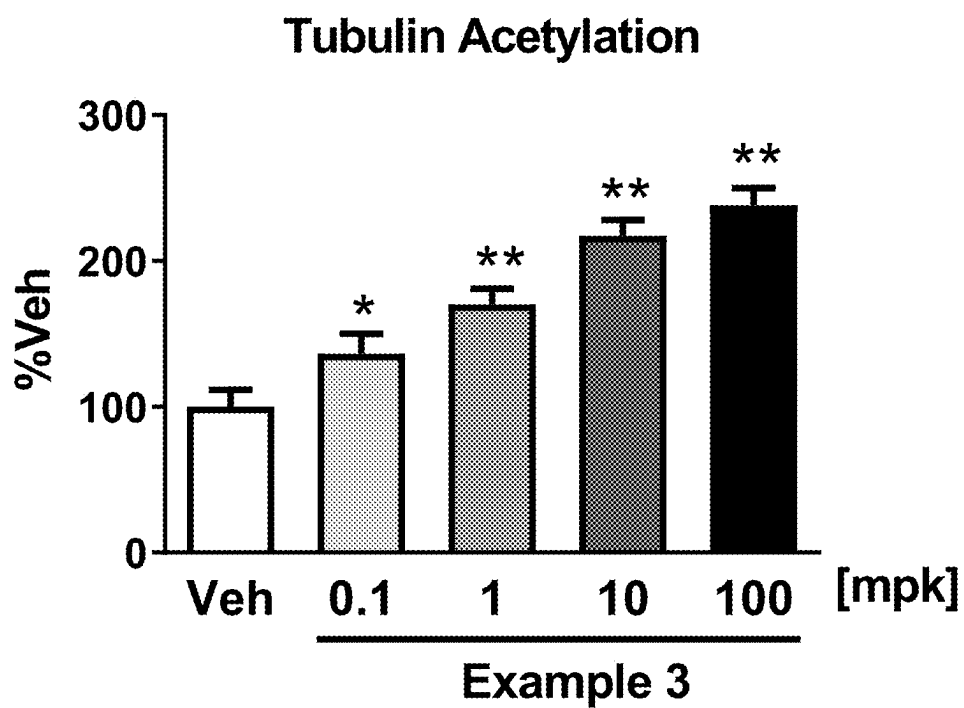


Fig. 3

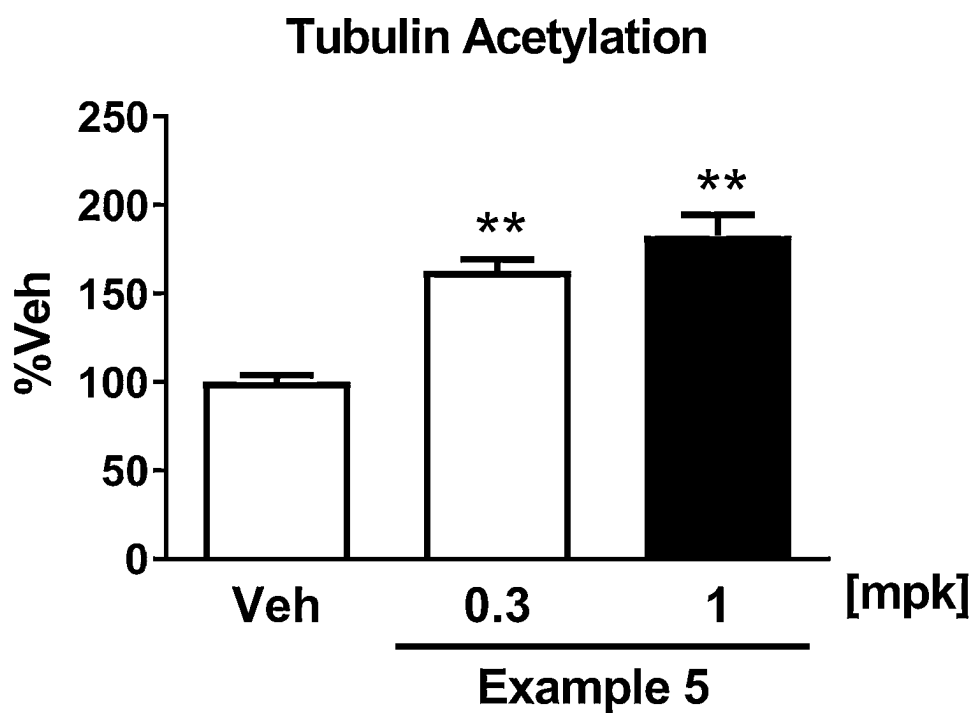


Fig. 4

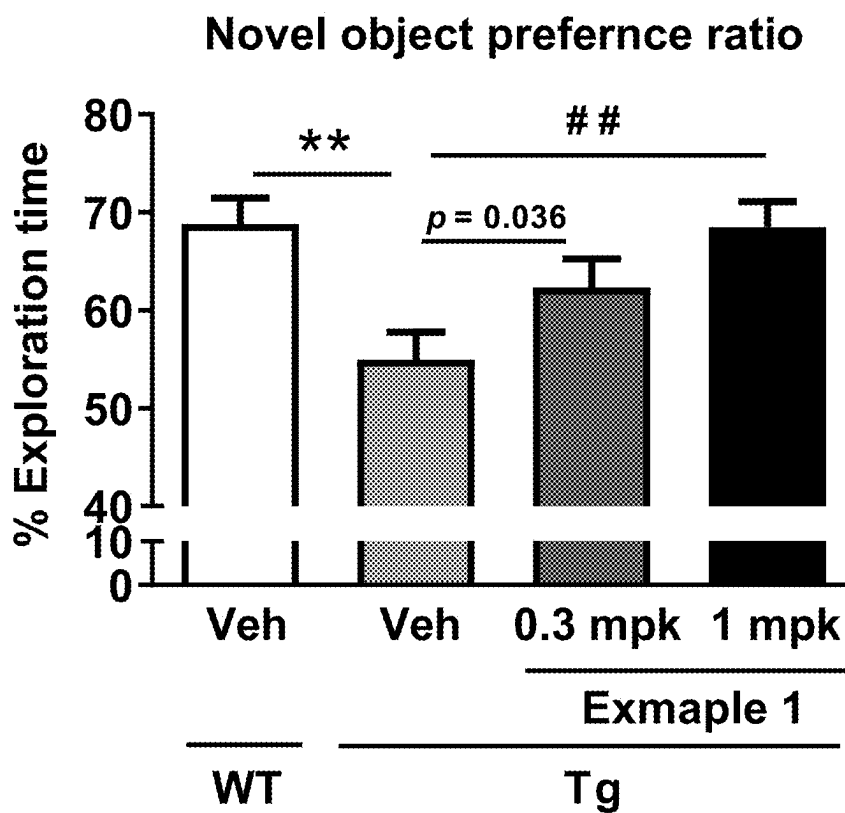
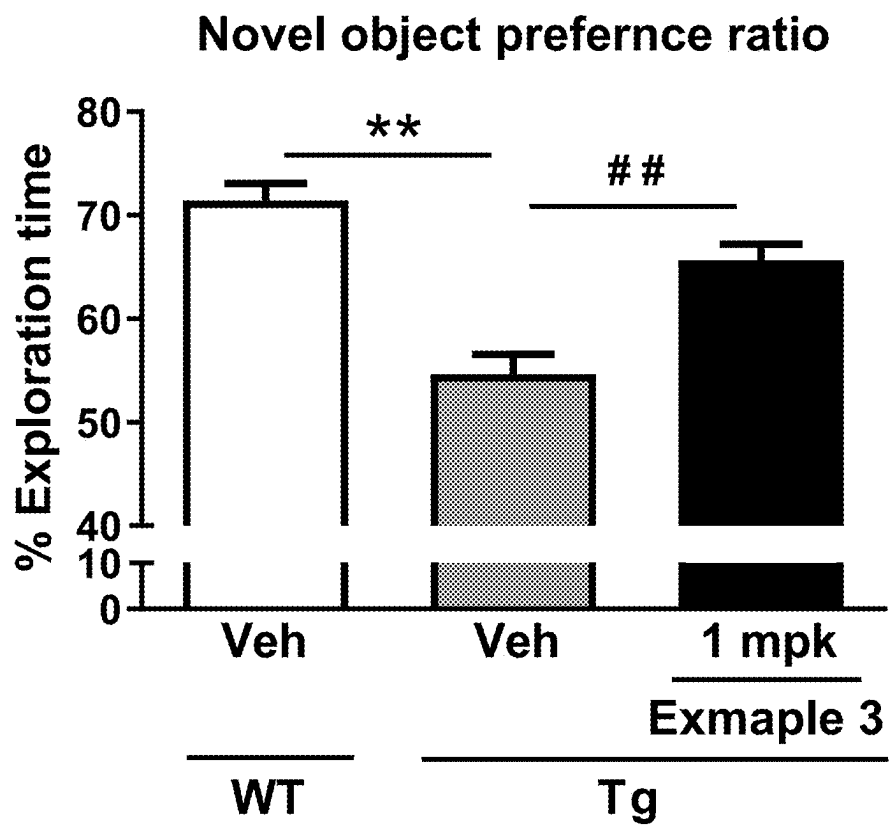


Fig. 5



1

HETEROCYCLIC COMPOUND**CROSS-REFERENCE TO RELATED APPLICATIONS**

This application is a Divisional of U.S. application Ser. No. 17/952,546, filed Sep. 26, 2022, now U.S. Pat. No. 11,958,845, which is a Divisional of U.S. application Ser. No. 17/030,504, filed Sep. 24, 2020, now U.S. Pat. No. 11,453,661, which claims priority to JP 2019-177815, filed Sep. 27, 2019.

TECHNICAL FIELD

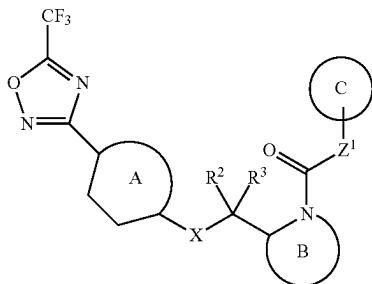
The present invention relates to a heterocyclic compound having a histone deacetylase (hereinafter sometimes to be referred to as "HDAC") inhibitory activity, preferably a class II HDAC inhibitory activity, more preferably a HDAC6 inhibitory activity, which may be useful for the treatment of central nervous system diseases including neurodegenerative diseases (Alzheimer's disease, progressive supranuclear palsy, etc.) and the like, a medicament comprising the compound, and the like.

BACKGROUND OF THE INVENTION

Nerve axon is known to play an important role in transport of nutritional factor, neurotransmitter, organelle and the like in nerve cell, and axon function disorder, axonal degeneration and intracellular accumulation of axon binding protein tau are observed in various neurodegenerative diseases (Non-Patent Document 1 and Non-Patent Document 2). Diseases characterized by intracellular tau accumulation are collectively called pathologically as tauopathy, and they encompass Alzheimer's disease, progressive supranuclear palsy and the like (Non-Patent Document 3). HDAC6 is an enzyme which plays a role in deacetylation of axon component, tubulin (Non-Patent Document 4), and microtubule containing acetylated tubulin is known to contribute to stability (Non-Patent Document 5). In addition, it is reported that Tubastatin A having a HDAC6 inhibitory activity increases acetylation of tubulin in tauopathy mouse model, and shows therapeutic effectiveness (Non-Patent Document 6). Therefore, the above-mentioned reports suggest that HDAC6 inhibitor has the potential to be a therapeutic drug for Alzheimer's disease and progressive supranuclear palsy via stabilization of axon.

As heterocyclic compounds, for example, the following compounds are known.

(1) Patent Document 1 discloses a compound represented by the following formula:

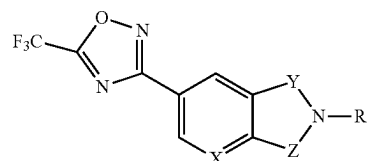


wherein each symbol is as defined in the document, which is a HDAC inhibitor, and is effective for the treatment of

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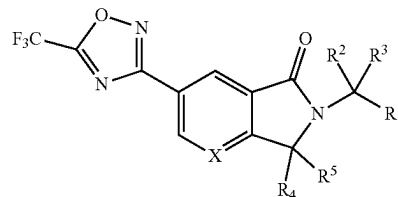
autoimmune diseases, inflammatory diseases, metabolic/ostearticular degenerative diseases, neurodegenerative diseases/central diseases (e.g., schizophrenia, Alzheimer's disease (dementia of Alzheimer type), Parkinson's disease, Huntington's disease, Rubinstein-Taybi syndrome, muscular dystrophy, Rett syndrome, Charcot-Marie-Tooth disease, depression), neoplastic diseases and the like.

(2) Patent Document 2 discloses a compound represented by the following formula:



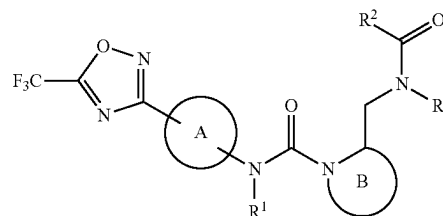
wherein each symbol is as defined in the document, which is a HDAC inhibitor, and is effective for the treatment of autoimmune diseases, inflammatory diseases, metabolic/ostearticular degenerative diseases, neurodegenerative diseases/central diseases (e.g., schizophrenia, Alzheimer's disease (dementia of Alzheimer type), Parkinson's disease, Huntington's disease, Rubinstein-Taybi syndrome, muscular dystrophy, Rett syndrome, Charcot-Marie-Tooth disease, depression), neoplastic diseases and the like.

(3) Patent Document 3 discloses a compound represented by the following formula:



wherein each symbol is as defined in the document, which is a HDAC inhibitor, and is effective for the treatment of autoimmune diseases, inflammatory diseases, metabolic/ostearticular degenerative diseases, neurodegenerative diseases/central diseases (e.g., schizophrenia, Alzheimer's disease (dementia of Alzheimer type), Parkinson's disease, Huntington's disease, Rubinstein-Taybi syndrome, muscular dystrophy, Rett syndrome, Charcot-Marie-Tooth disease, depression), neoplastic diseases and the like.

(4) Patent Document 4 discloses a compound represented by the following formula:

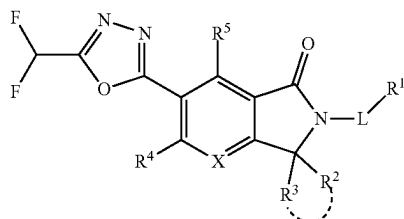


wherein each symbol is as defined in the document, which is a HDAC inhibitor, and is effective for the treatment of autoimmune diseases, inflammatory diseases, metabolic/ostearticular degenerative diseases, neurodegenerative dis-

3

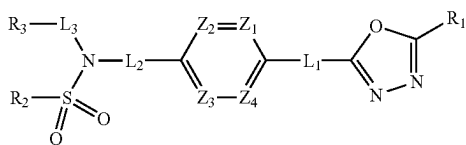
eases/central diseases (e.g., schizophrenia, Alzheimer's disease (dementia of Alzheimer type), Parkinson's disease, Huntington's disease, Rubinstein-Taybi syndrome, muscular dystrophy, Rett syndrome, Charcot-Marie-Tooth disease, depression), neoplastic diseases and the like.

(5) Patent Document 5 discloses a compound represented by the following formula:



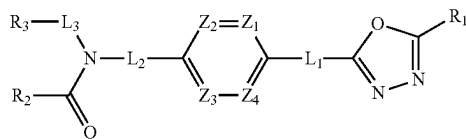
wherein each symbol is as defined in the document, which is a HDAC inhibitor, and is effective for the treatment of autoimmune diseases, inflammatory diseases, metabolic/osteoarticular degenerative diseases, neurodegenerative diseases/central diseases (e.g., schizophrenia, Alzheimer's disease (dementia of Alzheimer type), Parkinson's disease, Huntington's disease, Rubinstein-Taybi syndrome, muscular dystrophy, Rett syndrome, Charcot-Marie-Tooth disease, depression), neoplastic diseases and the like.

(6) Patent Document 6 discloses a compound represented by the following formula:



wherein each symbol is as defined in the document, which is a HDAC6 inhibitor, and is effective for the treatment of HDAC6-mediated diseases (e.g., infections, tumor, endocrine/nutritional/metabolic diseases, mental and behavioral disorders, neurological diseases, diseases of the eye and adnexa, cardiovascular diseases, respiratory diseases, digestive diseases, diseases of the skin and subcutaneous tissue, diseases of the musculoskeletal system and connective tissue, or congenital malformations, deformations and chromosomal abnormalities).

(7) Patent Document 7 discloses a compound represented by the following formula:

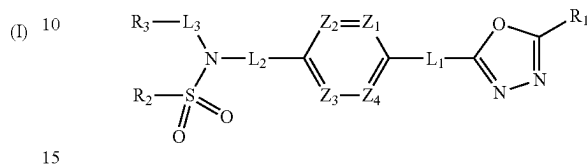


wherein each symbol is as defined in the document, which is a HDAC6 inhibitor, and is effective for the treatment of HDAC6-mediated diseases (e.g., infections, tumor, endocrine/nutritional/metabolic diseases, mental and behavioral disorders, neurological diseases, diseases of the eye and adnexa, cardiovascular diseases, respiratory diseases, diges-

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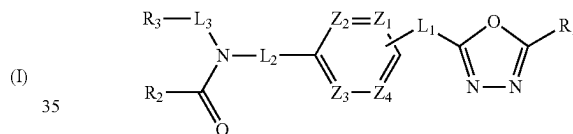
tive diseases, diseases of the skin and subcutaneous tissue, diseases of the musculoskeletal system and connective tissue, or congenital malformations, deformations and chromosomal abnormalities).

(8) Patent Document 8 discloses a compound represented by the following formula:



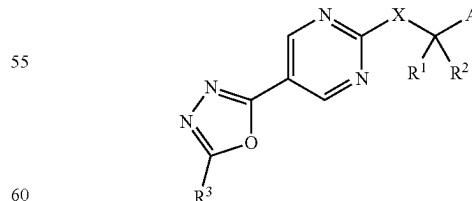
wherein each symbol is as defined in the document, which is a HDAC6 inhibitor, and is effective for the treatment of HDAC6-mediated diseases (e.g., infections, tumor, endocrine/nutritional/metabolic diseases, mental and behavioral disorders, neurological diseases, diseases of the eye and adnexa, cardiovascular diseases, respiratory diseases, digestive diseases, diseases of the skin and subcutaneous tissue, diseases of the musculoskeletal system and connective tissue, or congenital malformations, deformations and chromosomal abnormalities).

(9) Patent Document 9 discloses a compound represented by the following formula:



wherein each symbol is as defined in the document, which is a HDAC6 inhibitor, and is effective for the treatment of HDAC6-mediated diseases (e.g., infections, tumor, endocrine/nutritional/metabolic diseases, mental and behavioral disorders, neurological diseases, diseases of the eye and adnexa, cardiovascular diseases, respiratory diseases, digestive diseases, diseases of the skin and subcutaneous tissue, diseases of the musculoskeletal system and connective tissue, or congenital malformations, deformations and chromosomal abnormalities).

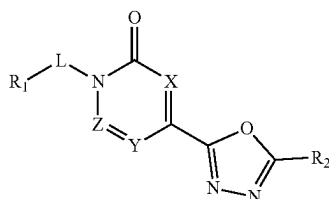
(10) Patent Document 10 discloses a compound represented by the following formula:



wherein each symbol is as defined in the document, which is a HDAC inhibitor, and is effective for the treatment of epilepsy, attentional deficit disorder, depression, anxiety, Alzheimer's disease, Parkinson's disease, Huntington's disease and the like.

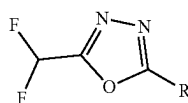
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(11) Patent Document 11 discloses a compound represented by the following formula:



wherein each symbol is as defined in the document, which is a HDAC inhibitor, and is effective for the treatment of infectious diseases, tumors, endocrine diseases, nutritional and metabolic diseases and the like.

(12) Patent Document 12 discloses a compound represented by the following formula:



wherein each symbol is as defined in the document, which is a HDAC inhibitor, and is effective for the treatment of autoimmune diseases, inflammatory diseases, metabolic/osteoarticular degenerative diseases, neurodegenerative diseases/central diseases (e.g., schizophrenia, Alzheimer's disease (dementia of Alzheimer type), Parkinson's disease, Huntington's disease, Rubinstein-Taybi syndrome, muscular dystrophy, Rett syndrome, Charcot-Marie-Tooth disease, depression), neoplastic diseases and the like.

DOCUMENT LIST

Patent Document

Patent Document 1: WO 2016/031815
 Patent Document 2: WO 2017/014321
 Patent Document 3: WO 2017/014170
 Patent Document 4: WO 2017/033946
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 Patent Document 7: WO 2017/018804
 Patent Document 8: WO 2017/018805
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 Patent Document 10: WO 2018/165520
 Patent Document 11: WO 2020/022794
 Patent Document 12: WO 2020/158762

Non-Patent Document

Non-Patent Document 1: Front Cell Neurosci. 9, 343, (2015).
 Non-Patent Document 2: Neuromolecular Med. 2: 89-99, (2002).
 Non-Patent Document 3: Cold Spring Harb Perspect Med. 2: a006254 (2012).
 Non-Patent Document 4: Nature. 417: 455-458, (2002).
 Non-Patent Document 5: Proc Natl Acad Sci USA. 107: 21238-21239, (2010).
 Non-Patent Document 6: Alzheimers Res Ther. 6: 12, (2014).

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SUMMARY OF THE INVENTION

Problems to be Solved by the Invention

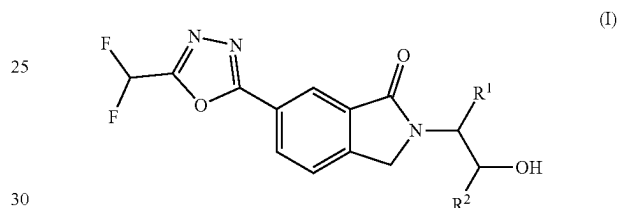
The present invention aims to provide a heterocyclic compound having a HDAC inhibitory action, which is useful for the treatment of central nervous system diseases including neurodegenerative diseases (Alzheimer's disease, progressive supranuclear palsy, etc.) and the like, and a medicament comprising the compound.

Means of Solving the Problems

The present inventors have conducted intensive studies to solve the above-mentioned problems, and have found that a compound represented by the following formula (I) has a superior HDAC inhibitory action, and completed the present invention based on these findings.

Accordingly, the present invention provides the following.

[1] A compound represented by the formula (I):



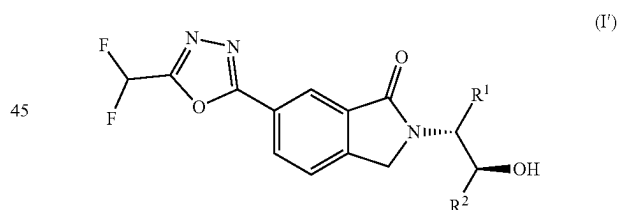
wherein

R¹ is an optionally substituted cyclic group, and

R² is an optionally substituted cyclic group,

or a salt thereof (hereinafter sometimes to be referred to as "compound (I)").

[2] The compound or salt according to the above-mentioned [1], which is represented by the formula (I'):



wherein each symbol is as defined in the above-mentioned [1].

[3] The compound or salt according to the above-mentioned [1],

wherein

R¹ is

(1) a C₃₋₁₀ cycloalkyl group optionally substituted by 1 to 3 halogen atoms,

(2) a C₆₋₁₄ aryl group optionally substituted by 1 to 3 substituents selected from (i) a halogen atom, (ii) a cyano group, and (iii) a C₁₋₆ alkyl group optionally substituted by 1 to 3 halogen atoms,

(3) a 5- or 6-membered monocyclic aromatic heterocyclic group optionally substituted by 1 to 3 substituents selected from (i) a halogen atom, (ii) a C₁₋₆ alkyl group optionally substituted by 1 to 3 halogen atoms, and (iii) a C₁₋₆ alkoxy group, or

(4) a 3- to 8-membered monocyclic non-aromatic heterocyclic group, and
R² is

- (1) a C₃₋₁₀ cycloalkyl group optionally substituted by 1 to 3 halogen atoms,
- (2) a C₆₋₁₄ aryl group optionally substituted by 1 to 3 substituents selected from (i) a halogen atom, and (ii) a C₁₋₆ alkyl group optionally substituted by 1 to 3 halogen atoms,
- (3) a 5- or 6-membered monocyclic aromatic heterocyclic group optionally substituted by 1 to 3 substituents selected from (i) a halogen atom, (ii) a C₁₋₆ alkyl group optionally substituted by 1 to 3 halogen atoms, and (iii) a C₁₋₆ alkoxy group, or
- (4) a 3- to 8-membered monocyclic non-aromatic heterocyclic group.

[4] The compound or salt according to the above-mentioned [1],

wherein

R¹ is

- (1) a C₃₋₁₀ cycloalkyl group optionally substituted by 1 to 3 halogen atoms, or
- (2) a 5- or 6-membered monocyclic aromatic heterocyclic group optionally substituted by 1 to 3 substituents selected from (i) a halogen atom, (ii) a C₁₋₆ alkyl group optionally substituted by 1 to 3 halogen atoms, and (iii) a C₁₋₆ alkoxy group, and

R² is

- (1) a C₆₋₁₄ aryl group optionally substituted by 1 to 3 substituents selected from (i) a halogen atom, and (ii) a C₁₋₆ alkyl group optionally substituted by 1 to 3 halogen atoms, or
- (2) a 5- or 6-membered monocyclic aromatic heterocyclic group optionally substituted by 1 to 3 substituents selected from (i) a halogen atom, (ii) a C₁₋₆ alkyl group optionally substituted by 1 to 3 halogen atoms, and (iii) a C₁₋₆ alkoxy group.

[5] The compound or salt according to the above-mentioned [1],

wherein

R¹ is

- (1) a C₃₋₁₀ cycloalkyl group, or
- (2) a 6-membered monocyclic aromatic heterocyclic group optionally substituted by 1 to 3 halogen atoms, and

R² is

- (1) a C₆₋₁₄ aryl group optionally substituted by 1 to 3 halogen atoms, or
- (2) a 6-membered monocyclic aromatic heterocyclic group optionally substituted by 1 to 3 halogen atoms.

[6] 6-[5-(Difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R, 2R)-2-hydroxy-1,2-di(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isindol-1-one or a salt thereof.

[7] 6-[5-(Difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R, 2R)-2-(5-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isindol-1-one or a salt thereof.

[8] 2-[(1R,2R)-1,2-Bis(5-fluoropyridin-2-yl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isindol-1-one or a salt thereof.

[9] A medicament comprising the compound or salt according to the above-mentioned [1].

[10] The medicament according to the above-mentioned [9], which is a histone deacetylase 6 inhibitor.

[11] The medicament according to the above-mentioned [9], which is an agent for the prophylaxis or treatment of Alzheimer's disease or progressive supranuclear palsy.

[12] The compound or salt according to the above-mentioned [1] for use in the prophylaxis or treatment of Alzheimer's disease or progressive supranuclear palsy.

[13] A method of inhibiting histone deacetylase 6 in a mammal, which comprises administering an effective amount of the compound or salt according to the above-mentioned [1] to the mammal.

[14] A method for the prophylaxis or treatment of Alzheimer's disease or progressive supranuclear palsy in a mammal, which comprises administering an effective amount of the compound or salt according to the above-mentioned [1] to the mammal.

[15] Use of the compound or salt according to the above-mentioned [1] for the production of an agent for the prophylaxis or treatment of Alzheimer's disease or progressive supranuclear palsy.

Effect of the Invention

Compound (I) has a HDAC6 inhibitory activity, and may be useful for the treatment of central nervous system diseases including neurodegenerative diseases (Alzheimer's disease, progressive supranuclear palsy, etc.) and the like.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a graph showing increase in acetylated tubulin in mouse brain by the compound of Example 1. The vertical axis is relative tubulin acetylation level, and the horizontal axis is the dose (mg/kg).

FIG. 2 is a graph showing increase in acetylated tubulin in mouse brain by the compound of Example 3. The vertical axis is relative tubulin acetylation level, and the horizontal axis is the dose (mg/kg).

FIG. 3 is a graph showing increase in acetylated tubulin in mouse brain by the compound of Example 5. The vertical axis is relative tubulin acetylation level, and the horizontal axis is the dose (mg/kg).

FIG. 4 is a graph showing cognitive improvement action in mouse by the compound of Example 1. The vertical axis is relative exploration time of the novelty discrimination (%), and the horizontal axis is the dose (mg/kg).

FIG. 5 is a graph showing cognitive improvement action in mouse by the compound of Example 3. The vertical axis is the novelty discrimination relative exploration time of (%), and the horizontal axis is the dose (mg/kg).

DETAILED DESCRIPTION OF THE INVENTION

The present invention is explained in detail in the following.

The definition of each symbol used in the formula (I) is explained.

R¹ and R² are the same or different and each is an optionally substituted cyclic group. Examples of the "cyclic group" of the "optionally substituted cyclic group" include a hydrocarbon ring group and a heterocyclic group.

Examples of the "hydrocarbon ring group" include a C₃₋₁₀ cycloalkyl group (e.g., cyclopropyl, cyclobutyl, cyclopentyl, cyclohexyl, cycloheptyl, cyclooctyl, bicyclo[2.2.1]heptyl, bicyclo[2.2.2]octyl, bicyclo[3.2.1]octyl, adamantyl, etc.), a C₃₋₁₀ cycloalkenyl group (e.g., cyclopropenyl, cyclobutenyl, cyclopentenyl, cyclohexenyl, cycloheptenyl, cyclooctenyl), a C₆₋₁₄ aryl group (e.g., phenyl, 1-naphthyl, 2-naphthyl, 1-anthryl, 2-anthryl, 9-anthryl) and the like.

Examples of the “heterocyclic group” include (i) an aromatic heterocyclic group, (ii) a non-aromatic heterocyclic group and (iii) a 7- to 10-membered bridged heterocyclic group, each containing, as a ring-constituting atom besides carbon atom, 1 to 4 heteroatoms selected from a nitrogen atom, a sulfur atom and an oxygen atom.

Examples of the “aromatic heterocyclic group” include a 5- to 14-membered (preferably 5- to 10-membered) aromatic heterocyclic group containing, as a ring-constituting atom besides carbon atom, 1 to 4 heteroatoms selected from a nitrogen atom, a sulfur atom and an oxygen atom.

Preferable examples of the “aromatic heterocyclic group” include 5- or 6-membered monocyclic aromatic heterocyclic groups such as thienyl, furyl, pyrrolyl, imidazolyl, pyrazolyl, thiazolyl, isothiazolyl, oxazolyl, isoxazolyl, pyridyl, pyrazinyl, pyrimidinyl, pyridazinyl, 1,2,4-oxadiazolyl, 1,3,4-oxadiazolyl, 1,2,4-thiadiazolyl, 1,3,4-thiadiazolyl, triazolyl, tetrazolyl, triazinyl and the like; and 8- to 14-membered fused polycyclic (preferably bi or tricyclic) aromatic heterocyclic groups such as benzothiophenyl, benzofuranyl, benzimidazolyl, benzoxazolyl, benzisoxazolyl, benzothiazolyl, benzisothiazolyl, benzotriazolyl, imidazopyridinyl, thienopyridinyl, furopyridinyl, pyrrolopyridinyl, pyrazolopyridinyl, oxazolopyridinyl, thiazolopyridinyl, imidazopyrazinyl, imidazopyrimidinyl, thienopyrimidinyl, furopyrimidinyl, pyrrolopyrimidinyl, pyrazolopyrimidinyl, oxazolopyrimidinyl, thiazolopyrimidinyl, pyrazolotriazinyl, naphtho[2,3-b]thienyl, phenoxathiinyl, indolyl, isoindolyl, 1H-indazolyl, purinyl, isoquinolyl, quinolyl, phthalazinyl, naphthyridinyl, quinoxalinyl, quinazolinyl, cinnolinyl, carbazolyl, β -carbolinyl, phenanthridinyl, acridinyl, phenazinyl, phenothiazinyl, phenoxazinyl and the like.

Examples of the “non-aromatic heterocyclic group” include a 3- to 14-membered (preferably 4- to 10-membered) non-aromatic heterocyclic group containing, as a ring-constituting atom besides carbon atom, 1 to 4 heteroatoms selected from a nitrogen atom, a sulfur atom and an oxygen atom.

Preferable examples of the “non-aromatic heterocyclic group” include 3- to 8-membered monocyclic non-aromatic heterocyclic groups such as aziridinyl, oxiranyl, thiranyl, azetidiny, oxetanyl, thietanyl, tetrahydrothienyl, tetrahydrofuranlyl, pyrrolinyl, pyrrolidinyl, imidazolinyl, imidazolidinyl, oxazolinyl, oxazolidinyl, pyrazolinyl, pyrazolidinyl, thiazolinyl, thiazolidinyl, tetrahydroisothiazolyl, tetrahydrooxazolyl, tetrahydroisooxazolyl, piperidinyl, piperazinyl, tetrahydropyridinyl, dihydropyridinyl, dihydrothiopyranyl, tetrahydropyrimidinyl, tetrahydropyridazinyl, dihydropyranlyl, tetrahydropyranlyl, tetrahydrothiopyranyl, morpholinyl, thiomorpholinyl, azepanyl, diazepanyl, azepinyl, oxepanyl, azocanyl, diazocanyl and the like; and 9- to 14-membered fused polycyclic (preferably bi or tricyclic) non-aromatic heterocyclic groups such as dihydrobenzofuranyl, dihydrobenzimidazolyl, dihydrobenzoxazolyl, dihydrobenzothiazolyl, dihydrobenzisothiazolyl, dihydronaphtho[2,3-b]thienyl, tetrahydroisoquinolyl, tetrahydroquinolyl, 4H-quinoliziny, indolinyl, isoindolinyl, tetrahydrothieno[2,3-c]pyridinyl, tetrahydrobenzazepinyl, tetrahydroquinoxalinyl, tetrahydrophenanthridinyl, hexahydrophenothiazinyl, hexahydrophenoxazinyl, tetrahydrophthalazinyl, tetrahydronaphthyridinyl, tetrahydroquinazolinyl, tetrahydrocinnolinyl, tetrahydrocarbazolyl, tetrahydro- β -carbolinyl, tetrahydroacrydinyl, tetrahydrophenazinyl, tetrahydrothioxanthenyl, octahydroisoquinolyl and the like.

Preferable examples of the “7- to 10-membered bridged heterocyclic group” include quinuclidinyl and 7-azabicyclo[2.2.1]heptanyl.

Examples of the “optionally substituted cyclic group” represented by R¹ or R² include a cyclic group optionally having substituent(s) selected from the following Substituent group A.

The number of the substituents in the “optionally substituted cyclic group” is, for example, 1 to 3. When the number of the substituents is two or more, the respective substituents may be the same or different.

[Substituent Group A]

- (1) a halogen atom (e.g., fluorine, chlorine, bromine, iodine),
- (2) a nitro group,
- (3) a cyano group,
- (4) an oxo group,
- (5) a hydroxy group,
- (6) an optionally halogenated C₁₋₆ alkoxy group (e.g., methoxy, difluoromethoxy, trifluoromethoxy, ethoxy, 2,2,2-trifluoroethoxy, propoxy, isopropoxy, butoxy, 4,4,4-trifluorobutoxy, isobutoxy, sec-butoxy, pentyloxy, hexyloxy),
- (7) a C₆₋₁₄ aryloxy group (e.g., phenoxy, naphthoxy),
- (8) a C₇₋₁₆ aralkyloxy group (e.g., benzyloxy),
- (9) a 5- to 14-membered aromatic heterocycloxy group (e.g., pyridyloxy),
- (10) a 3- to 14-membered non-aromatic heterocycloxy group (e.g., morpholinyl, piperidinyl),
- (11) a C₁₋₆ alkyl-carbonyloxy group (e.g., acetoxy, propanoyloxy),
- (12) a C₆₋₁₄ aryl-carbonyloxy group (e.g., benzyloxy, 1-naphthoyloxy, 2-naphthoyloxy),
- (13) a C₁₋₆ alkoxy-carbonyloxy group (e.g., methoxycarbonyloxy, ethoxycarbonyloxy, propoxycarbonyloxy, butoxycarbonyloxy),
- (14) a mono- or di-C₁₋₆ alkyl-carbamoyloxy group (e.g., methylcarbamoyloxy, ethylcarbamoyloxy, dimethylcarbamoyloxy, diethylcarbamoyloxy),
- (15) a C₆₋₁₄ aryl-carbamoyloxy group (e.g., phenylcarbamoyloxy, naphthylcarbamoyloxy),
- (16) a 5- to 14-membered aromatic heterocyclylcarbonyloxy group (e.g., nicotinoyloxy),
- (17) a 3- to 14-membered non-aromatic heterocyclylcarbonyloxy group (e.g., morpholinylcarbonyloxy, piperidinylcarbonyloxy),
- (18) an optionally halogenated C₁₋₆ alkylsulfonyloxy group (e.g., methylsulfonyloxy, trifluoromethylsulfonyloxy),
- (19) a C₆₋₁₄ arylsulfonyloxy group optionally substituted by C₁₋₆ alkyl group(s) (e.g., phenylsulfonyloxy, toluenesulfonyloxy),
- (20) an optionally halogenated C₁₋₆ alkylthio group (e.g., methylthio, difluoromethylthio, trifluoromethylthio, ethylthio, propylthio, isopropylthio, butylthio, 4,4,4-trifluorobutylthio, pentylthio, hexylthio),
- (21) a 5- to 14-membered aromatic heterocyclic group (e.g., thienyl, furyl, pyrrolyl, imidazolyl, pyrazolyl, thiazolyl, isothiazolyl, oxazolyl, isoxazolyl, pyridyl, pyrazinyl, pyrimidinyl, pyridazinyl, 1,2,4-oxadiazolyl, 1,3,4-oxadiazolyl, 1,2,4-thiadiazolyl, 1,3,4-thiadiazolyl, triazolyl, tetrazolyl, triazinyl, benzothiophenyl, benzofuranyl, benzimidazolyl, benzoxazolyl, benzisoxazolyl, benzothiazolyl, benzisothiazolyl, benzotriazolyl, imidazopyridinyl, thienopyridinyl, furopyridinyl, pyrrolopyridinyl, pyrazolopyridinyl, oxazolopyridinyl, thiazolopyridinyl, imidazopyrazinyl, imidazopyrimidinyl, thienopyrimidinyl, furopyrimidinyl, pyrrolopyrimidinyl, pyrazolopyrimidinyl, oxazolopyrimidinyl, thiazolopyrimidinyl, pyrazolotriazinyl,

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- naphtho[2,3-b]thienyl, phenoxathiinyl, indolyl, isoin-
dolyl, 1H-indazolyl, purinyl, isoquinolyl, quinolyl,
phthalazinyl, naphthyridinyl, quinoxalinyl, quinazoli-
nyl, cinnolinyl, carbazolyl, β -carbolinyl, phenanthridi-
nyl, acridinyl, phenazinyl, phenothiazinyl, phenoxazi-
n-yl),
- (22) a 3- to 14-membered non-aromatic heterocyclic
group (e.g., aziridinyl, oxiranyl, thiiranyl, azetidiny-
l, oxetanyl, thietanyl, tetrahydrothienyl, tetrahydrofura-
n-yl, pyrrolinyl, pyrrolidinyl, imidazolinyl, imidazolidi-
n-yl, oxazoliny, oxazolidinyl, pyrazolinyl, pyrazolidi-
n-yl, thiazolinyl, thiazolidinyl, tetrahydroisothiazolyl,
tetrahydrooxazolyl, tetrahydroisoxazolyl, piperidinyl,
piperazinyl, tetrahydropyridinyl, dihydropyridinyl,
dihydrothiopyranyl, tetrahydropyrimidinyl, tetrahydro-
pyridazinyl, dihydropyranyl, tetrahydropyranyl, tetra-
hydrothiopyranyl, morpholinyl, thiomorpholinyl, aze-
panyl, diazepanyl, azepinyl, oxepanyl, azocanyl,
diazocanyl, dihydrobenzofuranyl, dihydrobenzimidaz-
olyl, dihydrobenzoxazolyl, dihydrobenzothiazolyl,
dihydrobenziso-thiazolyl, dihydronaphtho[2,3-b]thie-
n-yl, tetrahydroisoquinolyl, tetrahydroquinolyl, 4H-qui-
noliziny, indolinyl, isoindolinyl, tetrahydrothieno[2,3-
c]pyridinyl, tetrahydrobenzazepinyl,
tetrahydroquinoxalinyl, tetrahydrophenanthridinyl,
hexahydrophenothiazinyl, hexahydrophenoxazinyl,
tetrahydrophthalazinyl, tetrahydronaphthyridinyl, tetra-
hydroquinazoliny, tetrahydrocinnolinyl, tetrahydro-
carbazolyl, tetrahydro- β -carbolinyl, tetrahydroacridi-
n-yl, tetrahydrophenazinyl, tetrahydrothioxanthenyl,
octahydroisoquinolyl),
- (23) a formyl group,
(24) a carboxy group,
(25) an optionally halogenated C_{1-6} alkyl-carbonyl group
(e.g., acetyl, chloroacetyl, trifluoroacetyl, trichloro-
acetyl, propanoyl, butanoyl, pentanoyl, hexanoyl),
- (26) a C_{6-14} aryl-carbonyl group (e.g., benzoyl, 1-naph-
thoyl, 2-naphthoyl),
- (27) a 5- to 14-membered aromatic heterocyclylcarbonyl
group (e.g., nicotinoyl, isonicotinoyl, thenoyl, furoyl),
- (28) a 3- to 14-membered non-aromatic heterocyclylcar-
bonyl group (e.g., morpholinylcarbonyl, piperidinyl-
carbonyl, pyrrolidinylcarbonyl),
- (29) a C_{1-6} alkoxy-carbonyl group (e.g., methoxycarbo-
nyl, ethoxycarbonyl, propoxycarbonyl, isopropoxycar-
bonyl, butoxycarbonyl, isobutoxycarbonyl, sec-bu-
toxycarbonyl, tert-butoxycarbonyl, pentyloxycarbonyl,
hexyloxycarbonyl),
- (30) a C_{6-14} aryloxy-carbonyl group (e.g., phenyloxycar-
bonyl, 1-naphthylloxycarbonyl, 2-naphthylloxycarbo-
n-yl),
- (31) a C_{7-16} aralkyloxy-carbonyl group (e.g., benzyloxy-
carbonyl, phenethyloxycarbonyl),
- (32) a carbamoyl group,
(33) a thiocarbamoyl group,
- (34) a mono- or di- C_{1-6} alkyl-carbamoyl group (e.g.,
methylcarbamoyl, ethylcarbamoyl, dimethylcarbam-
oyl, diethylcarbamoyl, N-ethyl-N-methylcarbamoyl),
- (35) a C_{6-14} aryl-carbamoyl group (e.g., phenylcarbam-
oyl),
- (36) a 5- to 14-membered aromatic heterocyclylcarbam-
oyl group (e.g., pyridylcarbamoyl, thienylcarbamoyl),
- (37) a 3- to 14-membered non-aromatic heterocyclylcar-
bamoyl group (e.g., morpholinylcarbamoyl, piperidi-
n-ylcarbamoyl),
- (38) an optionally halogenated C_{1-6} alkylsulfonyl group
(e.g., methylsulfonyl, difluoromethylsulfonyl, trifluo-

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- romethylsulfonyl, ethylsulfonyl, propylsulfonyl, iso-
propylsulfonyl, butylsulfonyl, 4,4,4-trifluorobu-
t-ylsulfonyl, pentylsulfonyl, hexylsulfonyl),
- (39) a C_{6-14} arylsulfonyl group (e.g., phenylsulfonyl,
1-naphthylsulfonyl, 2-naphthylsulfonyl),
- (40) a 5- to 14-membered aromatic heterocyclylsulfonyl
group (e.g., pyridylsulfonyl, thienylsulfonyl),
- (41) an optionally halogenated C_{1-6} alkylsulfinyl group
(e.g., methylsulfinyl, ethylsulfinyl),
- (42) a C_{6-14} aryl sulfinyl group (e.g., phenylsulfinyl,
1-naphthyl sulfinyl, 2-naphthyl sulfinyl),
- (43) a 5- to 14-membered aromatic heterocyclylsulfinyl
group (e.g., pyridyl sulfinyl, thienylsulfinyl),
- (44) an amino group,
(45) a mono- or di- C_{1-6} alkylamino group (e.g., methyl-
amino, ethylamino, propylamino, isopropylamino,
butylamino, dimethylamino, diethylamino, dipropyl-
amino, dibutylamino, N-ethyl-N-methylamino),
- (46) a mono- or di- C_{6-14} arylamino group (e.g., phe-
nylamino),
- (47) a 5- to 14-membered aromatic heterocyclylamino
group (e.g., pyridylamino),
- (48) a C_{7-16} aralkylamino group (e.g., benzylamino),
- (49) a formylamino group,
(50) a C_{1-6} alkyl-carbonylamino group (e.g., acetylamino,
propanoylamino, butanoylamino),
- (51) a (C_{1-6} alkyl) (C_{1-6} alkyl-carbonyl) amino group
(e.g., N-acetyl-N-methylamino),
- (52) a C_{6-14} aryl-carbonylamino group (e.g., phenylcar-
bonylamino, naphthylcarbonylamino),
- (53) a C_{1-6} alkoxy-carbonylamino group (e.g., methoxy-
carbonylamino, ethoxycarbonylamino, propoxycarbo-
nylamino, butoxycarbonylamino, tert-butoxycarbo-
nylamino),
- (54) a C_{7-16} aralkyloxy-carbonylamino group (e.g., ben-
zyloxycarbonylamino),
- (55) a C_{1-6} alkylsulfonylamino group (e.g., methylsulfo-
nylamino, ethylsulfonylamino),
- (56) a C_{6-14} arylsulfonylamino group optionally substi-
tuted by C_{1-6} alkyl group(s) (e.g., phenylsulfo-
nylamino, toluenesulfonylamino),
- (57) an optionally halogenated C_{1-6} alkyl group (e.g.,
methyl, chloromethyl, difluoromethyl, trichloromethyl,
trifluoromethyl, ethyl, 2-bromoethyl, 2,2,2-trifluoro-
ethyl, tetrafluoroethyl, pentafluoroethyl, propyl, 2,2-
difluoropropyl, 3,3,3-trifluoropropyl, isopropyl, butyl,
4,4,4-trifluorobutyl, isobutyl, sec-butyl, tert-butyl, pen-
tyl, isopentyl, neo-pentyl, 5,5,5-trifluoropentyl, hexyl,
6,6,6-trifluorohexyl),
- (58) a C_{2-6} alkenyl group (e.g., ethenyl, 1-propenyl,
2-propenyl, 2-methyl-1-propenyl, 1-butenyl, 2-butenyl,
3-butenyl, 3-methyl-2-butenyl, 1-pentenyl, 2-pentenyl,
3-pentenyl, 4-pentenyl, 4-methyl-3-pentenyl, 1-hex-
enyl, 3-hexenyl, 5-hexenyl),
- (59) a C_{2-6} alkynyl group (e.g., ethynyl, 1-propynyl,
2-propynyl, 1-butyne, 2-butyne, 3-butyne, 1-penty-
n-yl, 2-pentyne, 3-pentyne, 4-pentyne, 1-hexynyl,
2-hexynyl, 3-hexynyl, 4-hexynyl, 5-hexynyl, 4-methyl-
2-pentyne),
- (60) a C_{3-10} cycloalkyl group (e.g., cyclopropyl,
cyclobutyl, cyclopentyl, cyclohexyl, cycloheptyl,
cyclooctyl, bicyclo[2.2.1]heptyl, bicyclo[2.2.2]octyl,
bicyclo[3.2.1]octyl, adamantyl),
- (61) a C_{3-10} cycloalkenyl group (e.g., cyclopropenyl,
cyclobutenyl, cyclopentenyl, cyclohexenyl, cyclohep-
tenyl, cyclooctenyl), and

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(62) a C₆₋₁₄ aryl group (e.g., phenyl, 1-naphthyl, 2-naphthyl, 1-anthryl, 2-anthryl, 9-anthryl).

R¹ is preferably

- (1) a C₃₋₁₀ cycloalkyl group (e.g., cyclopropyl, cyclohexyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom),
- (2) a C₆₋₁₄ aryl group (e.g., phenyl) optionally substituted by 1 to 3 substituents selected from
 - (i) a halogen atom (e.g., a fluorine atom),
 - (ii) a cyano group, and
 - (iii) a C₁₋₆ alkyl group (e.g., methyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom),
- (3) a 5- or 6-membered monocyclic aromatic heterocyclic group (e.g., pyrazolyl, pyridyl, pyrimidinyl, pyrazinyl, pyridazinyl) optionally substituted by 1 to 3 substituents selected from
 - (i) a halogen atom (e.g., a fluorine atom),
 - (ii) a C₁₋₆ alkyl group (e.g., methyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom), and
 - (iii) a C₁₋₆ alkoxy group (e.g., ethoxy), or
- (4) a 3- to 8-membered monocyclic non-aromatic heterocyclic group (e.g., tetrahydropyranyl).

R¹ is more preferably

- (1) a C₃₋₁₀ cycloalkyl group (e.g., cyclopropyl, cyclohexyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom), or
- (2) a 5- or 6-membered monocyclic aromatic heterocyclic group (e.g., pyrazolyl, pyridyl, pyrimidinyl, pyrazinyl, pyridazinyl) optionally substituted by 1 to 3 substituents selected from
 - (i) a halogen atom (e.g., a fluorine atom),
 - (ii) a C₁₋₆ alkyl group (e.g., methyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom), and
 - (iii) a C₁₋₆ alkoxy group (e.g., ethoxy).

R¹ is particularly preferably

- (1) a C₃₋₁₀ cycloalkyl group (e.g., cyclopropyl, cyclohexyl), or
- (2) a 6-membered monocyclic aromatic heterocyclic group (e.g., pyridyl, pyrimidinyl, pyrazinyl, pyridazinyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom).

R² is preferably

- (1) a C₃₋₁₀ cycloalkyl group (e.g., cyclopropyl, cyclohexyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom),
- (2) a C₆₋₁₄ aryl group (e.g., phenyl) optionally substituted by 1 to 3 substituents selected from
 - (i) a halogen atom (e.g., a fluorine atom, a chlorine atom), and
 - (ii) a C₁₋₆ alkyl group (e.g., methyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom),
- (3) a 5- or 6-membered monocyclic aromatic heterocyclic group (e.g., pyridyl, pyrimidinyl) optionally substituted by 1 to 3 substituents selected from
 - (i) a halogen atom (e.g., a fluorine atom, a chlorine atom),
 - (ii) a C₁₋₆ alkyl group (e.g., methyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom), and
 - (iii) a C₁₋₆ alkoxy group (e.g., ethoxy), or
- (4) a 3- to 8-membered monocyclic non-aromatic heterocyclic group (e.g., tetrahydropyranyl).

R² is more preferably

- (1) a C₆₋₁₄ aryl group (e.g., phenyl) optionally substituted by 1 to 3 substituents selected from

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(i) a halogen atom (e.g., a fluorine atom, a chlorine atom), and

(ii) a C₁₋₆ alkyl group (e.g., methyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom), or

(2) a 5- or 6-membered monocyclic aromatic heterocyclic group (e.g., pyridyl, pyrimidinyl) optionally substituted by 1 to 3 substituents selected from

(i) a halogen atom (e.g., a fluorine atom, a chlorine atom),

(ii) a C₁₋₆ alkyl group (e.g., methyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom), and

(iii) a C₁₋₆ alkoxy group (e.g., ethoxy).

R² is particularly preferably

(1) a C₆₋₁₄ aryl group (e.g., phenyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom, a chlorine atom), or

(2) a 6-membered monocyclic aromatic heterocyclic group (e.g., pyridyl, pyrimidinyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom, a chlorine atom).

Preferred compound is compound (I) wherein

R¹ is

(1) a C₃₋₁₀ cycloalkyl group (e.g., cyclopropyl, cyclohexyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom),

(2) a C₆₋₁₄ aryl group (e.g., phenyl) optionally substituted by 1 to 3 substituents selected from

(i) a halogen atom (e.g., a fluorine atom),

(ii) a cyano group, and

(iii) a C₁₋₆ alkyl group (e.g., methyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom),

(3) a 5- or 6-membered monocyclic aromatic heterocyclic group (e.g., pyrazolyl, pyridyl, pyrimidinyl, pyrazinyl, pyridazinyl) optionally substituted by 1 to 3 substituents selected from

(i) a halogen atom (e.g., a fluorine atom),

(ii) a C₁₋₆ alkyl group (e.g., methyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom), and

(iii) a C₁₋₆ alkoxy group (e.g., ethoxy), or

(4) a 3- to 8-membered monocyclic non-aromatic heterocyclic group (e.g., tetrahydropyranyl), and R² is

(1) a C₃₋₁₀ cycloalkyl group (e.g., cyclopropyl, cyclohexyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom),

(2) a C₆₋₁₄ aryl group (e.g., phenyl) optionally substituted by 1 to 3 substituents selected from

(i) a halogen atom (e.g., a fluorine atom, a chlorine atom), and

(ii) a C₁₋₆ alkyl group (e.g., methyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom),

(3) a 5- or 6-membered monocyclic aromatic heterocyclic group (e.g., pyridyl, pyrimidinyl) optionally substituted by 1 to 3 substituents selected from

(i) a halogen atom (e.g., a fluorine atom, a chlorine atom),

(ii) a C₁₋₆ alkyl group (e.g., methyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom), and

(iii) a C₁₋₆ alkoxy group (e.g., ethoxy), or

(4) a 3- to 8-membered monocyclic non-aromatic heterocyclic group (e.g., tetrahydropyranyl).

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More preferred compound is compound (I) wherein

R¹ is

- (1) a C₃₋₁₀ cycloalkyl group (e.g., cyclopropyl, cyclohexyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom), or
- (2) a 5- or 6-membered monocyclic aromatic heterocyclic group (e.g., pyrazolyl, pyridyl, pyrimidinyl, pyrazinyl, pyridazinyl) optionally substituted by 1 to 3 substituents selected from
 - (i) a halogen atom (e.g., a fluorine atom),
 - (ii) a C₁₋₆ alkyl group (e.g., methyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom), and
 - (iii) a C₁₋₆ alkoxy group (e.g., ethoxy), and

R² is

- (1) a C₆₋₁₄ aryl group (e.g., phenyl) optionally substituted by 1 to 3 substituents selected from
 - (i) a halogen atom (e.g., a fluorine atom, a chlorine atom), and
 - (ii) a C₁₋₆ alkyl group (e.g., methyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom), or
- (2) a 5- or 6-membered monocyclic aromatic heterocyclic group (e.g., pyridyl, pyrimidinyl) optionally substituted by 1 to 3 substituents selected from
 - (i) a halogen atom (e.g., a fluorine atom, a chlorine atom),
 - (ii) a C₁₋₆ alkyl group (e.g., methyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom), and
 - (iii) a C₁₋₆ alkoxy group (e.g., ethoxy).

Particularly preferred compound is compound (I) wherein

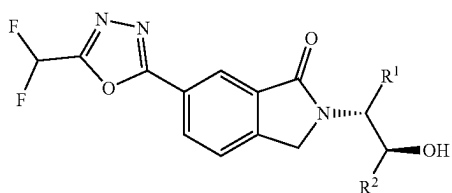
R¹ is

- (1) a C₃₋₁₀ cycloalkyl group (e.g., cyclopropyl, cyclohexyl), or
- (2) a 6-membered monocyclic aromatic heterocyclic group (e.g., pyridyl, pyrimidinyl, pyrazinyl, pyridazinyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom), and

R² is

- (1) a C₆₋₁₄ aryl group (e.g., phenyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom, a chlorine atom), or
- (2) a 6-membered monocyclic aromatic heterocyclic group (e.g., pyridyl, pyrimidinyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom, a chlorine atom).

The configuration of compound (I) is preferably represented by the formula (I'):



wherein each symbol is as defined above, (hereinafter sometimes to be referred to as compound (I')).

As embodiments of compound (I), the following compounds (compound (I'-1) and compound (I'-2)) are exemplified.

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Compound (I'-1)

Compound (I') wherein

R¹ is an optionally substituted hydrocarbon ring group, and

R² is an optionally substituted hydrocarbon ring group.

Compound (I'-1a)

Compound (I') wherein

R¹ is

- (1) a C₃₋₁₀ cycloalkyl group (e.g., cyclopropyl, cyclohexyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom), or
- (2) a C₆₋₁₄ aryl group (e.g., phenyl) optionally substituted by 1 to 3 substituents selected from
 - (i) a halogen atom (e.g., a fluorine atom),
 - (ii) a cyano group, and
 - (iii) a C₁₋₆ alkyl group (e.g., methyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom), and

R² is

- (1) a C₃₋₁₀ cycloalkyl group (e.g., cyclopropyl, cyclohexyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom), or
- (2) a C₆₋₁₄ aryl group (e.g., phenyl) optionally substituted by 1 to 3 substituents selected from
 - (i) a halogen atom (e.g., a fluorine atom, a chlorine atom), and
 - (ii) a C₁₋₆ alkyl group (e.g., methyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom).

Compound (I'-1b)

Compound (I') wherein

R¹ is a C₃₋₁₀ cycloalkyl group (e.g., cyclopropyl, cyclohexyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom), and

R² is a C₆₋₁₄ aryl group (e.g., phenyl) optionally substituted by 1 to 3 substituents selected from (i) a halogen atom (e.g., a fluorine atom, a chlorine atom), and (ii) a C₁₋₆ alkyl group (e.g., methyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom).

Compound (I'-1c)

Compound (I') wherein

R¹ is a C₃₋₁₀ cycloalkyl group (e.g., cyclopropyl, cyclohexyl), and

R² is a C₆₋₁₄ aryl group (e.g., phenyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom, a chlorine atom).

Compound (I'-2)

Compound (I') wherein

R¹ is an optionally substituted heterocyclic group, and R² is an optionally substituted heterocyclic group.

Compound (I'-2a)

Compound (I') wherein

R¹ is

- (1) a 5- or 6-membered monocyclic aromatic heterocyclic group (e.g., pyrazolyl, pyridyl, pyrimidinyl, pyrazinyl, pyridazinyl) optionally substituted by 1 to 3 substituents selected from
 - (i) a halogen atom (e.g., a fluorine atom),
 - (ii) a C₁₋₆ alkyl group (e.g., methyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom), and
 - (iii) a C₁₋₆ alkoxy group (e.g., ethoxy), or
- (2) a 3- to 8-membered monocyclic non-aromatic heterocyclic group (e.g., tetrahydropyranyl), and

R² is

- (1) a 5- or 6-membered monocyclic aromatic heterocyclic group (e.g., pyridyl, pyrimidinyl) optionally substituted by 1 to 3 substituents selected from

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- (i) a halogen atom (e.g., a fluorine atom, a chlorine atom),
- (ii) a C₁₋₆ alkyl group (e.g., methyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom), and
- (iii) a C₁₋₆ alkoxy group (e.g., ethoxy), or
- (2) a 3- to 8-membered monocyclic non-aromatic heterocyclic group (e.g., tetrahydropyranyl).

Compound (I'-2b)

Compound (I') wherein

R¹ is a 5- or 6-membered monocyclic aromatic heterocyclic group (e.g., pyrazolyl, pyridyl, pyrimidinyl, pyrazinyl, pyridazinyl) optionally substituted by 1 to 3 substituents selected from (i) a halogen atom (e.g., a fluorine atom), (ii) a C₁₋₆ alkyl group (e.g., methyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom), and (iii) a C₁₋₆ alkoxy group (e.g., ethoxy), and

R² is a 5- or 6-membered monocyclic aromatic heterocyclic group (e.g., pyridyl, pyrimidinyl) optionally substituted by 1 to 3 substituents selected from (i) a halogen atom (e.g., a fluorine atom, a chlorine atom), (ii) a C₁₋₆ alkyl group (e.g., methyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom), and (iii) a C₁₋₆ alkoxy group (e.g., ethoxy).

Compound (I'-2c)

Compound (I') wherein

R¹ is a 6-membered monocyclic aromatic heterocyclic group (e.g., pyridyl, pyrimidinyl, pyrazinyl, pyridazinyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom), and

R² is a 6-membered monocyclic aromatic heterocyclic group (e.g., pyridyl, pyrimidinyl) optionally substituted by 1 to 3 halogen atoms (e.g., a fluorine atom, a chlorine atom).

Specific examples of compound (I) include the compounds of Examples 1 to 150. Among them, compound (I) is preferably 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2R)-2-hydroxy-1,2-di(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Example 1) or a salt thereof, 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2R)-2-(5-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Example 3) or a salt thereof, 2-[(1R,2R)-1,2-bis(5-fluoropyridin-2-yl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Example 5) or a salt thereof, 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2R)-2-(6-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Example 7) or a salt thereof, or 2-[(1R,2S)-1-cyclopropyl-2-(4-fluorophenyl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Example 9) or a salt thereof, more preferably 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2R)-2-hydroxy-1,2-di(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Example 1) or a salt thereof, 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2R)-2-(5-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Example 3) or a salt thereof, or 2-[(1R,2R)-1,2-bis(5-fluoropyridin-2-yl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Example 5) or a salt thereof.

When compound (I) is in a form of a salt, examples thereof include metal salts, an ammonium salt, salts with organic base, salts with inorganic acid, salts with organic acid, salts with basic or acidic amino acid, and the like. Preferable examples of the metal salt include alkali metal salts such as sodium salt, potassium salt and the like;

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alkaline earth metal salts such as calcium salt, magnesium salt, barium salt and the like; an aluminum salt, and the like. Preferable examples of the salt with organic base include salts with trimethylamine, triethylamine, pyridine, picoline, 2,6-lutidine, ethanolamine, diethanolamine, triethanolamine, cyclohexylamine, dicyclohexylamine, N,N'-dibenzylethylenediamine and the like. Preferable examples of the salt with inorganic acid include salts with hydrochloric acid, hydrobromic acid, nitric acid, sulfuric acid, phosphoric acid and the like. Preferable examples of the salt with organic acid include salts with formic acid, acetic acid, trifluoroacetic acid, phthalic acid, fumaric acid, oxalic acid, tartaric acid, maleic acid, citric acid, succinic acid, malic acid, methanesulfonic acid, benzenesulfonic acid, p-toluenesulfonic acid and the like. Preferable examples of the salt with basic amino acid include salts with arginine, lysine, ornithine and the like. Preferable examples of the salt with acidic amino acid include salts with aspartic acid, glutamic acid and the like.

Among them, a pharmaceutically acceptable salt is preferable. For example, when a compound has an acidic functional group, examples thereof include inorganic salts such as alkali metal salts (e.g., sodium salt, potassium salt etc.), alkaline earth metal salts (e.g., calcium salt, magnesium salt etc.) and the like, ammonium salt etc., and when a compound has a basic functional group, examples thereof include salts with inorganic acid such as hydrochloric acid, hydrobromic acid, nitric acid, sulfuric acid, phosphoric acid and the like, and salts with organic acid such as acetic acid, phthalic acid, fumaric acid, oxalic acid, tartaric acid, maleic acid, citric acid, succinic acid, methanesulfonic acid, benzenesulfonic acid, p-toluenesulfonic acid and the like.

[Production Method]

The production method of compound (I) is explained below.

The raw material compound and reagent used and the compound obtained in each step in the following production method may be each in a form of a salt, and examples of such salt include those similar to the salts of compound (I) and the like.

When the compound obtained in each step is a free form, it can be converted to the objective salt according to a method known per se. When the compound obtained in each step is a salt, it can be converted to the objective free form or the other salt according to a method known per se.

The compound obtained in each step can be used directly as the reaction mixture or as a crude product for the next reaction. Alternatively, the compound obtained in each step can be isolated and purified from a reaction mixture according to a method known per se, for example, a separation means such as concentration, crystallization, recrystallization, distillation, solvent extraction, fractional distillation, chromatography and the like.

When the raw material compound and reagent used in each step are commercially available, the commercially available product can also be used directly.

In the reaction in each step, while the reaction time varies depending on the kind of the reagent and solvent to be used, it is generally 1 min-48 hr, preferably 10 min-8 hr, unless otherwise specified.

In the reaction in each step, while the reaction temperature varies depending on the kind of the reagent and solvent to be used, it is generally -78° C.-300° C., preferably -78° C.-150° C., unless otherwise specified.

In the reaction in each step, while the pressure varies depending on the kind of the reagent and solvent to be used, it is generally 1 atm-20 atm, preferably 1 atm-3 atm, unless otherwise specified.

Microwave synthesizer such as Initiator manufactured by Biotage and the like may be used for the reaction in each step. While the reaction temperature varies depending on the kind of the reagent and solvent to be used, it is generally room temperature—300° C., preferably 50° C.-250° C., unless otherwise specified. While the reaction time varies depending on the kind of the reagent and solvent to be used, it is generally 1 min-48 hr, preferably 1 min-8 hr, unless otherwise specified.

In the reaction in each step, the reagent is used in an amount of 0.5 equivalents-20 equivalents, preferably 0.8 equivalents-5 equivalents, relative to the substrate, unless otherwise specified. When the reagent is used as a catalyst, the reagent is used in an amount of 0.001 equivalent-1 equivalent, preferably 0.01 equivalent-0.2 equivalent, relative to the substrate. When the reagent is used as a reaction solvent, the reagent is used in a solvent amount.

Unless otherwise specified, the reaction in each step is carried out without solvent, or by dissolving or suspending the raw material compound in a suitable solvent. Examples of the solvent include those described in Examples and the following solvents.

alcohols: methanol, ethanol, tert-butyl alcohol, 2-methoxyethanol and the like;
ethers: diethyl ether, diphenyl ether, tetrahydrofuran, 1,2-dimethoxyethane and the like;
aromatic hydrocarbons: chlorobenzene, toluene, xylene and the like;
saturated hydrocarbons: cyclohexane, hexane and the like; amides: N,N-dimethylformamide, N-methylpyrrolidone and the like;
halogenated hydrocarbons: dichloromethane, carbon tetrachloride and the like;
nitriles: acetonitrile and the like;
sulfoxides: dimethyl sulfoxide and the like;
aromatic organic bases: pyridine and the like;
anhydrides: acetic anhydride and the like;
organic acids: formic acid, acetic acid, trifluoroacetic acid and the like;
inorganic acids: hydrochloric acid, sulfuric acid and the like; esters: ethyl acetate and the like;

ketones: acetone, methyl ethyl ketone and the like; water. The above-mentioned solvent can be used in a mixture of two or more kinds thereof in an appropriate ratio.

When a base is used for the reaction in each step, examples thereof include those described in Examples and the following bases.

inorganic bases: sodium hydroxide, magnesium hydroxide, sodium carbonate, calcium carbonate, sodium hydrogencarbonate, cesium carbonate, potassium acetate and the like;
organic bases: triethylamine, diethylamine, N,N-diisopropylethylamine (DIPEA), pyridine, 4-dimethylaminopyridine, N,N-dimethylaniline, 1,4-diazabicyclo[2.2.2]octane, 1,8-diazabicyclo[5.4.0]-7-undecene, imidazole, piperidine and the like;
metal alkoxides: sodium ethoxide, potassium tert-butoxide and the like;
alkali metal hydrides: sodium hydride and the like; metal amides: sodium amide, lithium diisopropylamide, lithium hexamethyldisilazide and the like;
organic lithiums: n-butyllithium and the like.

When an acid or an acid catalyst is used for the reaction in each step, examples thereof include those described in Examples and the following acids and acid catalysts.

inorganic acids: hydrochloric acid, sulfuric acid, nitric acid, hydrobromic acid, phosphoric acid and the like;
organic acids: acetic acid, trifluoroacetic acid, citric acid, p-toluenesulfonic acid, 10-camphorsulfonic acid and the like; Lewis acid: boron trifluoride diethyl ether complex, zinc iodide, anhydrous aluminium chloride, anhydrous zinc chloride, anhydrous iron chloride and the like.

Unless otherwise specified, the reaction in each step is carried out according to a method known per se, for example, the method described in Jikken Kagaku Kouza, 5th Edition, vol. 13-19 (the Chemical Society of Japan ed.); Shin Jikken Kagaku Kouza, vol. 14-15 (the Chemical Society of Japan ed.); Fine Organic Chemistry, Revised 2nd Edition (L. F. Fieser, Th. Eicher, Nankodo); Organic Name Reactions, the Reaction Mechanism and Essence, Revised Edition (Hideo Togo, Kodansha); ORGANIC SYNTHESIS Collective Volume I-VII (John Wiley & Sons Inc.); Modern Organic Synthesis in the Laboratory A Collection of Standard Experimental Procedures (Jie Jack Li, OXFORD UNIVERSITY); Comprehensive Heterocyclic Chemistry III, Vol. 1-Vol. 14 (Elsevier Japan); Strategic Applications of Named Reactions in Organic Synthesis (translated by Kiyoshi Tomioka, Kagakudojin); Comprehensive Organic Transformations (VCH Publishers Inc.), 1989, or the like, or the method described in Examples.

In each step, the protection or deprotection reaction of a functional group is carried out according to a method known per se, for example, the method described in "Protective Groups in Organic Synthesis, 4th Ed", Wiley-Interscience, Inc., 2007 (Theodora W. Greene, Peter G. M. Wuts); "Protecting Groups 3rd Ed." Thieme, 2004 (P. J. Kocienski), or the like, or the method described in Examples.

Examples of the protecting group for a hydroxy group of an alcohol and the like and a phenolic hydroxy group include ether-type protecting groups such as methoxymethyl ether, benzyl ether, tert-butyldimethylsilyl ether, tetrahydropyranyl ether and the like; carboxylate ester-type protecting groups such as acetate ester and the like; sulfonate ester-type protecting groups such as methanesulfonate ester and the like; carbonate ester-type protecting groups such as tert-butylcarbonate and the like, and the like.

Examples of the protecting group for a carbonyl group of an aldehyde include acetal-type protecting groups such as dimethylacetal and the like; cyclic acetal-type protecting groups such as 1,3-dioxane and the like, and the like.

Examples of the protecting group for a carbonyl group of a ketone include ketal-type protecting groups such as dimethylketal and the like; cyclic ketal-type protecting groups such as 1,3-dioxane and the like; oxime-type protecting groups such as O-methyloxime and the like; hydrazone-type protecting groups such as N,N-dimethylhydrazone and the like, and the like.

Examples of the protecting group for a carboxyl group include ester-type protecting groups such as methyl ester and the like; amide-type protecting groups such as N,N-dimethylamide and the like, and the like.

Examples of the protecting group for a thiol include ether-type protecting groups such as benzyl thioether and the like; ester-type protecting groups such as thioacetate ester, thiocarbonate, thiocarbamate and the like, and the like.

Examples of the protecting group for an amino group and an aromatic heterocycle such as imidazole, pyrrole, indole and the like include carbamate-type protecting groups such

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as benzyl carbamate and the like; amide-type protecting groups such as acetamide and the like; alkyl amine-type protecting groups such as N-triphenylmethylamine and the like; sulfonamide-type protecting groups such as methane-sulfonamide and the like, and the like.

The protecting groups can be removed according to a method known per se, for example, by employing a method using acid, base, ultraviolet rays, hydrazine, phenylhydrazine, sodium N-methyldithiocarbamate, tetrabutylammonium fluoride, palladium acetate, trialkylsilyl halide (e.g., trimethylsilyl iodide, trimethylsilyl bromide) and the like, a reduction method, and the like.

When reduction reaction is carried out in each step, examples of the reducing agent to be used include metal hydrides such as lithium aluminium hydride, sodium triacetoxyborohydride, sodium cyanoborohydride, diisobutylaluminium hydride (DIBAL-H), sodium borohydride, tetramethylammonium triacetoxyborohydride and the like; boranes such as borane tetrahydrofuran complex and the like; Raney nickel; Raney cobalt; hydrogen; formic acid; triethylsilane and the like. When carbon-carbon double bond or triple bond is reduced, a method using a catalyst such as palladium-carbon, Lindlar's catalyst and the like may be employed.

When oxidation reaction is carried out in each step, examples of the oxidizing agent to be used include peroxides such as m-chloroperbenzoic acid (mCPBA), hydrogen peroxide, tert-butylhydroperoxide and the like; perchlorates such as tetrabutylammonium perchlorate and the like; chlorates such as sodium chlorate and the like; chlorites such as sodium chlorite and the like; periodates such as sodium periodate and the like; hypervalent iodine reagents such as iodosylbenzene and the like; reagents containing manganese such as manganese dioxide, potassium permanganate and the like; leads such as lead tetraacetate and the like; reagents containing chromium such as pyridinium chlorochromate (PCC), pyridinium dichromate (PDC), Jones reagent and the like; halogen compounds such as N-bromosuccinimide (NBS) and the like; oxygen; ozone; sulfur trioxide-pyridine complex; osmium tetroxide; selenium dioxide; 2,3-dichloro-5,6-dicyano-1,4-benzoquinone (DDQ) and the like.

When radical cyclization reaction is carried out in each step, examples of the radical initiator to be used include azo compounds such as azobisisobutyronitrile (AIBN) and the like; water-soluble radical initiators such as 4-4'-azobis-4-cyanopentanoic acid (ACPA) and the like; triethylboron in the presence of air or oxygen; benzoyl peroxide and the like. Examples of the radical reagent to be used include tributylstannane, tris(trimethylsilyl)silane, 1,1,2,2-tetraphenyldisilane, diphenylsilane, samarium iodide and the like.

When Wittig reaction is carried out in each step, examples of the Wittig reagent to be used include alkylidene phosphoranes and the like. The alkylidene phosphoranes can be prepared according to a method known per se, for example, by reacting a phosphonium salt with a strong base.

When Horner-Emmons reaction is carried out in each step, examples of the reagent to be used include phosphonoacetates such as methyl dimethylphosphonoacetate, ethyl diethylphosphonoacetate and the like; and bases such as alkali metal hydrides, organic lithiums and the like.

When Friedel-Crafts reaction is carried out in each step, examples of the reagent to be used include a combination of a Lewis acid and an acid chloride or a combination of a Lewis acid and an alkylating agent (e.g., an alkyl halide, an alcohol, an olefin etc.). Alternatively, an organic acid or an inorganic acid can also be used instead of a Lewis acid, and

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an anhydride such as acetic anhydride and the like can also be used instead of an acid chloride.

When aromatic nucleophilic substitution reaction is carried out in each step, a nucleophile (e.g., an amine, imidazole etc.) and a base (e.g., an organic base etc.) are used as reagents.

When nucleophilic addition reaction by a carbo anion, nucleophilic 1,4-addition reaction (Michael addition reaction) by a carbo anion or nucleophilic substitution reaction by a carbo anion is carried out in each step, examples of the base to be used for generation of the carbo anion include organic lithiums, metal alkoxides, inorganic bases, organic bases and the like.

When Grignard reaction is carried out in each step, examples of the Grignard reagent to be used include arylmagnesium halides such as phenylmagnesium bromide and the like; and alkylmagnesium halides such as methylmagnesium bromide and the like. The Grignard reagent can be prepared according to a method known per se, for example, by reacting an alkyl halide or an aryl halide with a metal magnesium or isopropylmagnesium chloride-lithium chloride complex in an ether, tetrahydrofuran and the like as a solvent.

When Knoevenagel condensation reaction is carried out in each step, a compound having an activated methylene group with two electron withdrawing groups (e.g., malonic acid, diethyl malonate, malononitrile etc.) and a base (e.g., an organic base, a metal alkoxide, an inorganic base) are used as a reagent.

When Vilsmeier-Haack reaction is carried out in each step, phosphoryl chloride and an amide derivative (e.g., N,N-dimethylformamide etc.) are used as a reagent.

When azidation reaction of an alcohol, an alkyl halide or a sulfonate is carried out in each step, examples of the azidating agent to be used include diphenylphosphorylazide (DPPA), trimethylsilylazide, sodium azide and the like. For example, for the azidation reaction of an alcohol, a method using diphenylphosphorylazide and 1,8-diazabicyclo[5.4.0]undec-7-ene (DBU), a method using trimethylsilylazide and a Lewis acid, a method using diphenylphosphorylazide, triphenylphosphine and azodicarboxylate, and the like are employed.

When reductive amination reaction is carried out in each step, examples of the reducing agent to be used include sodium triacetoxyborohydride, sodium cyanoborohydride, hydrogen, formic acid and the like. When the substrate is an amine compound, examples of the carbonyl compound to be used include paraformaldehyde, aldehydes such as acetaldehyde and the like, and ketones such as cyclohexanone and the like. When the substrate is a carbonyl compound, examples of the amine to be used include ammonia, primary amines such as methylamine and the like; secondary amines such as dimethylamine and the like, and the like.

When Mitsunobu reaction is carried out in each step, an azodicarboxylate (e.g., diethyl azodicarboxylate (DEAD), diisopropyl azodicarboxylate (DIAD) etc.) and triphenylphosphine are used as a reagent.

When esterification reaction, amidation reaction or ureation reaction is carried out in each step, examples of the reagent to be used include acyl halides such as acid chlorides, acid bromides and the like; activated carboxylic acids such as anhydrides, activated esters, sulfates and the like. Examples of the activating agent of the carboxylic acid include carbodiimide condensing agents such as 1-ethyl-3-(3-dimethylaminopropyl)carbodiimide hydrochloride (EDC HCl) and the like; triazine condensing agents such as 4-(4,6-dimethoxy-1,3,5-triazin-2-yl)-4-methylmorpho-

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linium chloride n-hydrate (DMT-MM) and the like; carbon-
ate condensing agents such as 1,1-carbonyldiimidazole
(CDI) and the like; diphenylphosphorylazide (DPPA); benz-
otriazol-1-yloxy-trisdimethylaminophosphonium salt
(BOP reagent); 2-chloro-1-methyl-pyridinium iodide (Mu-
kaiyama reagent); thionyl chloride; lower alkyl haloform-
ates such as ethyl chloroformate and the like; O-(7-
azabenzotriazol-1-yl)-N,N,N',N'-tetramethyluronium
hexafluorophosphate (HATU); sulfuric acid; combinations
thereof and the like. When carbodiimide condensing agent is
used, an additive such as 1-hydroxybenzotriazole (HOBt),
N-hydroxysuccinimide (HOSu), dimethylaminopyridine
(DMAP) and the like may be added to the reaction system.

When coupling reaction is carried out in each step,
examples of the metal catalyst to be used include palladium
compounds such as palladium(II) acetate, tetrakis(triphenylphosphine)palladium(0), dichlorobis(triphenylphosphine)palladium(0), dichlorobis(triethylphosphine)palladium(II), tris(dibenzylideneacetone)dipalladium(0), 1,1'-bis(diphenylphosphino)ferrocene palladium(II) chloride and the like; nickel compounds such as tetrakis(triphenylphosphine)nickel(0) and the like; rhodium compounds such as tris(triphenylphosphine)rhodium(III) chloride and the like; cobalt compounds; copper compounds such as copper oxide, copper(I) iodide and the like; platinum compounds and the like. In addition, a base can be added to the reaction system, and examples thereof include inorganic bases and the like.

When thiocarbonylation reaction is carried out in each
step, phosphorus pentasulfide is typically used as the thio-
carbonylating agent. Alternatively, a reagent having a 1,3,
2,4-dithiadiphosphetane-2,4-disulfide structure (e.g., 2,4-bis
(4-methoxyphenyl)-1,3,2,4-dithiadiphosphetane-2,4-
disulfide (Lawesson reagent) etc.) can also be used instead
of phosphorus pentasulfide.

When halogenation reaction is carried out in each step,
examples of the halogenating agent to be used include
N-iodosuccinimide, N-bromosuccinimide (NBS), N-chloro-
succinimide (NCS), bromine, sulfuryl chloride, trichloroi-
socyanuric acid and the like. In addition, the reaction can be
accelerated by subjecting a radical initiator such as heat,
light, benzoyl peroxide, azobisisobutyronitrile and the like
to the reaction system reaction.

When halogenation reaction of a hydroxy group is carried
out in each step, examples of the halogenating agent to be
used include hydrohalic acids and acid halides of inorganic
acids, specifically, hydrochloric acid, thionyl chloride, phos-
phorus oxychloride and the like for chlorination, 48% hyd-
robromic acid and the like for bromination. In addition, a
method of producing an alkyl halide by reacting an alcohol
with triphenylphosphine and carbon tetrachloride or carbon
tetrabromide or the like can be employed. Alternatively, a
method of producing an alkyl halide via two steps compris-
ing converting an alcohol to the corresponding sulfonate,
and then reacting the sulfonate with lithium bromide,
lithium chloride or sodium iodide can also be employed.

When Arbuzov reaction is carried out in each step,
examples of the reagent to be used include alkyl halides such
as ethyl bromoacetate and the like; and phosphites such as
triethyl phosphite, tri(isopropyl) phosphite and the like.

When sulfonate esterification reaction is carried out in
each step, examples of the sulfonating agent to be used
include methanesulfonyl chloride, p-toluenesulfonyl chlo-
ride, methanesulfonic anhydride, p-toluenesulfonic anhy-
dride and the like.

When hydrolysis reaction is carried out in each step, an
acid or a base is used as a reagent. For acid hydrolysis

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reaction of tert-butyl ester, formic acid, triethylsilane and the
like may be added to reductively-trap tert-butyl cation which
is by-produced.

When dehydration reaction is carried out in each step,
examples of the dehydrating agent to be used include
sulfuric acid, diphosphorus pentaoxide, phosphorus oxy-
chloride, N,N'-dicyclohexylcarbodiimide, alumina, poly-
phosphoric acid and the like.

When alkylation reaction is carried out in each step, an
electrophile (e.g., an alkyl halide etc.) and a base (e.g., an
organic base, an inorganic base, a metal alkoxide, a metal
amide etc.) are used as reagents.

When cyanidation reaction is carried out in each step,
examples of the reagent to be used include metal cyanides
such as copper cyanide, potassium cyanide and the like.

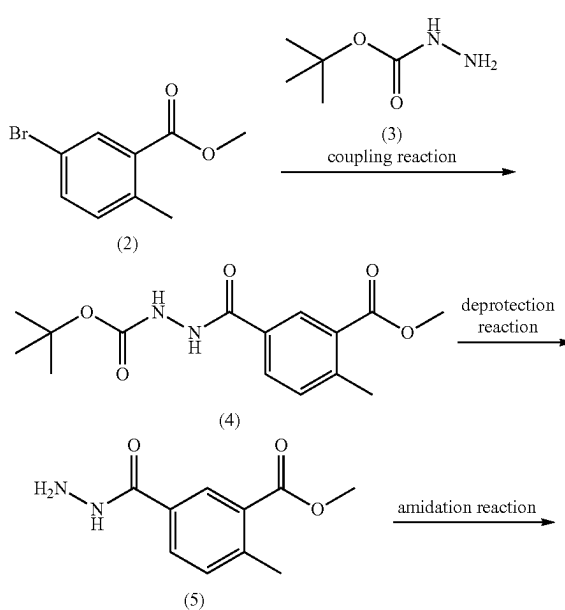
Compound (I) can be produced according to the produc-
tion methods shown in the following Scheme 1 or Scheme
2. Each symbol in the formulas of the schemes is as defined
above, unless otherwise specified.

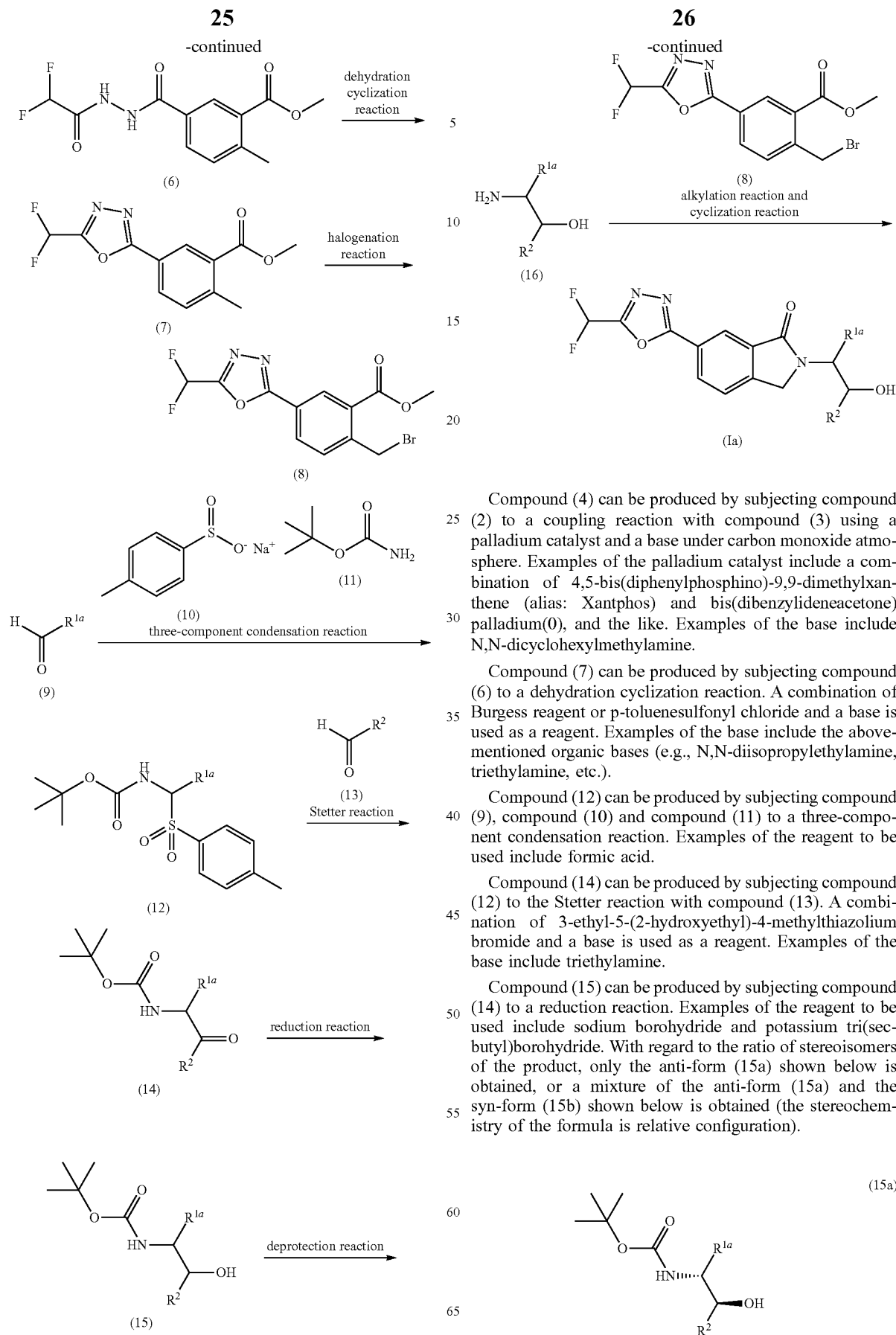
Moreover, compound (I) can be produced by carrying out
protection reaction, deprotection reaction, amidation reac-
tion, sulfonamidation reaction, ureation reaction, carbamoy-
lation reaction, alkylation reaction, Mitsunobu reaction,
hydrogenation reaction, oxidation reaction, reduction reac-
tion, halogenation reaction, coupling reaction, nucleophilic
addition reaction by a carbo anion, Grignard reaction,
deoxofluorination reaction, dehydration reaction and the like
singly or two or more thereof in combination.

The below-mentioned compound (Ia) (compound (I)
wherein

R^1 is R^{1a}) can be produced according to the method
shown in the following Scheme 1. R^{1a} is an optionally
substituted cyclopropyl group, an optionally substi-
tuted C_{6-14} aryl group, or an optionally substituted 5- or
6-membered monocyclic aromatic heterocyclic group.
The other symbols are as defined above.

Scheme 1





Compound (4) can be produced by subjecting compound (2) to a coupling reaction with compound (3) using a palladium catalyst and a base under carbon monoxide atmosphere. Examples of the palladium catalyst include a combination of 4,5-bis(diphenylphosphino)-9,9-dimethylxanthene (alias: Xantphos) and bis(dibenzylideneacetone) palladium(0), and the like. Examples of the base include N,N-dicyclohexylmethylamine.

Compound (7) can be produced by subjecting compound (6) to a dehydration cyclization reaction. A combination of Burgess reagent or p-toluenesulfonyl chloride and a base is used as a reagent. Examples of the base include the above-mentioned organic bases (e.g., N,N-diisopropylethylamine, triethylamine, etc.).

Compound (12) can be produced by subjecting compound (9), compound (10) and compound (11) to a three-component condensation reaction. Examples of the reagent to be used include formic acid.

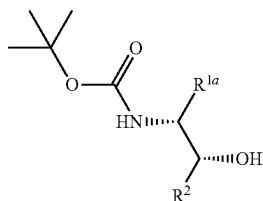
Compound (14) can be produced by subjecting compound (12) to the Stetter reaction with compound (13). A combination of 3-ethyl-5-(2-hydroxyethyl)-4-methylthiazolium bromide and a base is used as a reagent. Examples of the base include triethylamine.

Compound (15) can be produced by subjecting compound (14) to a reduction reaction. Examples of the reagent to be used include sodium borohydride and potassium tri(sec-butyl)borohydride. With regard to the ratio of stereoisomers of the product, only the anti-form (15a) shown below is obtained, or a mixture of the anti-form (15a) and the syn-form (15b) shown below is obtained (the stereochemistry of the formula is relative configuration).

(15a)

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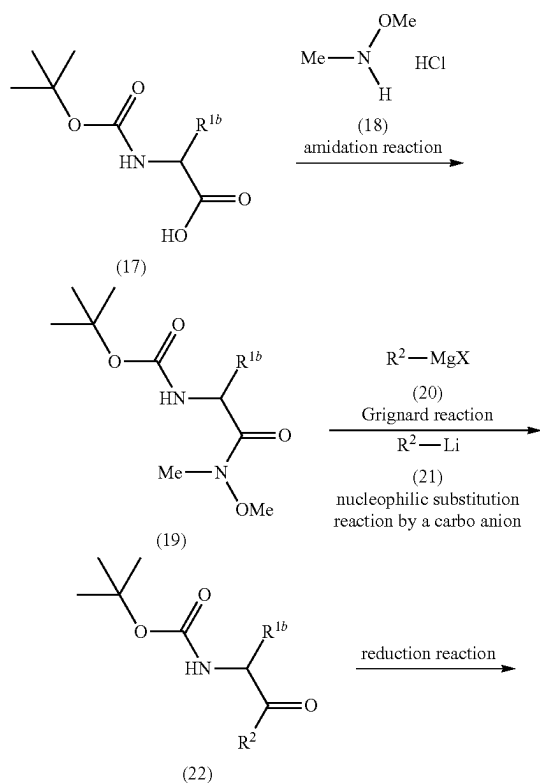


Compound (1a) can be produced by subjecting compound (16) to an alkylation reaction with compound (8), followed by a cyclization reaction. The cyclization reaction follows the alkylation reaction, but it may be carried out in stages. In this case, the cyclization reaction can be carried out under basic or acidic condition. Examples of the base include triethylamine, DIPEA, potassium acetate and the like. Examples of the acid to be used include acetic acid and the like.

The below-mentioned compound (1b) (compound (I) wherein

R^1 is R^{1b}) can be produced from compound (17) according to the method shown in the following Scheme 2. R^{1b} is an optionally substituted cycloalkyl group, an optionally substituted C_{6-14} aryl group, an optionally substituted 5- or 6-membered monocyclic aromatic heterocyclic group, or an optionally substituted 3- to 8-membered monocyclic non-aromatic heterocyclic group. X is a halogen atom. The other symbols are as defined above.

Scheme 2



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-continued

(15b)

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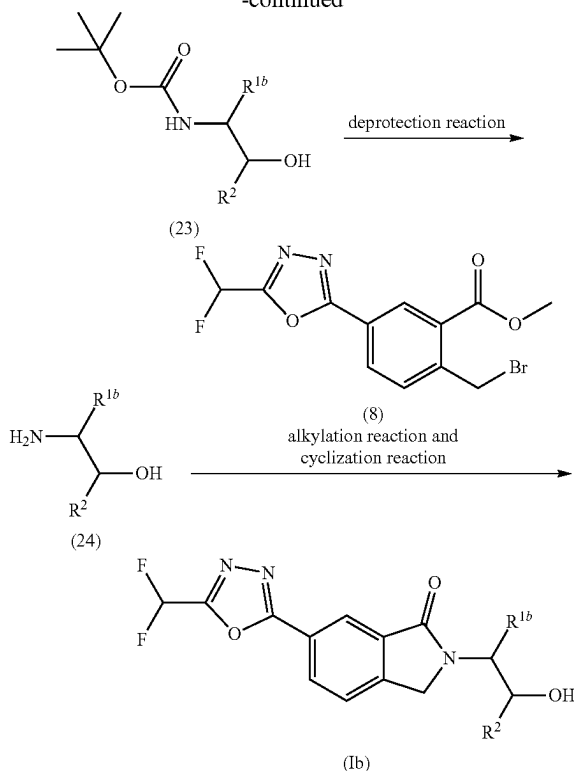
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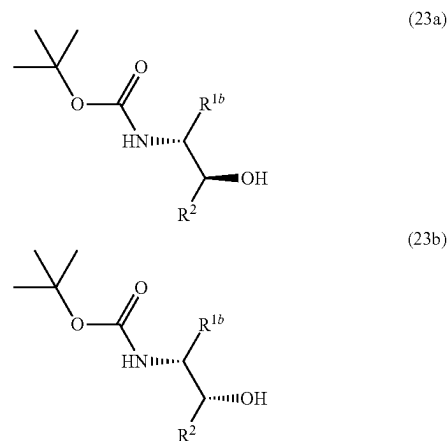
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Compound (23) can be produced by subjecting compound (22) to a reduction reaction. Examples of the reagent to be used include sodium borohydride and potassium tri(sec-butyl)borohydride. Alternatively, a combination of aluminium triisopropoxide and 2-propanol may also be used. With regard to the ratio of stereoisomers of the product, only the anti-form (23a) shown below is obtained, or a mixture of the anti-form (23a) and the syn-form (23b) shown below is obtained (the stereochemistry of the formula is relative configuration).



Compound (1b) can be produced by subjecting compound (24) to an alkylation reaction with compound (8), followed by a cyclization reaction. The cyclization reaction follows the alkylation reaction, but it may be carried out in stages. In this case, the cyclization reaction can be carried out under

basic or acidic condition. Examples of the base include triethylamine, DIPEA, potassium acetate and the like. Examples of the acid to be used include acetic acid and the like.

Compounds (2), (3), (9), (10), (11), (13), (17), (18), (20) and (21) which are used as raw materials in each production method may be commercially easily available or can be produced according to a method known per se.

As for the configurational isomers (E, Z forms) of compound (I), they can be isolated and purified when isomerization occurs, for example, according to a conventional separation means such as extraction, recrystallization, distillation, chromatography and the like to obtain a pure compound. In addition, the corresponding pure isomer can also be obtained by isomerizing a double bond using heating, an acid catalyst, a transition metal complex, a metal catalyst, a radical catalyst, light irradiation, a strong base catalyst and the like, according to the method described in Shin Jikken Kagaku Kouza 14 (The Chemical Society of Japan ed.), pages 251 to 253, or 4th Edition Jikken Kagaku Kouza 19 (The Chemical Society of Japan ed.), pages 273 to 274, or a method analogous thereto.

Compound (I) contains a stereoisomer depending on the kind of a substituent, and each stereoisomer and a mixture thereof are encompassed in the present invention.

Compound (I) may be a hydrate or a non-hydrate.

When the objective product is obtained as a free form by the above-mentioned reaction, it can be converted to a salt according to a conventional method, or when the objective product is obtained as a salt, it can be converted to a free form or other salt according to a conventional method. The thus-obtained compound (I) can also be isolated and purified from a reaction mixture according to a known method such as phase transfer, concentration, solvent extraction, distillation, crystallization, recrystallization, chromatography and the like.

When compound (I) contains a configurational isomer, a diastereomer, a conformer and the like, each can be isolated according to the above-mentioned separation and purification methods, if desired. In addition, when compound (I) is racemic, d-form and l-form can be isolated according to a conventional optical resolution such as preparative high performance liquid chromatography (preparative HPLC), supercritical fluid chromatography (preparative SFC) and the like.

The thus-obtained compound (I), other reaction intermediate therefor and starting compounds thereof can be isolated and purified from a reaction mixture according to a method known per se, for example, extraction, concentration, neutralization, filtration, distillation, recrystallization, column chromatography, thin layer chromatography, preparative high performance liquid chromatography (preparative HPLC), moderate-pressure preparative liquid chromatography (moderate-pressure preparative LC) and the like.

A salt of compound (I) can be produced according to a method known per se. For example, when compound (I) is a basic compound, it can be produced by adding an inorganic acid or organic acid, or when compound (I) is an acidic compound, by adding an organic base or inorganic base.

When compound (I) contains an optical isomer, each optical isomer and a mixture thereof are encompassed in the scope of the present invention, and these isomers can be subjected to optical resolution or can be produced respectively, according to a method known per se, if desired.

Compound (I) may be a crystal.

The crystal of compound (I) can be produced according to a crystallization method known per se.

Examples of the crystallization method include crystallization method from a solution, crystallization method from vapor, crystallization method from a melt, and the like.

The "crystallization method from a solution" is typically a method of shifting a non-saturated state to supersaturated state by varying factors involved in solubility of compounds (solvent composition, pH, temperature, ionic strength, redox state, etc.) or the amount of solvent. Specific examples thereof include a concentration method, a slow cooling method, a reaction method (a diffusion method, an electrolysis method), a hydrothermal growth method, a flux method and the like. Examples of the solvent to be used include aromatic hydrocarbons (e.g., benzene, toluene, xylene, etc.), halogenated hydrocarbons (e.g., dichloromethane, chloroform, etc.), saturated hydrocarbons (e.g., hexane, heptane, cyclohexane, etc.), ethers (e.g., diethyl ether, diisopropyl ether, tetrahydrofuran, dioxane, etc.), nitriles (e.g., acetonitrile, etc.), ketones (e.g., acetone, etc.), sulfoxides (e.g., dimethyl sulfoxide, etc.), acid amides (e.g., N,N-dimethylformamide, etc.), esters (e.g., ethyl acetate, isopropyl acetate, etc.), alcohols (e.g., methanol, ethanol, 2-propanol, etc.), water and the like. These solvents are used alone or in a combination of two or more at a suitable ratio (e.g., 1:1 to 1:100 (a volume ratio)). Where necessary, a seed crystal can be used.

The "crystallization method from vapor" is, for example, a vaporization method (a sealed tube method, a gas stream method), a gas phase reaction method, a chemical transportation method and the like.

The "crystallization method from a melt" is, for example, a normal freezing method (a pulling method, a temperature gradient method, a Bridgman method), a zone melting method (a zone leveling method, a floating zone method), a special growth method (a VLS method, a liquid phase epitaxy method) and the like.

Preferable examples of the crystallization method include a method comprising dissolving compound (I) in a suitable solvent (e.g., alcohols such as methanol, ethanol etc.) at 20° C. to 120° C., and cooling the obtained solution to a temperature (e.g., 0 to 50° C., preferably 0 to 20° C.) not higher than the dissolution temperature, and the like.

The thus-obtained crystals of the present invention can be isolated, for example, by filtration and the like.

An analysis method of the obtained crystal is generally a method of crystal analysis by powder X-ray diffraction. As a method of determining crystal orientation, a mechanical method or an optical method and the like can also be used.

The crystal of compound (I) obtained by the above-mentioned production method may have high purity, high quality, and low hygroscopicity, may not be denatured even after a long-term preservation under general conditions, and may be expected to be superior in the stability. In addition, it may be also superior in the biological properties (e.g., pharmacokinetics (absorption, distribution, metabolism, excretion), efficacy expression etc.) and may be extremely useful as a medicament.

Compound (I) may be a prodrug. The prodrug of compound (I) means a compound which is converted to compound (I) with a reaction due to an enzyme, gastric acid and the like under the physiological condition in the living body, that is, a compound which is converted to compound (I) by enzymatic oxidation, reduction, hydrolysis and the like; a compound which is converted to compound (I) by hydrolysis and the like due to gastric acid, and the like. Examples of the prodrug for compound (I) include a compound obtained by subjecting an amino group in compound (I) to acylation, alkylation or phosphorylation (e.g., a compound

obtained by subjecting an amino group in compound (I) to eicosanoylation, alanylation, pentylaminocarbonylation, (5-methyl-2-oxo-1,3-dioxolen-4-yl)methoxycarbonylation, tetrahydrofurylation, pyrrolidylmethylation, pivaloyloxymethylation or tert-butylation); a compound obtained by subjecting a hydroxy group in compound (I) to acylation, alkylation, phosphorylation or boration (e.g., a compound obtained by subjecting a hydroxy group in compound (I) to acetylation, palmitoylation, propanoylation, pivaloylation, succinylation, fumarylation, alanylation or dimethylaminomethylcarbonylation); a compound obtained by subjecting a carboxyl group in compound (I) to esterification or amidation (e.g., a compound obtained by subjecting a carboxyl group in compound (I) to ethyl esterification, phenyl esterification, carboxymethyl esterification, dimethylaminomethyl esterification, pivaloyloxymethyl esterification, ethoxycarbonyloxyethyl esterification, phthalidyl esterification, (5-methyl-2-oxo-1,3-dioxolen-4-yl)methyl esterification, cyclohexyloxycarbonyl ethyl esterification or methylamidation) and the like. These compounds can be produced from compound (I) according to a method known per se.

The prodrug of compound (I) may also be one which is converted to compound (I) under physiological conditions as described in "IYAKUHIN no KAIHATSU (Development of Pharmaceuticals)", Vol. 7, Design of Molecules, p. 163-198, Published by HIROKAWA SHOTEN (1990).

In the present specification, compound (I) and the prodrug of compound (I) are sometimes collectively abbreviated as "the compound of the present invention".

Compound (I) may be a hydrate, a non-hydrate, a solvate or a non-solvate.

In addition, compound (I) may be a compound labeled or substituted with an isotope (e.g., ^2H , ^3H , ^{11}C , ^{14}C , ^{18}F , ^{35}S , ^{125}I) and the like. The compound labeled or substituted with an isotope may be used, for example, as a tracer (PET tracer) used in positron emission tomography (PET), and useful in the field of medical diagnosis and the like.

Compound (I) also encompasses a deuterium conversion form wherein ^1H is converted to $^2\text{H}(\text{D})$.

Compound (I) also encompasses a tautomer thereof.

Compound (I) may be a pharmaceutically acceptable cocrystal or a salt thereof. The cocrystal or a salt thereof means a crystalline substance constituted with two or more special solids at room temperature, each having different physical properties (e.g., structure, melting point, melting heat, hygroscopicity, solubility and stability). The cocrystal or a salt thereof can be produced according to a cocrystallization a method known per se.

Compound (I) may also be used as a PET tracer.

Since the compound of the present invention has a superior HDAC inhibitory action, preferably class II HDAC inhibitory action, more preferably HDAC6 inhibitory action, it may be also useful as safe medicaments based on such action.

For example, the medicament of the present invention containing the compound of the present invention may be expected to show low toxicity (e.g., acute toxicity, chronic toxicity, genetic toxicity, hematotoxicity, reproductive toxicity, cardiotoxicity, carcinogenicity), and used for a mammal (e.g., mouse, rat, hamster, rabbit, cat, dog, bovine, sheep, monkey, human) as a prophylactic or therapeutic agent for HDAC-associated diseases, preferably class II HDAC-associated diseases, more preferably HDAC6-associated diseases, more specifically, the diseases described in (1)-(7) below.

Particularly, the compound of the present invention may be expected to show low genetic toxicity, and therefore, the medicament of the present invention may be expected to show low genetic toxicity.

- (1) inflammatory diseases (e.g., acute pancreatitis, chronic pancreatitis, asthma, adult respiratory distress syndrome, chronic obstructive pulmonary disease (COPD), idiopathic pulmonary fibrosis, inflammatory bone disease, inflammatory pulmonary disease, inflammatory bowel disease, celiac disease, hepatitis, systemic inflammatory response syndrome (SIRS), post-operative or posttraumatic inflammation, pneumonia, nephritis, meningitis, cystitis, pharyngolaryngitis, gastric mucosal injury, spondylitis, arthritis, dermatitis, chronic pneumonia, bronchitis, pulmonary infarction, silicosis, pulmonary sarcoidosis, diabetic nephropathy, uveitis, suppurative hidradenitis etc.),
- (2) autoimmune diseases (e.g., rheumatoid arthritis, psoriasis, inflammatory bowel disease (e.g., Crohn's disease, ulcerative colitis etc.), Sjogren's syndrome, Behcet's disease, multiple sclerosis, systemic lupus erythematosus, lupus nephritis, discoid lupus erythematosus, Castleman's disease, ankylopoietic spondylarthritis, polymyositis, dermatomyositis (DM), polyarteritis nodosa (PN), mixed connective tissue disease (MCTD), scleroderma, profundus lupus erythematosus, chronic thyroiditis, Graves' disease, autoimmune gastritis, type I diabetes, autoimmune hemolytic anemia, autoimmune neutropenia, thrombocytopenia, atopic dermatitis, pemphigus, chronic active hepatitis, myasthenia gravis, graft versus host disease, Addison's disease, abnormal immunoresponse, arthritis, dermatitis, radiodermatitis, primary biliary cirrhosis etc.),
- (3) osteoarticular degenerative disease (e.g., rheumatoid arthritis, osteoporosis, osteoarthritis etc.),
- (4) neoplastic diseases [e.g., malignant tumor, angiogenesis glaucoma, infantile hemangioma, multiple myeloma, chronic sarcoma, metastasis melanoma, Kaposi's sarcoma, vascular proliferation, cachexia, metastasis of the breast cancer, cancer (e.g., colorectal cancer (e.g., familial colorectal cancer, hereditary non-polyposis colorectal cancer, gastrointestinal stromal tumor etc.), lung cancer (e.g., non-small cell lung cancer, small cell lung cancer, malignant mesothelioma etc.), mesothelioma, pancreatic cancer (e.g., pancreatic duct cancer etc.), gastric cancer (e.g., papillary adenocarcinoma, mucinous adenocarcinoma, adenosquamous carcinoma, etc.), breast cancer (e.g., invasive ductal carcinoma, ductal carcinoma in situ, inflammatory breast cancer etc.), ovarian cancer (e.g., ovarian epithelial carcinoma, extragonadal germ cell tumor, ovarian germ cell tumor, ovarian low malignant potential tumor etc.), prostate cancer (e.g., hormone-dependent prostate cancer, non-hormone dependent prostate cancer etc.), liver cancer (e.g., primary liver cancer, extrahepatic bile duct cancer etc.), thyroid cancer (e.g., medullary thyroid carcinoma etc.), kidney cancer (e.g., renal cell carcinoma, transitional cell carcinoma in kidney and urinary duct etc.), uterine cancer, brain tumor (e.g., pineal astrocytoma, pilocytic astrocytoma, diffuse astrocytoma, anaplastic astrocytoma etc.), melanoma, sarcoma, urinary bladder cancer, hematologic cancer and the like including multiple myeloma, hypophyseal adenoma, glioma, acoustic neuroma, retinoblastoma, pharyngeal cancer, laryngeal cancer, cancer of the tongue, thymoma, esophagus cancer, duodenal cancer, colorectal cancer, rectal cancer, hepa-

- toma, pancreatic endocrine tumor, bile duct cancer, gallbladder cancer, penile cancer, urinary duct cancer, testis tumor, vulvar cancer, cervix cancer, endometrial cancer, uterus sarcoma, cholinergic disease, vaginal cancer, skin cancer, fungoid mycosis, basal cell tumor, soft tissue sarcoma, malignant lymphoma, Hodgkin's disease, myelodysplastic syndrome, adult T cell leukemia, chronic bone marrow proliferative disease, pancreatic endocrine, tumor fibrous histiocytoma, leiomyosarcoma, rhabdomyosarcoma, cancer of unknown primary), leukemia (e.g., acute leukemia (e.g., acute lymphatic leukemia, acute myelocytic leukemia etc.), chronic leukemia (e.g., chronic lymphatic leukemia, chronic myelocytic leukemia etc.), myelodysplastic syndrome), uterine sarcoma (e.g., mixed mesodermal tumor, uterine leiomyosarcoma, endometrial stromal tumor etc.), myelofibrosis etc.],
- (5) neurodegenerative diseases and/or central diseases
- (i) psychiatric diseases [e.g., depression, major depression, bipolar depression, dysthymic disorder, emotional disorder (seasonal affective disorder and the like), recurrent depression, postpartum depression, stress disorder, depression symptom, mania, anxiety, generalized anxiety disorder, anxiety syndrome, panic disorder, phobia, social phobia, social anxiety disorder, obsessive disorder, post-traumatic stress syndrome, post-traumatic stress disorder, Tourette syndrome, autism, autistic spectrum syndrome, fragile X syndrome, Rett syndrome, adjustment disorder, bipolar disorder, neurosis, schizophrenia (e.g., positive symptom, negative symptom, cognitive symptom), cognitive dysfunction associated with schizophrenia, chronic fatigue syndrome, anxiety neurosis, compulsive neurosis, epilepsy, anxiety, anxious mental state, emotional abnormality, cyclothymia, nervous erethism, faint, addiction, low sex drive, attention deficit hyperactivity disorder (ADHD), psychotic major depression, refractory major depression, treatment-resistant depression, hereditary praplegia],
- (ii) neurodegenerative diseases [e.g., Alzheimer's disease, dementia of Alzheimer type, Alzheimer-type senile dementia, Parkinson's disease, muscular dystrophy, Parkinson's disease associated with dementia, Huntington's disease, multi-infarct dementia, frontotemporal lobar degeneration [progressive supranuclear palsy, corticobasal degeneration, frontotemporal dementia and parkinsonism linked to MAPT mutation (FTDP-17), frontotemporal dementia, Pick's disease, argyrophilic grain dementia etc.], Parkinson's type dementia, Niemann-Pick syndrome, Down's disease, vascular dementia, postencephalitic parkinsonism, Lewy body dementia, Rubinstein-Taybi syndrome, HIV dementia, amyotrophic lateral sclerosis (ALS), motor neurogenesis disease (MND), Creutzfeldt-Jakob disease or prion disease, cerebral palsy, multiple sclerosis, Riley-Day syndrome],
- (iii) age-related cognition memory disorders [e.g., age-related memory disorders, senile dementia],
- (iv) sleep disorders [e.g., intrinsic sleep disorders (e.g., psychophysiological insomnia and the like), extrinsic sleep disorder, circadian rhythm disorders (e.g., time zone change syndrome (jet lag), shift work sleep disorder, irregular sleep-wake pattern, delayed sleep phase syndrome, advanced sleep phase syndrome, non-24-hour sleep-wake and the like), parasomnia, sleep disorders associated with internal medical or psychiatric disorder (e.g., chronic obstructive pulmonary diseases,

- Alzheimer's disease, Parkinson's disease, cerebrovascular dementia, schizophrenia, depression, anxiety neurosis), stress insomnia, insomnia, insomniac neurosis, sleep apnea syndrome],
- (v) respiratory depression caused by anesthetics, traumatic disease, or neurodegenerative disease and the like,
- (vi) traumatic brain injury, cerebral apoplexy, neurotic anorexia, eating disorder, anorexia nervosa, hyperorexia, other eating disorder, alcohol dependence, alcohol abuse, alcoholic amnesia, alcohol paranoia, alcohol preference, alcohol withdrawal, alcoholic insanity, alcohol poisoning, alcoholic jealousy, alcoholic mania, alcohol-dependent psychiatric disorder, alcoholic insanity, pharmacophilia, pharmacophobia, pharmacomania, drug withdrawal, migraine, stress headache, catatonic headache, diabetic neuropathy, obesity, diabetes, muscular spasm, Meniere's disease, autonomic ataxia, alopecia, glaucoma, hypertension, cardiac disease, tachycardia, congestive cardiac failure, hyperventilation, bronchial asthma, apnea, sudden infant death syndrome, inflammatory bowel disease, allergic disease, impotence, climacteric disorder, infertility, cancer, immunodeficiency syndrome caused by HIV infection, immunodeficiency syndrome caused by stress, cerebrospinal meningitis, acromegaly, incontinence, metabolic syndrome, osteoporosis, peptic ulcer, irritable bowel syndrome, inflammatory bowel disease, ulcerative colitis, Crohn's disease, stress gastrointestinal disorder, stress vomiting, peptic ulcer, diarrhea, constipation, postoperative ileus,
- (vii) pain (pain, cancer pain, acute pain caused by inflammation, pain associated with chronic inflammation, postoperative pain (incision pain, deep pain, visceral pain, chronic pain after operation and the like), muscular pain (muscular pain associated with chronic pain disease, stiff shoulder and the like), arthralgia, toothache, temporomandibular joint pain, headache (migraine, catatonic headache, headache associated with fever, headache associated with hypertension), visceral pain (cardiac pain, angina pain, abdominal pain, renal pain, urinary tract pain, bladder pain), obstetric and gynecologic pain (mittelschmerz, dysmenorrhea, labor pain), neuropathic pain (hernia of intervertebral disk, nerve root pain, neuralgia after herpes zoster, trigeminal neuralgia, lumbago and the like), peripheral neuropathy (CIPN) derived from anticancer drugs (taxane anti-cancer agent (e.g., paclitaxel (taxol), docetaxel), vinca alkaloid anti-cancer agent (e.g., vincristine, vinblastine), platinum preparation (e.g., cisplatin, carboplatin, oxaliplatin), molecularly targeted drug (e.g., bortezomib) and the like) and neurological symptoms associated therewith (chemotherapy-induced neuropathic pain (CINP))),
- (6) chronic heart failure or acute heart failure, acute decompensated heart failure, ischemic heart disease, cardiomyopathy, myocarditis, valvular disease,
- (7) peripheral neuropathy and the like (e.g., demyelinating diseases and neuropathy (multiple sclerosis, Guillain-Barre syndrome, Fisher syndrome, chronic inflammatory demyelinating polyneuropathy (CIDP), multifocal motor neuropathy (MMN), Charcot-Marie-Tooth disease, hereditary sensory and autonomic neuropathy, familial amyloidotic polyneuropathy)).
- The medicament of the present invention may be preferably used as an agent for the prophylaxis or treatment of autoimmune disease, inflammatory disease, osteoarticular

degenerative disease, neurodegenerative disease, central disease, neoplastic disease, or peripheral neuropathy, more preferably inflammatory bowel disease (preferably Crohn's disease or ulcerative colitis), systemic lupus erythematosus, rheumatoid arthritis, psoriasis, Sjogren's syndrome, Behcet's syndrome, multiple sclerosis, graft versus host disease, Alzheimer's disease (preferably dementia of Alzheimer type), schizophrenia, dementia with Lewy Bodies, frontotemporal lobar degeneration [progressive supranuclear palsy, corticobasal degeneration, frontotemporal dementia and parkinsonism linked to MAPT mutation (FTDP-17), frontotemporal dementia, Pick's disease, argyrophilic grain dementia etc.], Parkinson's disease, Huntington's disease, Rubinstein-Taybi Syndrome, muscular dystrophy, Rett Syndrome, amyotrophic lateral sclerosis, Charcot-Marie-Tooth disease, depression, hereditary spastic paraplegia, Riley-Day syndrome, Castleman's disease, leukemia, uterine leiomyosarcoma, prostate cancer, colon cancer, multiple myeloma, cachexia or myelofibrosis, chronic heart failure or acute heart failure, acute decompensated heart failure, ischemic heart disease, cardiomyopathy, myocarditis, valvular disease, peripheral neuropathy and the like.

The medicament of the present invention may be more preferably used as an agent for the prophylaxis or treatment of Alzheimer's disease, frontotemporal lobar degeneration [progressive supranuclear palsy, corticobasal degeneration, frontotemporal dementia and parkinsonism linked to MAPT mutation (FTDP-17), frontotemporal dementia, Pick's disease, argyrophilic grain dementia etc.], Charcot-Marie-Tooth disease and the like, particularly Alzheimer's disease or progressive supranuclear palsy.

Here, the above-mentioned "prophylaxis" of a disease means, for example, administration of a medicament containing the compound of the present invention to patients who are expected to have a high risk of the onset due to some factor relating to the disease but have not developed the disease or patients who have developed the disease but do not have a subjective symptom, or administration of a medicament containing the compound of the present invention to patients who are feared to show recurrence of the disease after treatment of the disease.

A medicament containing the compound of the present invention may be safely administered solely or by mixing with a pharmacologically acceptable carrier according to a method known per se (e.g., the method described in the Japanese Pharmacopoeia etc.) as the production method of a pharmaceutical preparation, and in the form of, for example, tablet (including sugar-coated tablet, film-coated tablet, sublingual tablet, orally disintegrating tablet, buccal etc.), pill, powder, granule, capsule (including soft capsule, microcapsule), troche, syrup, liquid, emulsion, suspension, release control preparation (e.g., immediate-release preparation, sustained-release preparation, sustained-release microcapsule), aerosol, film (e.g., orally disintegrating film, oral mucosa-adhesive film), injection (e.g., subcutaneous injection, intravenous injection, intramuscular injection, intraperitoneal injection), drip infusion, transdermal absorption type preparation, cream, ointment, lotion, adhesive preparation, suppository (e.g., rectal suppository, vaginal suppository), pellet, nasal preparation, pulmonary preparation (inhalant), eye drop and the like, orally or parenterally (e.g., intravenous, intramuscular, subcutaneous, intraorgan, intranasal, intradermal, instillation, intracerebral, intrarectal, intravaginal, intraperitoneal and intratumor administrations, administration to the vicinity of tumor and administration to the lesion).

The content of the compound of the present invention in the medicament of the present invention is about 0.01 to 100% by weight of the entire medicament. The dose of the medicament of the present invention may vary depending on administration subject, administration route, disease and the like. For example, for oral administration to patients (body weight about 60 kg) with neurodegenerative disease (for example Alzheimer's disease, progressive supranuclear palsy, etc.), about 0.01 mg/kg body weight-about 50 mg/kg body weight, preferably about 0.05 mg/kg body weight-about 25 mg/kg body weight, more preferably about 0.1 mg/kg body weight-about 2 mg/kg body weight of an active ingredient (compound (I)) may be administered once to several portions per day.

Examples of the pharmaceutically acceptable carrier include various organic or inorganic carrier materials that are conventionally used as preparation materials, for example, excipient, lubricant, binding agent and disintegrant for solid preparations; or solvent, solubilizing agent, suspending agent, isotonic agent, buffering agent, soothing agent and the like for liquid preparations. Furthermore, when necessary, ordinary additives such as preservative, antioxidant, colorant, sweetening agent, adsorbing agent, wetting agent and the like can be also used as appropriate in an appropriate amount.

The dose of the medicament of the present invention as a sustained-release preparation varies depending on the kind and content of compound (I), dosage form, period of sustained drug release, subject animal of administration (e.g., mammals such as mouse, rat, hamster, guinea pig, rabbit, cat, dog, bovine, horse, swine, sheep, monkey, human etc.), and administration object. For example, for application by parenteral administration, about 0.1 to about 100 mg of compound (I) needs to be released from the administered preparation per 1 week.

Examples of the excipient include lactose, white sugar, D-mannitol, starch, corn starch, crystalline cellulose, and light anhydrous silicic acid.

Examples of the lubricant include magnesium stearate, calcium stearate, talc, and colloidal silica.

Examples of the binding agent include crystalline cellulose, white sugar, D-mannitol, dextrin, hydroxypropylcellulose, hydroxypropylmethylcellulose, polyvinylpyrrolidone, starch, sucrose, gelatin, methylcellulose, and carboxymethylcellulose sodium.

Examples of the disintegrant include starch, carboxymethylcellulose, carboxymethylcellulose calcium, carboxymethylstarch sodium, and L-hydroxypropylcellulose.

Examples of the solvent include water for injection, alcohol, propylene glycol, Macrogol, sesame oil, corn oil, and olive oil.

Examples of the solubilizing agent include polyethylene glycol, propylene glycol, D-mannitol, benzyl benzoate, ethanol, trisaminomethane, cholesterol, triethanolamine, and sodium carbonate, sodium citrate.

Examples of the suspending agent include surfactants such as stearyl triethanolamine, sodium lauryl sulfate, laurylaminopropionic acid, lecithin, benzalkonium chloride, benzetonium chloride, glycerin monostearate and the like; and hydrophilic polymers such as polyvinyl alcohol, polyvinylpyrrolidone, carboxymethylcellulose sodium, methylcellulose, hydroxymethylcellulose, hydroxyethylcellulose, hydroxypropylcellulose and the like.

Examples of the isotonic agent include glucose, D-sorbitol, sodium chloride, glycerin, and D-mannitol.

Examples of the buffering agent include buffer solutions such as phosphates, acetates, carbonates, and citrates.

therapeutic drug for Parkinson's disease associated with dementia (rivastigmine), therapeutic drug for Lewy body dementia (donepezil), therapeutic drug for ALS (riluzole, neurotrophic factor, etc.), therapeutic drug for hyperlipidemia such as cholesterol-lowering drug (statin series (pravastatin sodium, atorvastatin, simvastatin, rosuvastatin, etc.), fibrate (clofibrate etc.), squalene synthase inhibitor), therapeutic drug for abnormal behavior or dementia-related wandering (sedative drug, anti-anxiety drug, etc.), apoptosis inhibitor, anti-obesity drug, antidiabetic drug, therapeutic drug for hypertension, therapeutic drug for hypotension, therapeutic drug for rheumatism (DMARD), anticancer drug, therapeutic drug for hypoparathyroidism (PTH), calcium receptor antagonist, sex hormone or derivative thereof (pro-

gesterone, estradiol, estradiol benzoate etc.), neuron differentiation accelerator, neurogeneration promotor, non-steroidal anti-inflammatory drug (meloxicam, tenoxicam, indomethacin, ibuprofen, celecoxib, rofecoxib, aspirin etc.), steroid (dexamethasone, cortisone acetate, etc.), anti-cytokine drug (TNF inhibitor, MAP kinase inhibitor, etc.), antibody medicament, nucleic acid or nucleic acid derivative, aptamer and the like.

For combined use, the administration time of the compound of the present invention and the concomitant drug is not restricted, and the compound of the present invention or the concomitant drug can be administered to an administration subject simultaneously, or may be administered at different times. The dosage of the concomitant drug may be determined according to the dose clinically used, and can be appropriately selected depending on an administration subject, administration route, disease, combination and the like.

The administration form of the combined use is not particularly limited, and the compound of the present invention and a concomitant drug only need to be combined on administration. Examples of such administration mode include the following:

- (1) administration of a single preparation obtained by simultaneously processing the compound of the present invention and the concomitant drug, (2) simultaneous administration of two kinds of preparations of the compound of the present invention and the concomitant drug, which have been separately produced, by the same administration route, (3) administration of two kinds of preparations of the compound of the present invention and the concomitant drug, which have been separately produced, by the same administration route in a staggered manner, (4) simultaneous administration of two kinds of preparations of the compound of the present invention and the concomitant drug, which have been separately produced, by different administration routes, (5) administration of two kinds of preparations of the compound of the present invention and the concomitant drug, which have been separately produced, by different administration routes in a staggered manner (e.g., administration in the order of the compound of the present invention and the concomitant drug, or in the reverse order) and the like.

The mixing ratio of the compound of the present invention and a concomitant drug in the combination agent of the present invention can be appropriately selected based on the subject of administration, administration route, disease and the like.

For example, while the content of the compound of the present invention in the combination agent of the present invention varies depending on the preparation form, it is generally about 0.01-100 wt %, preferably about 0.1-50 wt %, more preferably about 0.5-20 wt %, of the whole preparation.

The content of the concomitant drug in the combination agent of the present invention varies depending on the preparation form, and generally about 0.01 to 100% by weight, preferably about 0.1 to 50% by weight, further preferably about 0.5 to 20% by weight, of the entire preparation.

While the content of the additive such as a carrier and the like in the combination agent of the present invention varies depending on the form of a preparation, it is generally about 1 to 99.99% by weight, preferably about 10 to 90% by weight, based on the preparation.

When the compound of the present invention and the concomitant drug are separately prepared, the same content may be adopted.

Any amount of the concomitant drug can be adopted as long as the side effects do not cause a problem. The daily dosage in terms of the concomitant drug varies depending on the severity, age, sex, body weight, sensitivity difference of the subject, administration period, interval, and nature, pharmacology, kind of the pharmaceutical preparation, kind of effective ingredient, and the like, and not particularly restricted, and the amount of a drug is, in the case of oral administration for example, generally about 0.001 to 2000 mg, preferably about 0.01 to 500 mg, further preferably about 0.1 to 100 mg, per 1 kg of a mammal and this is generally administered once to 4-times, divided in a day.

When the combination agent of the present invention is administered, the compound of the present invention and the concomitant drug can be administered simultaneously, or may be administered in a staggered manner. When administered at a time interval, the interval varies depending on the effective ingredient, dosage form and administration method, and, for example, when the concomitant drug is administered first, a method in which the compound of the present invention is administered within time range of from 1 minute to 3 days, preferably from 10 minutes to 1 day, more preferably from 15 minutes to 1 hour, after administration of the concomitant drug is an example. When the compound of the present invention is administered first, a method in which the concomitant drug is administered within time range of from 1 minute to 1 day, preferably from 10 minutes to 6 hours, more preferably from 15 minutes to 1 hour after administration of the compound of the present invention is an example.

EXAMPLES

The present invention is explained in detail in the following by referring to Examples, Experimental Examples and Formulation Examples, which are not to be construed as limitative, and the invention may be changed within the scope of the present invention.

In the following Examples, the "room temperature" generally means about 10° C. to about 35° C. The ratios indicated for mixed solvents are volume mixing ratios, unless otherwise specified. % means wt %, unless otherwise specified.

The elution by column chromatography in the Examples was performed under the observation by TLC (Thin Layer Chromatography) unless otherwise specified. In the observation by TLC, 60 F₂₅₄ manufactured by Merck was used as a TLC plate, the solvent used as an elution solvent in column chromatography was used as a developing solvent, and UV detector was used for the detection. In silica gel column chromatography, the indication of NH means use of amino-propylsilane-bonded silica gel, and the indication of Diol means use of 3-(2,3-dihydroxypropoxy)propylsilane-bonded silica gel. In preparative HPLC (high performance liquid chromatography), the indication of C18 means use of octadecyl-bonded silica gel. The ratio for elution solvent is, unless otherwise specified, a volume mixing ratio.

For the analysis of ¹H NMR, ACD/SpecManager (trade name) software and the like were used. Peaks of a hydroxyl group, an amino group and the like, having very mild proton peak, are not sometimes described.

MS (mass spectrum) was measured by LC/MS (liquid chromatograph mass spectrometer). As the ionization method, ESI (Electrospray Ionization) method, or APCI

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(Atmospheric Pressure Chemical Ionization) method was used. The data indicates actual measured value (found). While molecular ion peak is generally observed, a fragment ion peak or fragment ion peak of free form is generally observed. Elemental analysis value (Anal.) is described as calculated value (Calcd) and actual measured value (Found).

Powder X-RAY diffraction pattern was measured using Cu-K α characteristic radiation from Rigaku Ultima IV, and characteristic peaks were described.

In the following Examples, the following abbreviations are used.

MS: mass spectrum

M: mol concentration

CDCl₃: deuteriochloroform

DMSO-d₆: deuterodimethyl sulfoxide

¹H NMR: proton nuclear magnetic resonance

LC/MS: liquid chromatograph mass spectrometer

ESI: Electrospray Ionization

APCI: Atmospheric Pressure Chemical Ionization

AIBN: 2,2'-azobis(isobutyronitrile)

Boc: tert-butoxycarbonyl

CDI: 1,1'-carbonyldiimidazole

CO₂: carbon dioxide

DIPEA: N,N-diisopropylethylamine

DMF: N,N-dimethylformamide

DMSO: dimethyl sulfoxide

NBS: N-bromosuccinimide

TEA: triethylamine

THF: tetrahydrofuran

XANTPHOS: 4,5-bis(diphenylphosphino)-9,9-dimethyl-xanthene

Example 1 and Example 2

6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R, 2R)-2-hydroxy-1,2-di(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Compound of Example 1)

6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1S, 2S)-2-hydroxy-1,2-di(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Compound of Example 2)

A) tert-butyl 2-[3-(methoxycarbonyl)-4-methylbenzoyl]hydrazine-1-carboxylate

The following reaction was carried out by dividing the described scale into 5 reactions. A mixture of methyl 5-bromo-2-methylbenzoate (42.0 g), tert-butyl hydrazinecarboxylate (29.1 g), bis(dibenzylideneacetone)palladium (0) (5.27 g), XANTPHOS (5.30 g), N,N-dicyclohexylmethylamine (58.3 mL) and cyclopentyl methyl ether (1000 mL) was stirred under carbon monoxide atmosphere (0.5 MPa) at 95° C. for 5 hr. The five reaction mixtures were combined, and filtered through Celite, and the filtrate was concentrated under reduced pressure. The obtained residue was dissolved in a mixture of ethyl acetate and THF, and the solution was washed with water and saturated brine, dried over anhydrous sodium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (NH, ethyl acetate), and crystallized from hexane/THF to give the title compound (45.7 g). ¹H NMR (300 MHz, DMSO-d₆) δ 1.32-1.53 (9H, m), 2.57 (3H, s), 3.86 (3H, s), 7.46 (1H, d, J=7.9 Hz), 7.94 (1H, dd, J=7.9, 1.5 Hz), 8.33 (1H, s), 8.93 (1H, s), 10.30 (1H, s).

MS: [M-H]⁻ 307.0.

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B) methyl 5-(hydrazinecarbonyl)-2-methylbenzoate hydrochloride

To tert-butyl 2-[3-(methoxycarbonyl)-4-methylbenzoyl]hydrazine-1-carboxylate (45.7 g) was added 4 M hydrogen chloride cyclopentyl methyl ether solution (200 mL) at room temperature. The mixture was stirred at room temperature for 3 days. The precipitated solid was collected by filtration, and dried to give the title compound (36.3 g). ¹H NMR (300 MHz, DMSO-d₆) δ 2.59 (3H, s), 3.88 (3H, s), 7.52 (1H, d, J=8.1 Hz), 8.03 (1H, dd, J=7.9, 2.1 Hz), 8.36 (1H, d, J=1.9 Hz), 9.71-10.80 (2H, m), 11.64 (1H, br s).

MS: [M+H]⁺ 209.0.

C) methyl 5-(5-(difluoromethyl)-1,3,4-oxadiazol-2-yl)-2-methylbenzoate

To a mixture of methyl 5-(hydrazinecarbonyl)-2-methylbenzoate hydrochloride (36.3 g) and THF (750 mL) was added DIPEA (129 mL) at room temperature. To the mixture was added dropwise difluoroacetic acid anhydride (27.6 mL) at 0° C. The mixture was stirred at room temperature for 2 hr, and to the mixture was added 4-methylbenzenesulfonyl chloride (56.5 g) at room temperature. The mixture was stirred overnight at room temperature, to the mixture was added water at room temperature, and the mixture was extracted with ethyl acetate.

The organic layer was separated, washed with saturated aqueous sodium hydrogencarbonate solution, water and saturated brine, dried over anhydrous sodium sulfate, and concentrated under reduced pressure. The residue was crystallized from hexane/ethyl acetate to give the title compound (24.9 g).

¹H NMR (300 MHz, DMSO-d₆) δ 2.63 (3H, s), 3.90 (3H, s), 7.37-7.75 (2H, m), 8.15 (1H, dd, J=8.1, 2.1 Hz), 8.44 (1H, d, J=2.3 Hz).

D) methyl 2-(bromomethyl)-5-((5-(difluoromethyl)-1,3,4-oxadiazol-2-yl))benzoate

To a mixture of methyl 5-(5-(difluoromethyl)-1,3,4-oxadiazol-2-yl)-2-methylbenzoate (14.0 g) and benzotrifluoride (500 mL) were added NBS (16.7 g) and AIBN (0.857 g) at room temperature. The mixture was stirred under argon atmosphere at 90° C. for 1 hr. To the mixture was added water at room temperature, and the mixture was extracted with ethyl acetate. The organic layer was separated, washed with saturated brine, dried over anhydrous sodium sulfate, and concentrated under reduced pressure. To the mixture of the residue and THF (500 mL) was added DIPEA (10.0 mL) at room temperature. To the mixture was added dropwise diethyl phosphonate (7.40 mL) at 0° C., and the mixture was stirred overnight at room temperature. To the mixture was added water at room temperature, and the mixture was extracted with ethyl acetate. The organic layer was separated, washed with saturated aqueous ammonium chloride solution, water and saturated brine, dried over anhydrous sodium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (ethyl acetate/hexane), and crystallized from hexane/ethyl acetate to give the title compound (12.7 g).

¹H NMR (300 MHz, CDCl₃) δ 4.01 (3H, s), 5.01 (2H, s), 6.94 (1H, t, J=51.6 Hz), 7.69 (1H, d, J=8.3 Hz), 8.24 (1H, dd, J=7.9, 1.9 Hz), 8.71 (1H, d, J=1.9 Hz).

MS: [M+H]⁺ 347.0.

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E) tert-butyl [(4-methylbenzene-1-sulfonyl) (pyridin-2-yl)methyl]carbamate

To a mixture of sodium p-toluenesulfinate (25.0 g), tert-butyl carbamate (10.9 g) and water (140 mL) was added a mixture of pyridine-2-carbaldehyde (10.0 g) and methanol (70 mL) at room temperature. The mixture was stirred at room temperature for 30 min until the suspension became a clear solution. To the mixture was added formic acid (7.16 mL) at room temperature, and the mixture was stirred at room temperature for 3 days. The precipitated solid was collected by filtration, and washed with water and diisopropyl ether to give the title compound (23.0 g).

¹H NMR (300 MHz, DMSO-d₆) δ 1.02-1.31 (9H, m), 2.39 (3H, s), 6.06 (1H, br d, J=10.2 Hz), 7.31-7.52 (3H, m), 7.66 (2H, d, J=8.3 Hz), 7.77-7.84 (1H, m), 7.85-8.00 (1H, m), 8.15 (1H, br d, J=10.5 Hz), 8.49-8.63 (1H, m).

F) tert-butyl [2-oxo-1,2-di(pyridin-2-yl)ethyl]carbamate

A mixture of tert-butyl [(4-methylbenzene-1-sulfonyl) (pyridin-2-yl)methyl]carbamate (11.0 g), pyridine-2-carbaldehyde (3.41 g), 3-ethyl-5-(2-hydroxyethyl)-4-methylthiazolium bromide (1.53 g) and THF (121 mL) was degassed, and to the mixture was added dropwise triethylamine (63.3 mL) under argon atmosphere at room temperature. The mixture was again degassed, and the mixture was stirred overnight under argon atmosphere at 60° C. To the mixture was added saturated aqueous ammonium chloride solution at room temperature, and the mixture was partitioned between water and ethyl acetate. The separated organic layer was washed with water and saturated brine, dried over anhydrous sodium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (NH, ethyl acetate/hexane) to give the title compound (9.81 g) as a crude product. The obtained compound was used in the next step without further purification.

MS: [M+H-Boc]⁺ 214.0.

G) tert-butyl [(1RS,2RS)-2-hydroxy-1,2-di(pyridin-2-yl)ethyl]carbamate

To a mixture of tert-butyl [2-oxo-1,2-di(pyridin-2-yl)ethyl]carbamate (9.81 g), methanol (60 mL) and THF (60 mL) was added sodium borohydride (1.72 g) at 0° C., and the mixture was stirred overnight at room temperature. To the mixture was added water at 0° C., and the mixture was stirred at room temperature for 1 hr, and partitioned between water and ethyl acetate. The separated organic layer was washed with water and saturated brine. The combined aqueous layer was re-extracted with ethyl acetate, and the extract was washed with water and saturated brine. The combined organic layer was dried over anhydrous sodium sulfate, and concentrated under reduced pressure. The residue was solidified from ethyl acetate/hexane, and the obtained solid was recrystallized from ethyl acetate/hexane to give the title compound (6.64 g).

¹H NMR (300 MHz, DMSO-d₆) δ 1.11-1.35 (9H, m), 4.82-5.09 (2H, m), 5.61 (1H, d, J=5.1 Hz), 7.02 (1H, br d, J=8.7 Hz), 7.10 (1H, br d, J=7.9 Hz), 7.15-7.30 (3H, m), 7.57-7.80 (2H, m), 8.42 (1H, d, J=4.3 Hz), 8.52 (1H, br d, J=4.1 Hz).

MS: [M+H]⁺ 316.1.

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H) 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2RS)-2-hydroxy-1,2-di(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one

To a mixture of tert-butyl [(1RS,2RS)-2-hydroxy-1,2-di(pyridin-2-yl)ethyl]carbamate (2.21 g) and acetic acid (20 mL) was added dropwise hydrogen bromide (30% acetic acid solution (ca. 5.1 M), 11.0 mL) at room temperature, and the mixture was stirred overnight at room temperature. The mixture was concentrated under reduced pressure, and subjected to azeotropy with toluene and toluene/ethanol to give a white solid (3.54 g). To a mixture of the obtained solid (2.83 g), methyl 2-(bromomethyl)-5-(5-(difluoromethyl)-1,3,4-oxadiazol-2-yl)benzoate (2.15 g) and DMF (50 mL) was added dropwise DIPEA (6.46 mL) at 0° C., and the mixture was stirred at room temperature for 3 hr. To the mixture was added dropwise acetic acid (2.12 mL) at room temperature, and the mixture was stirred overnight at room temperature. The mixture was poured into water at room temperature, and extracted with ethyl acetate. The extract was washed with water and saturated brine, dried over anhydrous sodium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (methanol/ethyl acetate), and crystallized from ethyl acetate/diisopropyl ether to give the title compound (2.36 g).

¹H NMR (300 MHz, DMSO-d₆) δ 4.75 (1H, d, J=18.8 Hz), 5.03 (1H, d, J=18.8 Hz), 5.49 (1H, dd, J=8.8, 5.8 Hz), 5.76 (1H, d, J=9.0 Hz), 5.98 (1H, d, J=5.6 Hz), 7.17 (1H, ddd, J=7.3, 4.9, 1.3 Hz), 7.30-7.72 (4H, m), 7.72-7.89 (3H, m), 8.04 (1H, d, J=0.8 Hz), 8.17-8.24 (1H, m), 8.25-8.31 (1H, m), 8.60-8.66 (1H, m).

MS: [M+H]⁺ 450.1.

I) 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2R)-2-hydroxy-1,2-di(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Compound of Example 1) and

6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1S,2S)-2-hydroxy-1,2-di(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Compound of Example 2)

6-[5-(Difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2RS)-2-hydroxy-1,2-di(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one was optically resolved by preparative SFC (column: CHIRALCEL OD-H, 20 mmID×250 mmL, 5 μm, mobile phase: CO₂/ethanol=770/230 (v/v)).

The fraction of the optical isomer having a longer retention time was concentrated under reduced pressure, and the residue was crystallized from hexane/ethyl acetate to give a white solid (1.20 g). The obtained solid (1.17 g) was dissolved in ethyl acetate (12 mL) at 70° C., and to the mixture was added dropwise heptane (10 mL) at 70° C. The mixture was stirred at 70° C. for 30 min, and then to the mixture was added dropwise additional heptane (30 mL) at 50° C.-75° C. The mixture was stirred at 50° C. for 30 min, and then at room temperature for 1 hr. The precipitated solid was collected by filtration, and washed with heptane to give 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2R)-2-hydroxy-1,2-di(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (1.09 g, >99% e.e., retention time: 5.48 min (analysis column, column: CHIRALCEL OD-H, 4.6 mmID×150 mmL, 5 μm, mobile phase: CO₂/ethanol=770/

230 (v/v)). The absolute stereochemistry of the compound was determined by X-ray crystallography.

¹H NMR (300 MHz, DMSO-d₆) δ 4.75 (1H, d, J=18.8 Hz), 5.03 (1H, d, J=18.8 Hz), 5.49 (1H, dd, J=8.8, 5.5 Hz), 5.76 (1H, d, J=9.0 Hz), 5.98 (1H, d, J=5.6 Hz), 7.17 (1H, ddd, J=7.3, 4.9, 1.3 Hz), 7.28-7.72 (4H, m), 7.72-7.91 (3H, m), 8.04 (1H, d, J=0.8 Hz), 8.18-8.24 (1H, m), 8.25-8.32 (1H, m), 8.60-8.67 (1H, m).

MS: [M+H]⁺ 450.1.

Anal. Calcd for C₂₃H₁₇F₂N₅O₃: C, 61.47; H, 3.81; N, 15.58. Found: C, 61.68; H, 4.01; N, 15.80.

powder X-RAY crystal diffraction pattern (diffraction angle 2θ): 5.0°, 7.1°, 9.1°, 10.8°, 16.6°, 18.3°, 19.8°, 21.6°, 22.8°, 23.7°, 26.1°

The fraction of the optical isomer having a shorter retention time was concentrated under reduced pressure, and the residue was dissolved in ethyl acetate (15 mL) at 70° C. To the mixture was added dropwise diisopropyl ether (30 mL) at 50° C.-70° C., and the obtained suspension was stirred at 50° C. for 1 hr. To the mixture was added dropwise diisopropyl ether (30 mL) at 50° C., and the mixture was stirred at 50° C. for 30 min, and then at room temperature for 1 hr. The precipitated solid was collected by filtration, and washed with diisopropyl ether to give 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1S,2S)-2-hydroxy-1,2-di(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (1.02 g, >99% e.e., retention time: 4.52 min (analysis column: column: CHIRALCEL OD-H, 4.6 mmID×150 mmL, 5 μm, mobile phase: CO₂/ethanol=770/230 (v/v))).

MS: [M+H]⁺ 450.1.

Example 3 and Example 4

6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2R)-2-(5-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Compound of Example 3)

6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1S,2S)-2-(5-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Compound of Example 4)

A) tert-butyl [2-(5-fluoropyridin-2-yl)-2-oxo-1-(pyridin-2-yl)ethyl]carbamate

A mixture of tert-butyl [(4-methylbenzene-1-sulfonyl)(pyridin-2-yl)methyl]carbamate (12.1 g), 5-fluoropyridine-2-carbaldehyde (4.18 g), 3-ethyl-5-(2-hydroxyethyl)-4-methylthiazolium bromide (1.68 g) and THF (134 mL) was degassed, and to the mixture was added dropwise triethylamine (69.6 mL) under argon atmosphere at room temperature. The mixture was again degassed, and the mixture was stirred overnight under argon atmosphere at 60° C. To the mixture was added saturated aqueous ammonium chloride solution at 0° C., and the mixture was partitioned between water and ethyl acetate. The separated organic layer was washed with water and saturated brine, dried over anhydrous sodium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (NH, ethyl acetate/hexane), and crystallized from ethyl acetate/hexane to give the title compound (7.14 g).

¹H NMR (300 MHz, DMSO-d₆) δ 1.12-1.51 (9H, m), 6.76 (1H, d, J=8.7 Hz), 7.25 (1H, ddd, J=7.5, 4.9, 1.1 Hz), 7.45 (1H, br d, J=8.3 Hz), 7.53 (1H, d, J=7.9 Hz), 7.78 (1H,

td, J=7.7, 1.9 Hz), 7.90 (1H, td, J=8.7, 2.6 Hz), 8.11 (1H, dd, J=8.8, 4.7 Hz), 8.33 (1H, br d, J=4.5 Hz), 8.64 (1H, d, J=2.6 Hz).

B) tert-butyl [(1R,2RS)-2-(5-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]carbamate

To a mixture of tert-butyl [2-(5-fluoropyridin-2-yl)-2-oxo-1-(pyridin-2-yl)ethyl]carbamate (6.59 g) and THF (100 mL) was added dropwise potassium tri(sec-butyl)borohydride (1 M THF solution, 23.9 mL) under argon atmosphere at -78° C., and the mixture was stirred at -78° C. for 2 hr. To the mixture was added water (40 mL) at -78° C., and to the mixture were added 2 M aqueous sodium hydroxide solution (99 mL) and 30% aqueous hydrogen peroxide (20.3 mL) while keeping the internal temperature 0° C. or below. The mixture was stirred at room temperature for 2 hr. Then, sodium thiosulfate pentahydrate (49.4 g) was dissolved in water (190 mL), and the obtained solution was added dropwise to the mixture at 0° C. The mixture was stirred at room temperature for 2 hr, and extracted with ethyl acetate. The extract was washed with water and saturated brine, dried over anhydrous sodium sulfate, and concentrated under reduced pressure. The residue was recrystallized from ethyl acetate/heptane to give the title compound (5.62 g).

¹H NMR (300 MHz, DMSO-d₆) δ 1.02-1.46 (9H, m), 4.84-5.10 (2H, m), 5.69 (1H, br d, J=4.5 Hz), 6.98 (1H, br d, J=8.7 Hz), 7.13-7.25 (2H, m), 7.29 (1H, dd, J=8.7, 4.5 Hz), 7.50-7.78 (2H, m), 8.43 (1H, dd, J=4.9, 0.8 Hz), 8.51 (1H, d, J=2.6 Hz).

MS: [M+H-Boc]⁺ 234.0.

C) 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2RS)-2-(5-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one

To a mixture of tert-butyl [(1R,2RS)-2-(5-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]carbamate (4.09 g) and acetic acid (40 mL) was added hydrogen bromide (25% acetic acid solution, 10 mL) at room temperature. To the mixture was added acetic acid (20 mL) at room temperature, and the mixture was stirred at room temperature for 30 min. The mixture was concentrated under reduced pressure, and the residue was subjected three times to azeotropy with toluene, and dried. To a mixture of the obtained residue, DIPEA (13.5 mL) and DMF (29 mL) was added dropwise a solution prepared by dissolving methyl 2-(bromomethyl)-5-(5-(difluoromethyl)-1,3,4-oxadiazol-2-yl)benzoate (3.87 g) in DMF (10 mL), at room temperature, and the mixture was stirred at room temperature for 60 hr. To the mixture was added water, and the mixture was extracted with ethyl acetate. The separated organic layer was washed with water and saturated brine, dried over anhydrous magnesium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (ethyl acetate/hexane). The obtained solid was suspended in diisopropyl ether/ethyl acetate, and collected by filtration. The solid was suspended in hexane/ethyl acetate=1/1 (50 mL), sonicated, and stirred at 50° C. for 10 min. The solid was collected by filtration, and washed with hexane/ethyl acetate to give the title compound (4.09 g).

¹H NMR (300 MHz, DMSO-d₆) δ 4.70 (1H, d, J=19.6 Hz), 4.97 (1H, d, J=20.0 Hz), 5.47-5.57 (1H, m), 5.72 (1H, d, J=9.0 Hz), 6.05 (1H, d, J=5.3 Hz), 7.31-7.74 (5H, m), 7.77-7.87 (2H, m), 8.05 (1H, d, J=1.1 Hz), 8.22 (1H, dd, J=7.9, 1.5 Hz), 8.26 (1H, t, J=1.5 Hz), 8.57-8.69 (1H, m).

MS: [M+H]⁺ 468.1.

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D) 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2R)-2-(5-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Compound of Example 3) and

6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1S,2S)-2-(5-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Compound of Example 4)

6-[5-(Difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2RS)-2-(5-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (3.09 g) was optically resolved by preparative HPLC (column: CHIRALCEL OD, 50 mmIDx500 mmL, 20 μ m, mobile phase: hexane/ethanol=700/300 (v/v)).

The fraction of the optical isomer having a longer retention time was concentrated under reduced pressure, and the residue was crystallized from heptane/isopropyl acetate to give 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2R)-2-(5-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (1.35 g, 99% e.e., retention time: 9.16 min (analysis condition, column: CHIRALCEL OD-H, 4.6 mmIDx250 mmL, 5 μ m, mobile phase: hexane/ethanol=700/300 (v/v)). The absolute stereochemistry of the compound was determined by X-ray crystallography.

¹H NMR (300 MHz, DMSO-d₆) δ 4.71 (1H, d, J=18.8 Hz), 4.97 (1H, d, J=18.8 Hz), 5.52 (1H, dd, J=9.0, 5.3 Hz), 5.72 (1H, d, J=9.0 Hz), 6.05 (1H, d, J=5.3 Hz), 7.31-7.74 (5H, m), 7.77-7.89 (2H, m), 8.05 (1H, d, J=1.1 Hz), 8.22 (1H, dd, J=8.1, 1.7 Hz), 8.26 (1H, t, J=1.5 Hz), 8.53-8.72 (1H, m).

MS: [M+H]⁺ 468.1.

Anal. Calcd for C₂₃H₁₆F₃N₅O₃·3H₂O:C, 58.43; H, 3.54; N, 14.81. Found: C, 58.53; H, 3.53; N, 14.78.

powder X-RAY crystal diffraction pattern (diffraction angle 2 θ): 7.2°, 11.3°, 16.2°, 19.2°, 22.1°, 22.7°, 24.2°, 26.5°

The fraction of the optical isomer having a shorter retention time was concentrated under reduced pressure, and the residue was crystallized from heptane/isopropyl acetate to give 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1S,2S)-2-(5-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (1.37 g, 99% e.e., retention time: 7.35 min (analysis condition, column: CHIRALCEL OD-H, 4.6 mmIDx250 mmL, 5 μ m, mobile phase: hexane/ethanol=700/300 (v/v)).

MS: [M+H]⁺ 468.1.

Example 5 and Example 6

2-[(1R,2R)-1,2-bis(5-fluoropyridin-2-yl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Compound of Example 5)

2-[(1S,2S)-1,2-bis(5-fluoropyridin-2-yl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Compound of Example 6)

A) tert-butyl [(5-fluoropyridin-2-yl) (4-methylbenzene-1-sulfonyl)methyl]carbamate

To a mixture of sodium p-toluenesulfonate (21.4 g), tert-butyl carbamate (9.36 g) and water (120 mL) was added a solution prepared by dissolving 5-fluoropyridine-2-carbal-

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dehyde (10 g) in methanol (60 mL), at room temperature. The mixture was stirred at room temperature for 30 min until the suspension became a clear solution. To the mixture was added formic acid (6.13 mL) at room temperature, and the mixture was stirred at room temperature for 3 days. The precipitated solid was collected by filtration, washed with water and diisopropyl ether, and dried under reduced pressure to give the title compound (24.4 g).

¹H NMR (300 MHz, DMSO-d₆) 1.10-1.28 (9H, m), 2.39 (3H, s), 6.09 (1H, d, J=10.5 Hz), 7.44 (2H, d, J=7.9 Hz), 7.67 (2H, d, J=7.9 Hz), 7.81-7.97 (2H, m), 8.26 (1H, d, J=10.2 Hz), 8.56 (1H, d, J=2.3 Hz).

B) tert-butyl [1,2-bis(5-fluoropyridin-2-yl)-2-oxoethyl]carbamate

A mixture of tert-butyl [(5-fluoropyridin-2-yl) (4-methylbenzene-1-sulfonyl)methyl]carbamate (12.7 g), 5-fluoropyridine-2-carbaldehyde (4.18 g), 3-ethyl-5-(2-hydroxyethyl)-4-methylthiazolium bromide (1.68 g) and THF (134 mL) was degassed, and to the mixture was added triethylamine (69.6 mL) under argon atmosphere at room temperature. The mixture was again degassed, and the mixture was stirred overnight under argon atmosphere at 60° C. To the mixture was added saturated aqueous ammonium chloride solution at 0° C., and the mixture was partitioned between water and ethyl acetate. The separated organic layer was washed with water and saturated brine, dried over anhydrous sodium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (NH, ethyl acetate/hexane), and crystallized from hexane/ethyl acetate to give the title compound (7.68 g).

¹H NMR (300 MHz, DMSO-d₆) δ 1.13-1.58 (9H, m), 6.74 (1H, d, J=8.7 Hz), 7.54 (1H, br d, J=8.7 Hz), 7.58-7.67 (1H, m), 7.68-7.81 (1H, m), 7.91 (1H, td, J=8.8, 2.8 Hz), 8.12 (1H, dd, J=8.7, 4.9 Hz), 8.34 (1H, d, J=2.6 Hz), 8.63 (1H, d, J=3.0 Hz).

MS: [M+H-Boc]⁺250.0.

C) tert-butyl [(1RS,2RS)-1,2-bis(5-fluoropyridin-2-yl)-2-hydroxyethyl]carbamate

To a mixture of tert-butyl [1,2-bis(5-fluoropyridin-2-yl)-2-oxoethyl]carbamate (7.0 g) and THF (100 mL) was added dropwise potassium tri(sec-butyl)borohydride (1 M THF solution, 30.1 mL) under argon atmosphere at -78° C., and the mixture was stirred at -78° C. for 2 hr. To the mixture was added water (20 mL) at -78° C., and 2 M aqueous sodium hydroxide solution (100 mL) and 30% aqueous hydrogen peroxide (20.5 mL) were added thereto while keeping the internal temperature 0° C. or below. The mixture was stirred at room temperature for 4 hr. Sodium thiosulfate pentahydrate (49.7 g) was dissolved in water (200 mL), the obtained solution was added to the mixture at 0° C., and the mixture was stirred at room temperature for 2 hr. The mixture was extracted with ethyl acetate. The extract was washed with water and saturated brine, dried over anhydrous sodium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (ethyl acetate/hexane), and crystallized from hexane/ethyl acetate to give the title compound (5.97 g).

¹H NMR (300 MHz, DMSO-d₆) δ 1.01-1.48 (9H, m), 4.84-4.95 (1H, m), 4.96-5.10 (1H, m), 5.67 (1H, d, J=5.6 Hz), 7.05 (1H, br d, J=9.0 Hz), 7.22-7.41 (2H, m), 7.62 (2H, td, J=8.8, 2.8 Hz), 8.40 (1H, d, J=3.0 Hz), 8.51 (1H, d, J=2.6 Hz).

MS: [M+H-Boc]⁺252.0.

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D) 2-[(1RS,2RS)-1,2-bis(5-fluoropyridin-2-yl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one

To a mixture of tert-butyl [(1RS,2RS)-1,2-bis(5-fluoropyridin-2-yl)-2-hydroxyethyl]carbamate (4.34 g) and acetic acid (30.9 mL) was added dropwise hydrogen bromide (30% acetic acid solution (ca. 5.1 M), 19.4 mL) at room temperature, and the mixture was stirred at room temperature for 3 hr. The mixture was concentrated under reduced pressure, and the residue was subjected to azeotropy with toluene. To a mixture of the obtained residue, methyl 2-(bromomethyl)-5-(5-(difluoromethyl)-1,3,4-oxadiazol-2-yl)benzoate (4.29 g) and DMF (50 mL) was added dropwise DIPEA (21.5 mL) at 0° C., and the mixture was stirred overnight at room temperature. The mixture was poured into water, and extracted with ethyl acetate. The extract was washed with water and saturated brine, dried over anhydrous sodium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (ethyl acetate/hexane), and crystallized from diisopropyl ether/ethyl acetate to give the title compound (3.68 g).

¹H NMR (300 MHz, DMSO-d₆) δ 4.69 (1H, d, J=18.8 Hz), 4.95 (1H, d, J=18.8 Hz), 5.49 (1H, dd, J=9.0, 5.3 Hz), 5.75 (1H, d, J=9.0 Hz), 6.05 (1H, d, J=5.3 Hz), 7.33-7.79 (5H, m), 7.84 (1H, d, J=8.3 Hz), 8.06 (1H, d, J=1.1 Hz), 8.19-8.26 (1H, m), 8.26-8.31 (1H, m), 8.62 (1H, d, J=3.0 Hz).

MS: [M+H]⁺ 486.1.

E) 2-[(1R,2R)-1,2-bis(5-fluoropyridin-2-yl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Compound of Example 5) and

2-[(1S,2S)-1,2-bis(5-fluoropyridin-2-yl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Compound of Example 6)

2-[(1RS,2RS)-1,2-Bis(5-fluoropyridin-2-yl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (3.52 g) was optically resolved by preparative HPLC (column: CHIRALCEL OD, 50 mmID×500 mmL, 20 μm, mobile phase: hexane/ethanol=700/300 (v/v)).

The fraction of the optical isomer having a longer retention time was concentrated under reduced pressure, the residue was dissolved in ethyl acetate, and the solution was passed through short silica gel column. The filtrate was concentrated under reduced pressure, and the residue was crystallized from diisopropyl ether/ethyl acetate. The obtained solid was recrystallized from heptane/isopropyl acetate, and dried under reduced pressure at 55-60° C. to give a white solid (1.42 g). 593 mg of the white solid was recrystallized from pentane/isopropyl acetate to give 2-[(1R,2R)-1,2-bis(5-fluoropyridin-2-yl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (506 mg, 99% e.e., retention time: 7.77 min (analysis condition, column: CHIRALCEL OD-H, 4.6 mmID×250 mmL, 5 μm, mobile phase: hexane/ethanol/diethylamine=700/300/1 (v/v/v)). The absolute stereochemistry of the compound was determined by X-ray crystallography.

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¹H NMR (300 MHz, DMSO-d₆) δ 4.68 (1H, d, J=18.4 Hz), 4.95 (1H, d, J=18.8 Hz), 5.49 (1H, dd, J=9.0, 5.6 Hz), 5.75 (1H, d, J=9.0 Hz), 6.04 (1H, d, J=6.0 Hz), 7.34-7.80 (5H, m), 7.84 (1H, d, J=7.9 Hz), 8.05 (1H, d, J=1.1 Hz), 8.23 (1H, dd, J=7.9, 1.5 Hz), 8.27 (1H, t, J=1.5 Hz), 8.61 (1H, d, J=3.0 Hz).

MS: [M+H]⁺ 486.1

Anal. Calcd for C₂₃H₁₅F₄N₅O₃·1H₂O: C, 56.70; H, 3.14; N, 14.37. Found: C, 56.57; H, 3.21; N, 14.32.

powder X-RAY crystal diffraction pattern (diffraction angle 2θ): 6.1°, 9.0°, 15.7°, 16.4°, 18.1°, 19.9°, 21.7°, 22.5°, 24.8°, 25.3°

The fraction of the optical isomer having a shorter retention time was concentrated under reduced pressure, the residue was crystallized from diisopropyl ether/ethyl acetate, and the obtained solid was further recrystallized from heptane/isopropyl acetate, and dried under reduced pressure to give a white solid (1.54 g). 100 mg of the white solid was recrystallized from pentane/isopropyl acetate to give 2-[(1S,2S)-1,2-bis(5-fluoropyridin-2-yl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (83 mg, 99% e.e., retention time: 6.43 min (analysis condition, column: CHIRALCEL OD-H, 4.6 mmID×250 mmL, 5 μm, mobile phase: hexane/ethanol/diethylamine=700/300/1 (v/v/v)).

MS: [M+H]⁺ 486.1

Example 7 and Example 8

6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2R)-2-(6-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Compound of Example 7)

6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1S,2S)-2-(6-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Compound of Example 8)

A) tert-butyl [2-(6-fluoropyridin-2-yl)-2-oxo-1-(pyridin-2-yl)ethyl]carbamate

To a mixture of tert-butyl [(4-methylbenzene-1-sulfonyl)(pyridin-2-yl)methyl]carbamate (11.9 g), 3-ethyl-5-(2-hydroxyethyl)-4-methylthiazolium bromide (2.48 g) and THF (230 mL) was added 6-fluoropyridine-2-carbaldehyde (4.60 g) under argon atmosphere at room temperature. The mixture was degassed, and replaced with argon gas. To the mixture was added dropwise triethylamine (68.5 mL) at 60° C., and the mixture was stirred overnight under argon atmosphere at 60° C. To the mixture was added water at room temperature, and the mixture was extracted with ethyl acetate. The extract was washed with water and saturated brine, dried over anhydrous sodium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (ethyl acetate/hexane) to give the title compound (11.4 g) as a crude product. The obtained compound was used in the next step without further purification.

MS: [M+H]⁺ 332.1.

B) tert-butyl [(1RS,2RS)-2-(6-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]carbamate

To a mixture of tert-butyl [2-(6-fluoropyridin-2-yl)-2-oxo-1-(pyridin-2-yl)ethyl]carbamate (11.4 g), methanol (100 mL) and THF (100 mL) was added sodium borohydride

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(1.49 g) at 0° C. The mixture was stirred at room temperature for 2.5 hr. To the mixture was added water at 0° C., and the mixture was extracted with ethyl acetate. The separated organic layer was washed with water and saturated brine, dried over anhydrous sodium sulfate, and concentrated under reduced pressure. The residue was crystallized from toluene, collected by filtration, and washed with diisopropyl ether. The obtained solid was recrystallized from hexane/ethyl acetate to give the title compound (6.92 g).

¹H NMR (300 MHz, DMSO-d₆) δ 1.16-1.32 (9H, m), 4.78-4.88 (1H, m), 4.90-5.01 (1H, m), 5.74 (1H, d, J=5.3 Hz), 6.95-7.10 (2H, m), 7.16-7.30 (3H, m), 7.68 (1H, t, J=6.8 Hz), 7.82-7.96 (1H, m), 8.42 (1H, d, J=4.5 Hz).

MS: [M+H]⁺ 334.1.

C) 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2R)-2-(6-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one

To a mixture of tert-butyl [(1R,2R)-2-(6-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]carbamate (4.55 g) and acetic acid (80 mL) was added hydrogen bromide (30% acetic acid solution (ca. 5.1 M), 21.4 mL) at room temperature. The mixture was stirred at room temperature for 30 min, concentrated under reduced pressure, and subjected to azeotropy with toluene. To a mixture of the obtained residue and DMF (50 mL) was added dropwise DIPEA (13.6 mL) at 0° C., and then a solution prepared by dissolving methyl 2-(bromomethyl)-5-(5-(difluoromethyl)-1,3,4-oxadiazol-2-yl)benzoate (4.51 g) in DMF (10 mL) was added to the mixture at 0° C. The mixture was stirred at 0° C. to room temperature for 3 days. The mixture was concentrated, and to the residue was added water at room temperature. The mixture was extracted with ethyl acetate. The separated organic layer was washed with water and saturated brine, dried over anhydrous sodium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (hexane/ethyl acetate), and crystallized from hexane/ethyl acetate to give the title compound (4.29 g).

¹H NMR (300 MHz, DMSO-d₆) δ 4.68 (1H, d, J=18.8 Hz), 4.98 (1H, d, J=18.8 Hz), 5.39-5.48 (1H, m), 5.71 (1H, d, J=9.0 Hz), 6.11 (1H, d, J=4.0 Hz), 6.95 (1H, dd, J=8.1, 2.4 Hz), 7.32-7.73 (4H, m), 7.77-7.88 (2H, m), 7.89-8.01 (1H, m), 8.06 (1H, d, J=0.9 Hz), 8.23 (1H, dd, J=8.0, 1.6 Hz), 8.58-8.67 (1H, m).

MS: [M+H]⁺ 468.1.

D) 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2R)-2-(6-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Compound of Example 7) and

6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1S,2S)-2-(6-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Compound of Example 8)

6-[5-(Difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2R)-2-(6-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (4.27 g) was optically resolved by preparative HPLC (column: CHIRALPAK IA, 50 mmID×500 mmL, 20 μm, mobile phase: hexane/ethanol=200/800 (v/v)).

The fraction of the optical isomer having a shorter retention time was concentrated under reduced pressure, and the residue was recrystallized from heptane/isopropyl acetate to

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give 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2R)-2-(6-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (1.72 g, >99% e.e., retention time: 15.49 min (analysis condition, column: CHIRALPAK IA, 4.6 mmID×250 mmL, 5 μm, mobile phase: hexane/ethanol=200/800 (v/v)). The absolute stereochemistry of the compound was determined by X-ray crystallography.

¹H NMR (300 MHz, DMSO-d₆) δ 4.68 (1H, d, J=18.8 Hz), 4.98 (1H, d, J=18.8 Hz), 5.43 (1H, dd, J=8.8, 5.8 Hz), 5.71 (1H, d, J=8.7 Hz), 6.10 (1H, d, J=6.0 Hz), 6.95 (1H, dd, J=7.9, 2.6 Hz), 7.32-7.73 (4H, m), 7.76-7.89 (2H, m), 7.89-8.00 (1H, m), 8.07 (1H, d, J=1.1 Hz), 8.23 (1H, dd, J=7.9, 1.5 Hz), 8.56-8.66 (1H, m).

MS: [M+H]⁺ 468.1.

Anal. Calcd for C₂₃H₁₆F₃N₅O₃·2H₂O: C, 58.65; H, 3.51; N, 14.87. Found: C, 58.78; H, 3.75; N, 15.00.

powder X-RAY crystal diffraction pattern (diffraction angle 2θ): 5.7°, 12.1°, 13.0°, 16.7°, 19.0°, 20.2°, 21.8°, 24.2°

The fraction of the optical isomer having a longer retention time was concentrated under reduced pressure, and the residue was recrystallized from heptane/isopropyl acetate to give 6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1S,2S)-2-(6-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (1.84 g, 99% e.e., retention time: 19.48 min (analysis condition, column: CHIRALPAK IA, 4.6 mmID×250 mmL, 5 μm, mobile phase: hexane/ethanol=200/800 (v/v)).

MS: [M+H]⁺ 468.1.

Example 9

2-[(1R,2S)-1-cyclopropyl-2-(4-fluorophenyl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one

A) tert-butyl {(1R)-1-cyclopropyl-2-[methoxy(methyl)amino]-2-oxoethyl}carbamate

To a mixture of (2R)-((tert-butoxycarbonyl)amino) (cyclopropyl)acetic acid (10.2 g) and THF (150 mL) was added CDI (9.22 g) at 0° C., and the mixture was stirred at 0° C. for 1 hr. N-Methoxymethanamine hydrochloride (5.55 g) and DIPEA (12.4 mL) were dissolved in DMF (30 mL), and the obtained solution was added to the mixture at 0° C. The mixture was stirred overnight at room temperature. The mixture was concentrated under reduced pressure. To the residue was added water, and the mixture was extracted with ethyl acetate/THF. The separated organic layer was washed with water and saturated brine, dried over anhydrous sodium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (hexane/ethyl acetate) to give the title compound (12.1 g).

¹H NMR (300 MHz, DMSO-d₆) δ 0.21-0.32 (1H, m), 0.33-0.49 (3H, m), 0.93-1.09 (1H, m), 1.36 (9H, s), 3.11 (3H, s), 3.70 (3H, s), 3.90-4.09 (1H, m), 7.06 (1H, d, J=8.3 Hz).

MS: [M+H-Boc]⁺ 159.1.

B) tert-butyl [(1R)-1-cyclopropyl-2-(4-fluorophenyl)-2-oxoethyl]carbamate

To a mixture of tert-butyl {(1R)-1-cyclopropyl-2-[methoxy(methyl)amino]-2-oxoethyl}carbamate (8.55 g) and THF (130 mL) was added 4-fluorophenylmagnesium bromide (2 M diethyl ether solution, 42 mL) at 40° C. The

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mixture was stirred under nitrogen atmosphere at 40° C. for 1 hr. To the mixture was added 4-fluorophenylmagnesium bromide (2 M diethyl ether solution, 3 mL) at 40° C. The mixture was stirred under nitrogen atmosphere at 40° C. for 30 min. The mixture was poured into saturated aqueous ammonium chloride solution at 0° C. To the mixture was added water, and the mixture was extracted with ethyl acetate. The separated organic layer was washed with water and saturated brine, dried over anhydrous sodium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (hexane/ethyl acetate) to give the title compound (7.80 g).

¹H NMR (300 MHz, DMSO-d₆) δ 0.19-0.56 (4H, m), 0.91-1.10 (1H, m), 1.13-1.50 (9H, m), 4.09-4.50 (1H, m), 7.35 (2H, t, J=8.8 Hz), 7.46 (1H, br d, J=7.2 Hz), 8.04 (2H, dd, J=8.7, 5.6 Hz).

MS: [M+H-Boc]⁺194.1.

C) tert-butyl [(1R,2S)-1-cyclopropyl-2-(4-fluorophenyl)-2-hydroxyethyl]carbamate

To a mixture of tert-butyl [(1R)-1-cyclopropyl-2-(4-fluorophenyl)-2-oxoethyl]carbamate (7.80 g), toluene (22 mL) and 2-propanol (23 mL) was added aluminium triisopropoxide (1.09 g) at room temperature. The mixture was stirred at 70° C. 3 hr. The reaction mixture was added to a mixture of 0.5 M hydrochloric acid and ethyl acetate at 0° C., and the mixture was extracted with ethyl acetate. The separated organic layer was washed with saturated aqueous sodium hydrogencarbonate solution and saturated brine, dried over anhydrous sodium sulfate, and concentrated under reduced pressure to give the title compound (7.79 g).

¹H NMR (300 MHz, DMSO-d₆) δ -0.24--0.08 (1H, m), 0.09-0.37 (3H, m), 0.95-1.13 (1H, m), 1.26 (9H, s), 3.08-3.25 (1H, m), 4.50 (1H, t, J=5.6 Hz), 5.37 (1H, d, J=4.5 Hz), 6.45 (1H, br d, J=9.4 Hz), 7.09 (2H, t, J=9.0 Hz), 7.29-7.41 (2H, m).

D) (1S,2R)-2-amino-2-cyclopropyl-1-(4-fluorophenyl)ethan-1-ol hydrobromide

To a mixture of tert-butyl [(1R,2S)-1-cyclopropyl-2-(4-fluorophenyl)-2-hydroxyethyl]carbamate (4.79 g) and acetic acid (50 mL) was added hydrogen bromide (25% acetic acid solution, 12.5 mL) at room temperature. The mixture was stirred at room temperature for 30 min. The mixture was concentrated under reduced pressure. The residue was subjected three times to azeotropy with toluene. The residue was suspended in ethyl acetate (10 mL) and diisopropyl ether (40 mL), and the obtained suspension was stirred at

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room temperature for 30 min. The precipitated solid was collected by filtration, washed with diisopropyl ether/ethyl acetate (4:1), and dried to give the title compound (4.29 g).

¹H NMR (300 MHz, DMSO-d₆) δ -0.41--0.24 (1H, m), 0.12-0.34 (2H, m), 0.34-0.50 (1H, m), 0.77-0.97 (1H, m), 2.59 (1H, dd, J=10.5, 3.0 Hz), 4.94 (1H, br s), 6.15 (1H, d, J=3.8 Hz), 7.09-7.29 (2H, m), 7.37-7.52 (2H, m), 7.89 (3H, br s).

MS: [M+H]⁺ 196.1.

E) 2-[(1R,2S)-1-cyclopropyl-2-(4-fluorophenyl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one

To a mixture of (1S,2R)-2-amino-2-cyclopropyl-1-(4-fluorophenyl)ethan-1-ol hydrobromide (3.65 g), DIPEA (10.5 mL) and DMF (30 mL) was added dropwise a solution prepared by dissolving methyl 2-(bromomethyl)-5-(5-(difluoromethyl)-1,3,4-oxadiazol-2-yl)benzoate (4.17 g) in DMF (12 mL), at room temperature. The mixture was stirred overnight at room temperature. To the mixture was added water at room temperature, and the mixture was extracted with ethyl acetate. The separated organic layer was washed with saturated brine, dried over anhydrous sodium sulfate, and concentrated under reduced pressure. The residue was purified by silica gel column chromatography (hexane/ethyl acetate), the obtained solid was dissolved in ethyl acetate, and the solution was filtered to remove the insoluble substance. The filtrate was concentrated under reduced pressure, and the residue was recrystallized from heptane/ethyl acetate to give the title compound (3.75 g).

¹H NMR (300 MHz, DMSO-d₆) δ -0.24--0.11 (1H, m), -0.11--0.01 (1H, m), 0.28-0.51 (2H, m), 1.31-1.48 (1H, m), 3.51 (1H, dd, J=10.2, 4.1 Hz), 4.78 (1H, d, J=19.2 Hz), 4.88 (1H, d, J=19.2 Hz), 5.00 (1H, t, J=4.5 Hz), 5.74 (1H, d, J=4.9 Hz), 7.05-7.21 (2H, m), 7.35-7.78 (3H, m), 7.90 (1H, d, J=8.3 Hz), 8.21 (1H, d, J=0.8 Hz), 8.29 (1H, dd, J=7.9, 1.9 Hz).

MS: [M+H]⁺ 430.1.

Anal. Calcd for C₂₂H₁₈F₃N₃O₃·0.1H₂O: C, 61.28; H, 4.25; N, 9.75. Found: C, 61.28; H, 4.48; N, 9.77.

powder X-RAY crystal diffraction pattern (diffraction angle 2θ): 6.9°, 13.9°, 16.9°, 18.6°, 20.2°, 20.9°, 22.2°, 22.7°, 25.6°, 28.1°

The compounds of Examples are shown in the following tables. MS in the tables means actual measured value. The compounds of Examples 10-150 in the following tables were produced according to the methods described in the above-mentioned Examples, or methods analogous thereto.

TABLE 1-1

EXAMPLE	IUPAC NAME	Structure	MS
1	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2R)-2-hydroxy-1,2-di(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one		450.2

TABLE 1-1-continued

EXAMPLE	IUPAC NAME	Structure	MS
2	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2S)-2-hydroxy-1,2-di(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one		450.2
3	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2R)-2-(5-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one		468.1
4	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1S,2S)-2-(5-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one		468.2
5	2-[(1R,2R)-1,2-bis(5-fluoropyridin-2-yl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one		486.1

TABLE 1-1-continued

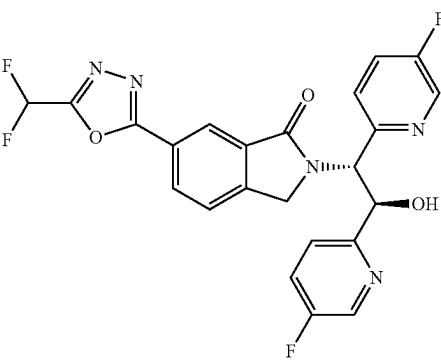
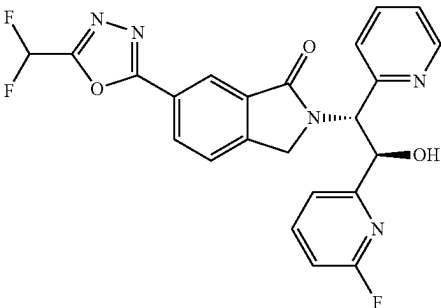
EXAMPLE	IUPAC NAME	Structure	MS
6	2-[(1S,2S)-1,2-bis(5-fluoropyridin-2-yl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isindol-1-one		486.1
7	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2R)-2-(6-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isindol-1-one		468.2

TABLE 1-2

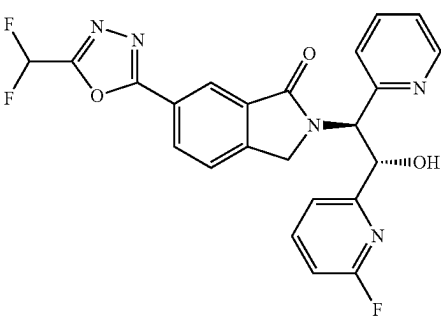
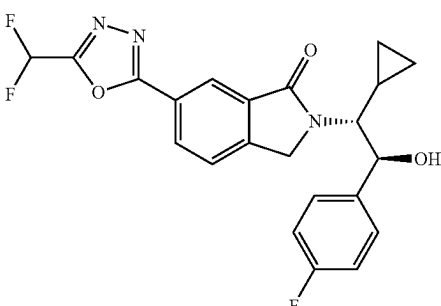
EXAMPLE	IUPAC NAME	Structure	MS
8	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1S,2S)-2-(6-fluoropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isindol-1-one		468.2
9	2-[(1R,2S)-1-cyclopropyl-2-(4-fluorophenyl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isindol-1-one		430.3

TABLE 1-2-continued

EXAMPLE	IUPAC NAME	Structure	MS
10	2-[(1R,2S)-1-cyclopropyl-2-(3-fluorophenyl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one		430.1
11	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2S)-2-hydroxy-1,2-diphenylethyl]-2,3-dihydro-1H-isoindol-1-one		446.2
12	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2R)-2-hydroxy-1,2-diphenylethyl]-2,3-dihydro-1H-isoindol-1-one		448.2
13	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1S,2S)-2-hydroxy-1,2-diphenylethyl]-2,3-dihydro-1H-isoindol-1-one		448.2
14	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1S,2R)-2-hydroxy-1,2-diphenylethyl]-2,3-dihydro-1H-isoindol-1-one		448.2

TABLE 1-3

EXAMPLE	IUPAC NAME	Structure	MS
15	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2RS)-2-hydroxy-2-(pyridin-2-yl)-1-(3,4,5-trifluorophenyl)ethyl]-2,3-dihydro-1H-isoindol-1-one		503.1
16	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2SR)-2-(3-fluorophenyl)-1-(4-fluorophenyl)-2-hydroxyethyl]-2,3-dihydro-1H-isoindol-1-one		484.2
17	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2SR)-2-hydroxy-2-phenyl-1-[6-(trifluoromethyl)pyridin-3-yl]ethyl]-2,3-dihydro-1H-isoindol-1-one		517.2
18	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2SR)-2-(4-fluorophenyl)-2-hydroxy-1-[6-(trifluoromethyl)pyridin-3-yl]ethyl]-2,3-dihydro-1H-isoindol-1-one		535.1

TABLE 1-3-continued

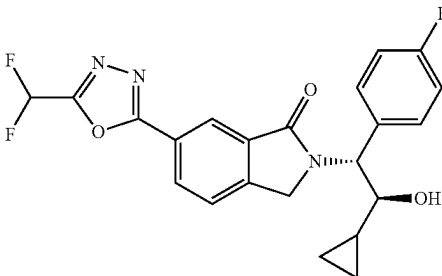
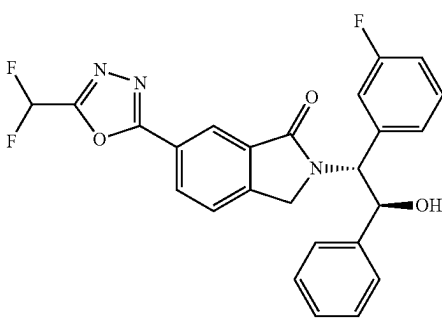
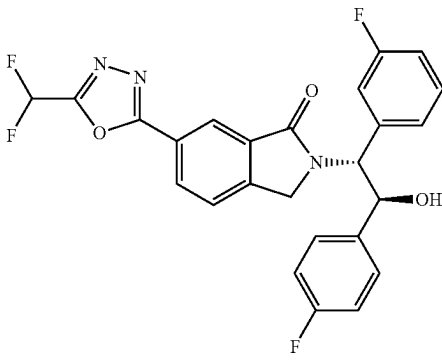
EXAMPLE	IUPAC NAME	Structure	MS
19	2-[(1RS,2SR)-2-cyclopropyl-1-(4-fluorophenyl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one		428.2
20	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2SR)-1-(3-fluorophenyl)-2-hydroxy-2-phenylethyl]-2,3-dihydro-1H-isoindol-1-one		466.1
21	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2SR)-1-(3-fluorophenyl)-2-(4-fluorophenyl)-2-hydroxyethyl]-2,3-dihydro-1H-isoindol-1-one		484.1

TABLE 1-4

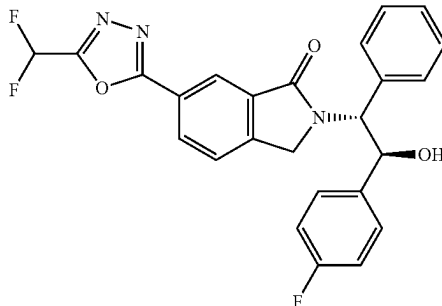
EXAMPLE	IUPAC NAME	Structure	MS
22	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2SR)-2-(4-fluorophenyl)-2-hydroxy-1-phenylethyl]-2,3-dihydro-1H-isoindol-1-one		466.1

TABLE 1-4-continued

EXAMPLE	IUPAC NAME	Structure	MS
23	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2SR)-2-(3-fluorophenyl)-2-hydroxy-1-phenylethyl]-2,3-dihydro-1H-isoindol-1-one		466.1
24	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2RS)-2-hydroxy-2-(pyridin-2-yl)-1-[3-(trifluoromethyl)phenyl]ethyl]-2,3-dihydro-1H-isoindol-1-one		517.0
25	2-[(1RS,2SR)-2-(4,4-difluorocyclohexyl)-2-hydroxy-1-phenylethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one		490.1
26	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2RS)-2-hydroxy-2-(pyridin-2-yl)-1-[4-(trifluoromethyl)phenyl]ethyl]-2,3-dihydro-1H-isoindol-1-one		517.1
27	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2SR)-1-(2-fluorophenyl)-2-hydroxy-2-phenylethyl]-2,3-dihydro-1H-isoindol-1-one		466.1

TABLE 1-4-continued

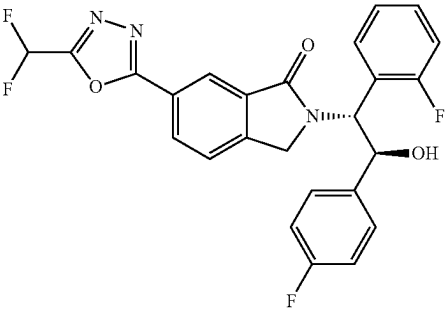
EXAMPLE	IUPAC NAME	Structure	MS
28	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2SR)-1-(2-fluorophenyl)-2-(4-fluorophenyl)-2-hydroxyethyl]-2,3-dihydro-1H-isoindol-1-one		484.1

TABLE 1-5

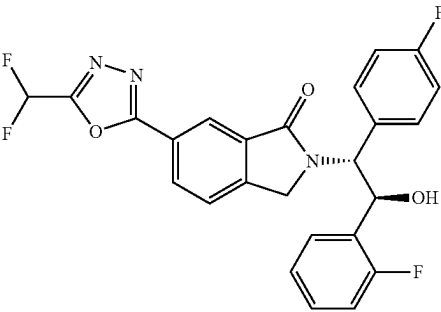
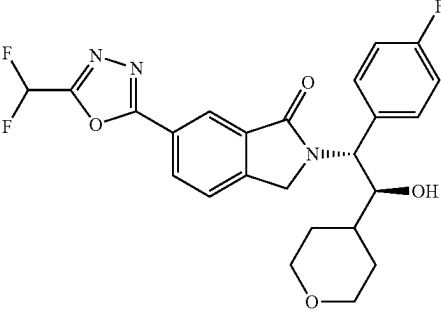
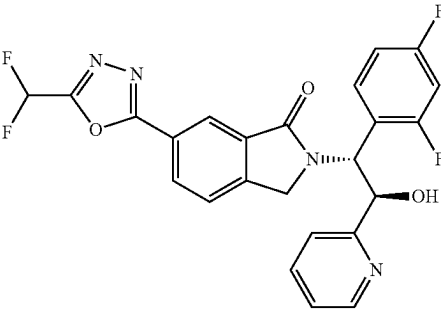
EXAMPLE	IUPAC NAME	Structure	MS
29	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2SR)-2-(2-fluorophenyl)-1-(4-fluorophenyl)-2-hydroxyethyl]-2,3-dihydro-1H-isoindol-1-one		484.1
30	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2SR)-1-(4-fluorophenyl)-2-hydroxy-2-(oxan-4-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one		474.2
31	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2SR)-1-(2,4-difluorophenyl)-2-hydroxy-2-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one		485.1

TABLE 1-5-continued

EXAMPLE	IUPAC NAME	Structure	MS
32	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-1-(2,4-difluorophenyl)-2-(4-fluorophenyl)-2-hydroxyethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)		500.1
33	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-1-(2,4-difluorophenyl)-2-(4-fluorophenyl)-2-hydroxyethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)		500.1
34	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-2-(4-fluorophenyl)-1-(6-fluoropyridin-3-yl)-2-hydroxyethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)		485.1
35	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-2-(4-fluorophenyl)-1-(6-fluoropyridin-3-yl)-2-hydroxyethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)		485.1

TABLE 1-6

EXAMPLE	IUPAC NAME	Structure	MS
36	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-2-(4-fluorophenyl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)		467.1
37	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-2-(4-fluorophenyl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)		467.1
38	2-[(1R,2R)-1-cyclopropyl-2-hydroxy-2-(pyridin-2-yl)ethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one		413.2
39	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-{(1RS,2RS)-2-hydroxy-2-(pyridin-2-yl)-1-[2-(trifluoromethyl)phenyl]ethyl}-2,3-dihydro-1H-isoindol-1-one		517.1
40	3-[(1RS,2SR)-1-{6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-1-oxo-1,3-dihydro-2H-isoindol-2-yl}-2-hydroxy-2-phenylethyl]benzonitrile		471.2

TABLE 1-6-continued

EXAMPLE	IUPAC NAME	Structure	MS
41	4-[(1RS,2SR)-1-{6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-1-oxo-1,3-dihydro-2H-isoindol-2-yl]-2-hydroxy-2-phenylethyl]benzonitrile		471.1
42	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2SR)-2-hydroxy-1-(1-methyl-1H-pyrazol-4-yl)-2-phenylethyl]-2,3-dihydro-1H-isoindol-1-one		452.1

TABLE 1-7

EXAMPLE	IUPAC NAME	Structure	MS
43	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2SR)-2-(4-fluorophenyl)-2-hydroxy-1-(1-methyl-1H-pyrazol-4-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one		468.2
44	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2SR)-2-(6-ethoxypyridin-3-yl)-2-hydroxy-1-phenylethyl]-2,3-dihydro-1H-isoindol-1-one		493.1

TABLE 1-7-continued

EXAMPLE	IUPAC NAME	Structure	MS
45	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-1-(4-fluorophenyl)-2-hydroxy-2-phenylethyl]-2,3-dihydro-1H-isindol-1-one (Enantiomer 1)		464.2
46	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-1-(4-fluorophenyl)-2-hydroxy-2-phenylethyl]-2,3-dihydro-1H-isindol-1-one (Enantiomer 2)		464.2
47	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2R*)-2-hydroxy-1-phenyl-2-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isindol-1-one (Enantiomer 1)		449.2
48	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2R*)-2-hydroxy-1-phenyl-2-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isindol-1-one (Enantiomer 2)		449.2
49	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-1-(4-fluorophenyl)-2-hydroxy-2-(pyridin-3-yl)ethyl]-2,3-dihydro-1H-isindol-1-one (Enantiomer 2)		467.1

TABLE 1-8

EXAMPLE	IUPAC NAME	Structure	MS
50	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2R*)-1-(3,4-difluorophenyl)-2-hydroxy-2-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)		485.1
51	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2R*)-1-(3,4-difluorophenyl)-2-hydroxy-2-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)		485.1
52	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-1-(4-fluorophenyl)-2-hydroxy-2-(pyridin-3-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)		467.1
53	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2SR)-2-(3,4-difluorophenyl)-1-(6-ethoxypyridin-3-yl)-2-hydroxyethyl]-2,3-dihydro-1H-isoindol-1-one		529.1

TABLE 1-8-continued

EXAMPLE	IUPAC NAME	Structure	MS
54	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-1-(4-fluorophenyl)-2-hydroxy-2-[6-(trifluoromethyl)pyridin-3-yl]ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)		533.1
55	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S)-1-(4-fluorophenyl)-2-hydroxy-2-[6-(trifluoromethyl)pyridin-3-yl]ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)		533.2
56	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2SR)-2-(3,4-difluorophenyl)-1-(6-fluoropyridin-3-yl)-2-hydroxyethyl]-2,3-dihydro-1H-isoindol-1-one		501.2

TABLE 1-9

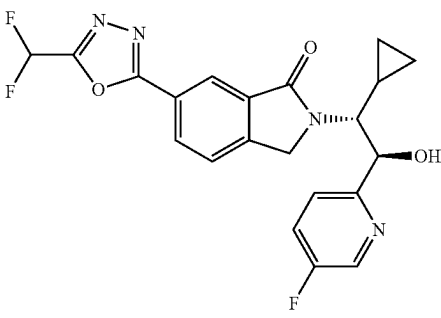
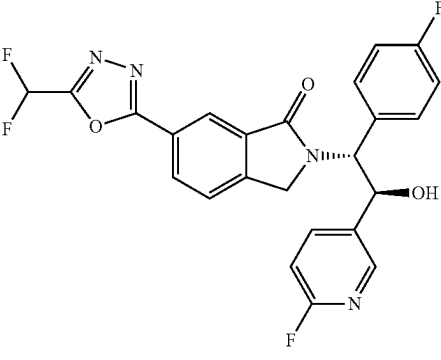
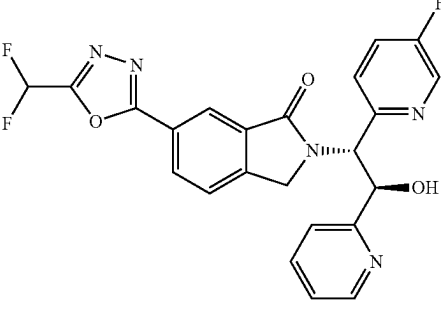
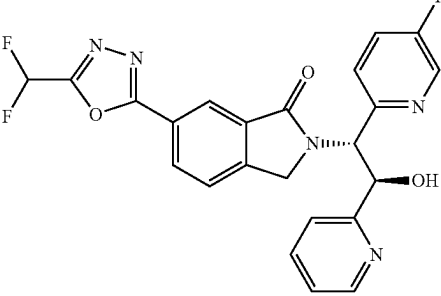
EXAMPLE	IUPAC NAME	Structure	MS
57	2-[(1R,2R)-1-cyclopropyl-2-(5-fluoropyridin-2-yl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one		431.2
58	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2SR)-1-(4-fluorophenyl)-2-(6-fluoropyridin-3-yl)-2-hydroxyethyl]-2,3-dihydro-1H-isoindol-1-one		483.2
59	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2R*)-1-(5-fluoropyridin-2-yl)-2-hydroxy-2-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)		468.2
60	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2R*)-2-(5-fluoropyridin-2-yl)-2-hydroxy-2-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)		468.2

TABLE 1-9-continued

EXAMPLE	IUPAC NAME	Structure	MS
61	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-2-(3-fluorophenyl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)		467.1
62	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-2-(3-fluorophenyl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)		467.1
63	2-[(1R,2S)-1-cyclopropyl-2-(6-fluoropyridin-3-yl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one		429.2

TABLE 1-10

EXAMPLE	IUPAC NAME	Structure	MS
64	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2S)-2-(3-fluorophenyl)-2-hydroxy-1-(oxan-4-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one		472.2

TABLE 1-10-continued

EXAMPLE	IUPAC NAME	Structure	MS
65	2-[(1R,2R)-2-(5-chloropyridin-2-yl)-1-cyclopropyl-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one		447.1
66	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2R*)-1-(5-fluoropyridin-2-yl)-2-(6-fluoropyridin-2-yl)-2-hydroxyethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)		486.1
67	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2R*)-1-(5-fluoropyridin-2-yl)-2-(6-fluoropyridin-2-yl)-2-hydroxyethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)		486.1
68	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-2-(3-fluorophenyl)-2-hydroxy-1-(pyrimidin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)		468.1

TABLE 1-10-continued

EXAMPLE	IUPAC NAME	Structure	MS
69	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-2-(3-fluorophenyl)-2-hydroxy-1-(pyrimidin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)		468.2
70	2-[(1R*,2S*)-2-(3-chlorophenyl)-2-hydroxy-1-(oxan-4-yl)ethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)		488.1

30

TABLE 1-11

EXAMPLE	IUPAC NAME	Structure	MS
71	2-[(1R*,2S*)-2-(3-chlorophenyl)-2-hydroxy-1-(oxan-4-yl)ethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)		488.1
72	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2S)-2-(4-fluorophenyl)-2-hydroxy-1-(oxan-4-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one		472.2

TABLE 1-11-continued

EXAMPLE	IUPAC NAME	Structure	MS
73	2-[(1R*,2R*)-1-cyclopropyl-2-hydroxy-2-(pyrimidin-2-yl)ethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)		414.2
74	2-[(1R*,2R*)-1-cyclopropyl-2-hydroxy-2-(pyrimidin-2-yl)ethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)		414.2
75	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2R*)-1-(6-fluoropyridin-2-yl)-2-hydroxy-2-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)		468.2
76	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2R*)-1-(6-fluoropyridin-2-yl)-2-hydroxy-2-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)		468.2
77	2-[(1R*,2S*)-1-cyclopropyl-2-hydroxy-2-[6-(trifluoromethyl)pyridin-3-yl]ethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)		479.1

TABLE 1-12

EXAMPLE	IUPAC NAME	Structure	MS
78	2-[(1R*,2S*)-1-cyclopropyl-2-hydroxy-2-[6-(trifluoromethyl)pyridin-3-yl]ethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)		479.1
79	2-[(1R*,2R*)-1-cyclopropyl-2-(6-fluoropyridin-2-yl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)		431.2
80	2-[(1R*,2R*)-1-cyclopropyl-2-(6-fluoropyridin-2-yl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)		431.2
81	2-[(1RS,2RS)-1-(4,4-difluorocyclohexyl)-2-hydroxy-2-(pyridin-2-yl)ethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one		491.1

TABLE 1-12-continued

EXAMPLE	IUPAC NAME	Structure	MS
82	2-[(1R,2SR)-1-(4,4-difluorocyclohexyl)-2-(6-fluoropyridin-3-yl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one		507.2
83	2-[(1R*,2S*)-2-(4-chlorophenyl)-2-hydroxy-1-(oxan-4-yl)ethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)		488.1
84	2-[(1R*,2S*)-2-(4-chlorophenyl)-2-hydroxy-1-(oxan-4-yl)ethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)		488.2

TABLE 1-13

EXAMPLE	IUPAC NAME	Structure	MS
85	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-2-(3,4-difluorophenyl)-2-hydroxy-1-(pyrazin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)		486.1

TABLE 1-13-continued

EXAMPLE	IUPAC NAME	Structure	MS
86	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-2-(3,4-difluorophenyl)-2-hydroxy-1-(pyrazin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)		486.1
87	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-2-(3,4-difluorophenyl)-2-hydroxy-1-(oxan-4-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)		490.2
88	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-2-(3,4-difluorophenyl)-2-hydroxy-1-(oxan-4-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)		490.1
89	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-2-(3-fluorophenyl)-1-(6-fluoropyridin-2-yl)-2-hydroxyethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)		485.1
90	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-2-(3-fluorophenyl)-1-(6-fluoropyridin-2-yl)-2-hydroxyethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)		485.0

TABLE 1-13-continued

EXAMPLE	IUPAC NAME	Structure	MS
91	2-[(1R*,2S*)-2-(3-chlorophenyl)-2-hydroxy-1-(pyrazin-2-yl)ethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)		484.1

TABLE 1-14

EXAMPLE	IUPAC NAME	Structure	MS
92	2-[(1R*,2S*)-2-(3-chlorophenyl)-2-hydroxy-1-(pyrazin-2-yl)ethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)		484.1
93	2-[(1R,2SR)-2-(4-chlorophenyl)-2-hydroxy-1-(pyrimidin-2-yl)ethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one		484.1
94	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-2-(3,4-difluorophenyl)-2-hydroxy-1-(pyrimidin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)		486.1

TABLE 1-14-continued

EXAMPLE	IUPAC NAME	Structure	MS
95	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-2-(3,4-difluorophenyl)-2-hydroxy-1-(pyrimidin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)		486.1
96	2-[(1R*,2S*)-2-(4-chlorophenyl)-2-hydroxy-1-(pyrazin-2-yl)ethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)		484.1
97	2-[(1R*,2S*)-2-(4-chlorophenyl)-2-hydroxy-1-(pyrazin-2-yl)ethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)		484.1
98	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-2-(4-fluorophenyl)-2-hydroxy-1-(pyrazin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)		468.2

TABLE 1-15

EXAMPLE	IUPAC NAME	Structure	MS
99	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-2-(4-fluorophenyl)-2-hydroxy-1-(pyrazin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)		468.2
100	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-2-(3-fluorophenyl)-2-hydroxy-1-(pyrazin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)		468.2
101	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-2-(3-fluorophenyl)-2-hydroxy-1-(pyrazin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)		468.2
102	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-2-hydroxy-2-phenyl-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)		449.2
103	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-2-hydroxy-2-phenyl-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)		449.2

TABLE 1-15-continued

EXAMPLE	IUPAC NAME	Structure	MS
104	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2R*)-2-hydroxy-1-(pyridin-2-yl)-2-[5-(trifluoromethyl)pyridin-2-yl]ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)		518.2
105	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2R*)-2-hydroxy-1-(pyridin-2-yl)-2-[5-(trifluoromethyl)pyridin-2-yl]ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)		518.2

TABLE 1-16

EXAMPLE	IUPAC NAME	Structure	MS
106	2-[(1RS,2SR)-2-(3-chloro-5-fluorophenyl)-2-hydroxy-1-(oxan-4-yl)ethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one		506.1
107	2-[(1RS,2SR)-2-(3-chloro-4-fluorophenyl)-2-hydroxy-1-(oxan-4-yl)ethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one		506.2

TABLE 1-16-continued

EXAMPLE	IUPAC NAME	Structure	MS
108	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-2-(3,4-difluorophenyl)-2-hydroxy-1-(pyridazin-3-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)		486.1
109	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-2-(3,4-difluorophenyl)-2-hydroxy-1-(pyridazin-3-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)		486.1
110	2-[(1R*,2S)-2-(4-chlorophenyl)-2-hydroxy-1-(pyridazin-3-yl)ethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)		484.1
111	2-[(1R*,2S*)-2-(4-chlorophenyl)-2-hydroxy-1-(pyridazin-3-yl)ethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)		484.1
112	2-[(1R*,2R*)-2-(5-chloropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)		484.1

TABLE 1-17

EXAMPLE	IUPAC NAME	Structure	MS
113	2-[(1R*,2R)-2-(5-chloropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)		484.1
114	2-[(1R*,2R*)-2-(6-chloropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)		484.1
115	2-[(1R*,2R*)-2-(6-chloropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)		484.0
116	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2SR)-2-hydroxy-1-(oxan-4-yl)-2-[3-(trifluoromethyl)phenyl]ethyl]-2,3-dihydro-1H-isoindol-1-one		522.1
117	2-[(1R*,2R*)-1,2-bis(6-fluoropyridin-2-yl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)		486.1

TABLE 1-17-continued

EXAMPLE	IUPAC NAME	Structure	MS
118	2-[(1R*,2R*)-1,2-bis(6-fluoropyridin-2-yl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)		486.1
119	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2R*)-2-hydroxy-1-(pyridin-2-yl)-2-[6-(trifluoromethyl)pyridin-2-yl]ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)		518.1

TABLE 1-18

EXAMPLE	IUPAC NAME	Structure	MS
120	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2R*)-2-hydroxy-1-(pyridin-2-yl)-2-[6-(trifluoromethyl)pyridin-2-yl]ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)		518.1
121	2-[(1R*,2S)-2-(3-chlorophenyl)-2-hydroxy-1-(pyridazin-3-yl)ethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)		484.1

TABLE 1-18-continued

EXAMPLE	IUPAC NAME	Structure	MS
122	2-[(1R*,2S*)-2-(3-chlorophenyl)-2-hydroxy-1-(pyridazin-3-yl)ethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)		484.1
123	2-[(1R*,2R*)-2-(4-chloropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)		484.1
124	2-[(1R*,2R*)-2-(4-chloropyridin-2-yl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)		484.1
125	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2R)-2-(5-fluoropyridin-2-yl)-1-(6-fluoropyridin-2-yl)-2-hydroxyethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)		486.1
126	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2R*)-2-(5-fluoropyridin-2-yl)-1-(6-fluoropyridin-2-yl)-2-hydroxyethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)		486.1

TABLE 1-19

EXAMPLE	IUPAC NAME	Structure	MS
127	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-2-hydroxy-2-phenyl-1-(pyrazin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)		450.1
128	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-2-hydroxy-2-phenyl-1-(pyrazin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)		450.1
129	2-[(1R*,2R*)-2-(6-chloropyridin-2-yl)-1-cyclopropyl-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)		447.1
130	2-[(1R*,2R*)-2-(6-chloropyridin-2-yl)-1-cyclopropyl-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)		447.1
131	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-1-(4-fluorophenyl)-2-(6-fluoropyridin-3-yl)-2-hydroxyethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 1)		485.1

TABLE 1-19-continued

EXAMPLE	IUPAC NAME	Structure	MS
132	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R*,2S*)-1-(4-fluorophenyl)-2-(6-fluoropyridin-3-yl)-2-hydroxyethyl]-2,3-dihydro-1H-isoindol-1-one (Enantiomer 2)		485.1
133	2-[(1RS,2SR)-2-(3-chlorophenyl)-2-hydroxy-1-(pyrimidin-2-yl)ethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one		484.1

TABLE 1-20

EXAMPLE	IUPAC NAME	Structure	MS
134	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2SR)-2-(4-fluorophenyl)-2-hydroxy-1-(pyrimidin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one		468.2
135	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2SR)-2-hydroxy-1,2-di(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one		450.1

TABLE 1-20-continued

EXAMPLE	IUPAC NAME	Structure	MS
136	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2R)-2-(3-fluorophenyl)-2-hydroxy-1-(pyrimidin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one		468.2
137	2-[(1R,2R)-1-cyclopropyl-2-(4-fluorophenyl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one		430.2
138	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2SR)-1-(4-fluorophenyl)-2-hydroxy-2-phenylethyl]-2,3-dihydro-1H-isoindol-1-one		446.1
139	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2RS)-2-(2-fluorophenyl)-2-hydroxy-1-phenylethyl]-2,3-dihydro-1H-isoindol-1-one		466.1
140	2-[(1RS,2RS)-2-cyclopropyl-1-(4-fluorophenyl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one		430.2

TABLE 1-21

EXAMPLE	IUPAC NAME	Structure	MS
141	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2RS)-2-(4-fluorophenyl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one		467.1
142	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2RS)-1-(4-fluorophenyl)-2-hydroxy-2-[6-(trifluoromethyl)pyridin-3-yl]ethyl]-2,3-dihydro-1H-isoindol-1-one		533.1
143	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2RS)-2-hydroxy-2-phenyl-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one		449.2
144	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2SR)-1-(5-fluoropyridin-2-yl)-2-hydroxy-2-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one		468.2

TABLE 1-21-continued

EXAMPLE	IUPAC NAME	Structure	MS
145	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1SR,2SR)-2-(3-fluorophenyl)-2-hydroxy-1-(pyridin-2-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one		467.1
146	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1SR,2SR)-1-(2,4-difluorophenyl)-2-(4-fluorophenyl)-2-hydroxyethyl]-2,3-dihydro-1H-isoindol-1-one		500.1
147	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1RS,2RS)-2-(3-fluorophenyl)-1-(4-fluorophenyl)-2-hydroxyethyl]-2,3-dihydro-1H-isoindol-1-one		484.1

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TABLE 1-22

EXAMPLE	IUPAC NAME	Structure	MS
148	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2R)-2-(4-fluorophenyl)-2-hydroxy-1-(oxan-4-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one		472.2

TABLE 1-22-continued

EXAMPLE	IUPAC NAME	Structure	MS
149	6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2-[(1R,2R)-2-(3-fluorophenyl)-2-hydroxy-1-(oxan-4-yl)ethyl]-2,3-dihydro-1H-isoindol-1-one		472.2
150	2-[(1R,2S)-1-cyclopropyl-2-(5-fluoropyridin-2-yl)-2-hydroxyethyl]-6-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-2,3-dihydro-1H-isoindol-1-one		431.2

Experimental Example 1 HDAC6 Enzyme Inhibitory Assay

HDAC6 enzyme prepared by transducing full length HDAC6 gene into Sf-9 insect cells and purifying by GST affinity column were purchased from SignalChem. Using this enzyme, HDAC6 enzyme inhibitory activity of the compound of the present invention were evaluated. Enzymes were used after preserved at -70°C . HDAC6 enzyme inhibitory activity of the compound of the present invention was measured using HDAC-Glo™ I/II Assay kit (Promega) according to the following experimental method. The test compound diluted with assay buffer (24 mM Tris-HCl (pH 7.5), 1 mM MgCl_2 , 0.35 mM KCl, 135 mM NaCl, 0.6 mM Glutathione, 0.01% Tween-20) was added to a 384-well plate by each 2 μL . Then, HDAC6 enzyme solution diluted with assay buffer was added thereto by each 4 μL , and the plate was incubated at room temperature for 60 min. After incubated, HDAC substrate-Developer solution prepared according to Promega protocol attached to the assay kit was added to the 384-well plate by each 2 μL , and the enzyme reaction was started. After reacting at room temperature for 20 min, luminescence level was measured using plate reader Envision (PerkinElmer). The inhibitory activity of each compound of Example was calculated as a relative activity value when luminescence level in wells without enzyme is considered as 100% inhibition. The results are shown in Table 2-1 and Table 2-2.

TABLE 2-1

Ex. No.	HDAC6 inhibitory rate (%) (10 μM)
1	101
2	100
3	101

TABLE 2-1-continued

Ex. No.	HDAC6 inhibitory rate (%) (10 μM)
4	100
5	99
6	100
7	102
8	99
9	100
10	100
11	99
12	100
13	99
14	100
15	99
16	99
17	100
18	99
19	99
20	100
21	100
22	99
23	100
24	99
25	100
26	99
27	101
28	99
29	99
30	99
31	97
32	99
33	100
34	100
35	100
36	101
37	100
38	100
39	101
40	100
41	100
42	100
43	100

125

TABLE 2-1-continued

Ex. No.	HDAC6 inhibitory rate (%) (10 μ M)
44	100
45	100
46	100
47	99
48	100
49	99
50	99
51	99
52	100
53	100
54	100
55	98
56	100
57	100
58	100
59	100
60	99
61	100
62	100
63	100
64	100
65	98
66	100
67	99
68	101
69	100
70	101
71	101
72	100
73	100
74	96
75	99
76	100
77	100
78	99
79	99
80	100
81	100
82	100
83	101
84	100
85	100
86	100
87	100
88	99
89	100
90	100

TABLE 2-2

Ex. No.	HDAC6 inhibitory rate (%) (10 μ M)
91	100
92	100
93	100
94	100
95	99
96	100
97	99
98	100
99	100
100	100
101	100
102	100
103	100
104	100
105	99
106	100
107	101
108	100
109	100
110	100
111	99

126

TABLE 2-2-continued

Ex. No.	HDAC6 inhibitory rate (%) (10 μ M)
112	99
113	100
114	99
115	100
116	101
117	101
118	102
119	101
120	101
121	101
122	100
123	101
124	100
125	101
126	101
127	99
128	99
129	100
130	98
131	99
132	99
133	100
134	101
135	99
136	99
137	100
138	99
139	100
140	98
141	99
142	100
143	100
144	100
145	99
146	95
147	99
148	101
149	100
150	99

As is clear from Table 2-1 and Table 2-2, the compound of the present invention has an excellent HDAC6 inhibitory activity.

Experimental Example 2 Increase in Acetylated Tubulin in Brain

A drug was suspended in 0.5% methyl cellulose to prepare a suspension for administration, having a concentration shown in Table 3. The suspension was orally administered into 8 to 10-week old male C57BL/6J mouse, and after the passage of the time shown in Table 3, the brain was taken out, and hippocampus was obtained. The hippocampus was homogenized under RIPA extract (FUJIFILM Wako Pure Chemical Corporation) to which protease inhibitor (Thermo Fisher Scientific) and phosphatase inhibitor (Thermo Fisher Scientific) were added, and centrifuged in 15,000 g for 15 min to prepare a protein extract. The acetylated tubulin and total tubulin in the extract were detected by the following ELISA method. Tubulin antibody (Sigma) was immobilized on 96-well plate (Microlite2+, Thermo Fisher Scientific), and the plate was stored at 4° C. Next day, the plate was washed four times with PBS-T, and blocking buffer (1% BSA/PBS-T) was added thereto, and the reaction was carried out at 37° C. for 2 hr. The plate was washed four times with PBS-T, and the above-mentioned protein extract was added thereto to capture the tubulin on the antibody. The plate was stored at 37° C. for 2 hr, and washed four times with PBS-T. Anti-acetylated tubulin antibody or total tubulin antibody (Cell Signaling Technology), each diluted with

blocking buffer, was added thereto, and the reaction was carried out at 37° C. for 1 hr. Then, the plate was washed four times with PBS-T, and the reaction was carried out using anti-mouse HRP (Cell Signaling Technology) at 37° C. for 30 min. The plate was washed, HRP substrate was added thereto, and the luminescence was measured using a plate reader. The acetylated tubulin and total tubulin were quantified from logistics curve based on dilution series using the mouse hippocampus tissue extract, and the amount of the acetylated tubulin relative to the total tubulin was calculated, and evaluated as an relative tubulin acetylation level. The test was performed using SAS system 8. After F test for vehicle group and drug group, the significant difference between the two groups was analyzed by Student t-test or Welch test (indicated as * $p < 0.05$, ** $p < 0.01$, n.s.: not significant). The graph was indicated by the mean $\pm$ standard error. The results are shown in FIGS. 1 to 3. In the mouse into which the compound of Example 1, 3 or 5 was administered, increase in significant tubulin acetylation level was observed.

TABLE 3

Ex. No.	concentration (mg/mL)	dose (mg/kg)	time (h) after administration of test compound	C57BL/6J old	C573L/6J number
1	0.01, 0.1, 1, 10	0.1, 1, 10, 100	4	9	6
3	0.01, 0.1, 1, 10	0.1, 1, 10, 100	4	8	6
5	0.03, 0.1	0.3, 1	4	10	6

Experimental Example 3 Cognitive Improvement Action in Neurodegenerative Disease Model Mouse

A drug was suspended in 0.5% methyl cellulose to prepare a suspension for administration. The suspension was repeatedly administered into six month-old male P301S mutant human tau (4R1N) Tg mouse once a day for three months, and novel object recognition test was performed by the following method. The test is comprised of memory acquisition trial of object on the first day, and memory retention trial of familiar object and novel object on the second day. Drug was administered into the mouse 1 hr before the trials on both days. For acquisition trial, The mouse was put in a test box (30 cm $\times$ 30 cm $\times$ 25 cm) in which two identical objects were placed, and contact frequency and contact duration with the objects for 5 min were measured under 50 lux. Herein, the contact means a behavior getting a sniff of an object (Sniffing). Next day, one of the objects was replaced with new object, and contact frequency and contact duration with each object for 5 min were measured. Novelty discrimination index (NDI) as an index of cognitive function was calculated by novel object contact frequency/(novel object contact duration+familiar object contact duration) %. The test was performed using SAS system 8. After F test for vehicle administration wild-type group and vehicle administration Tg group, the significant difference between the two groups was analyzed by Student t-test or Welch test (indicated as ** $p < 0.01$). In addition, Bartlett test for homogeneity of variance was performed for vehicle administration or drug administration Tg group, and the significant difference was analyzed by one-tailed Williams test (indicated as ## $p < 0.005$). Each group consisted of 14 or 15 mice. The graph was indicated by the mean $\pm$ standard error. The results are shown in FIGS. 4 and 5.

Experimental Example 4 Toxicity Test in Rat

A Two-week toxicity test in rats is conducted using the following test system.

Test System:

Rat, Sprague-Dawley [CrI:CD(SD)]

Administration Method/Period:

Oral administration, which is the scheduled route for clinical application, is selected as the administration route, and the administration period is 2 weeks. The administration frequency is once a day (7 days/week), which is generally used in repeated dose studies. For oral administration to rodents, the administration method is general oral gavage. The dose is 5 mL/kg, and it is administered by oral gavage using a flexible gastric tube (between 08:00 and 15:00). The vehicle is similarly administered to the vehicle control group. The dose of each animal is calculated based on the most recent body weight.

Vehicle and Preparation Method:

0.5 w/v % methylcellulose solution and the required amount of methylcellulose (METOLOSE SM-100) are weighed and gradually added to an appropriate amount of warm water for injection with stirring to disperse the methylcellulose. This is cooled to dissolve the methylcellulose, and water for injection is further added thereto to give a 0.5% solution.

A hematology protocol is described below as an example of toxicity assessment.

Materials:

At the time of planned autopsy, all animals in the test group that had been fasted overnight from the previous day (about 16 to 21 hours) are subjected to laparotomy under isoflurane inhalation anesthesia, and the blood (about 1 mL) is collected from the abdominal aorta into a blood sampling bottle containing EDTA-2K. The obtained blood is tested for the items and by the methods listed below.

Used Equipment:

Automatic blood cell counter (ADVIA2120i, Siemens Healthcare Diagnostics)

Test Items (Measurement Method):

Red blood cell count (2-angle laser flow cytometry method)

Hematocrit value (calculated from red blood cell count and MCV)

Hemoglobin concentration (modified cyanmethemoglobin method)

Average red blood cell hemoglobin amount (calculated from MCH, red blood cell count and hemoglobin concentration)

Average red blood cell hemoglobin concentration (calculated from MCHC, red blood cell count, MCV and hemoglobin concentration)

Average red blood cell volume (MCV, 2-angle laser flow cytometry method)

Red blood cell volume distribution width (RDW, 2-angle laser flow cytometry method)

Reticulocyte count (absolute count, laser flow cytometry by RNA staining)

Platelet count (2-angle laser flow cytometry method)

White blood cell count (2-angle laser flow cytometry method)

White blood cell classification (absolute count, flow cytometry by peroxidase staining+2-angle laser flow cytometry)

Erythrocyte morphology flag (MICRO: microcytic red blood cells, MACRO: macrocytic red blood cells, HYPO: hypochromic red blood cells, HYPER: hyper-

chromic red blood cells, ANISO: red blood cell size difference, HCVAR: hemoglobin concentration difference, RBCF: crushed red blood cells)

Experimental Example 5 Bacterial Reversion Assay

Materials and Methods

Test Strains:

Salmonella typhimurium TA100 or TA98

Metabolic Activation System:

S9 mix for Ames test prepared from livers of 7-week-old male Sprague-Dawley rats treated with phenobarbital and 5,6-benzoflavone

Doses:

78.1, 156.3, 312.5, 625, 1250, 2500 or 5000 µg/plate

Test Method:

After the bacterial cells are cultured at 37° C. for 8 hours, the optical density (at 660 nm) of cell suspension is measured by spectrophotometer to determine if bacteria grows well. The cell numbers calculated from the optical density are confirmed to be within the acceptable range (cell number: $\geq 1 \times 10^9$ /mL). Cells are exposed to test article using the pre-incubation method (37° C., 20 minutes) in the presence or absence of S9 mix. After mixing with top agar, cells are overlaid on plates. The plates are incubated at 37° C. for 48 hours. The top agar, solvent, and the test article solution are inspected for bacterial contamination.

Criteria for Mutagenicity:

The test article is determined to be positive if the test article induced a dose-dependent increase in the number of the revertant colonies to a level equal to or greater than 2 times of the mean negative control value in any test strain either in the presence or absence of S9 mix.

Experimental Example 6 In Vitro Micronucleus Assay

Materials and Methods

Test System:

Human lymphoblastoid cells derived TK6 cells

Culture Condition:

The cells are cultured in a 6-well plastic plate at 37° C. and 5% CO₂ with RPMI1640 supplemented with 10 vol % heat-inactivated horse serum, 2 mmol/L sodium pyruvate, 100 unit/mL penicillin and 100 µg/mL streptomycin.

Metabolic Activation System:

Rat liver S9 mix prepared from the livers of 7-week-old male Sprague-Dawley rats treated with phenobarbital and 5,6-benzoflavone.

Treatment Conditions:

(1) 3-hour treatment with S9 mix, followed by 21-hour culture, (2) 24-hour continuous treatment without S9 mix

Doses:

450.0 µg/mL to 12.6 µg/mL

Duplicate cultures are prepared for each concentration.

Cytotoxicity:

Cell numbers are counted, and the relative population doubling (RPD) as a cytotoxic evaluation index is calculated using the following formula:

Population doubling (PD) =

$$\frac{\log_{10}([\text{cell number at the end of culture}])}{\log_{10}([\text{cell number at the start of treatment}])} \times 1 \times 10^5 \text{ cells/mL}$$

$$\text{RPD (\%)} = \frac{\text{PD for the test article group}}{\text{PD for the concurrent negative control group}} \times 100$$

Micronucleus Observation:

For each treatment condition, the highest concentration for micronucleus analysis is selected based on precipitation of test articles and cytotoxicity, and two lower consecutive concentrations are examined. The number of micronucleated cells among 4,000 mononuclear cells with cytoplasm is counted using a microscope.

Statistical Analysis:

Fisher's exact test is performed at upper-tailed significance levels of 5% and 1% in order to compare the incidence of micronucleated cells in the test article groups or the positive control group with that of the negative control group for each treatment condition. If Fisher's exact test shows statistically significance in the test article groups, the exact Cochran-Armitage trend test is performed.

Judgment Criteria:

The test article is judged to be positive if the incidence of micronucleated cells between any test article groups satisfied both criteria (1) and (2) shown below.

- (1) A significant difference in the incidence of micronucleated cells between the negative control group is detected in the statistical analysis and dose-dependent increase is also detected in the statistical analysis.
- (2) The incidence of micronucleated cells is more than the historical control range (mean+2SD). Otherwise, the test article is considered to be negative.

Formulation Example 1 (Production of Capsule)

1) compound of Example 1	30 mg
2) fine powder cellulose	10 mg
3) lactose	19 mg
4) magnesium stearate	1 mg
Total	60 mg

1), 2), 3) and 4) are mixed and filled in a gelatin.

1) compound of Example 1	30 g
2) lactose	50 g
3) cornstarch	15 g
4) calcium carboxymethylcellulose	44 g
5) magnesium stearate	1 g
1000 tablets total	140 g

The total amount of 1), 2) and 3) and 4) (30 g) is kneaded with water, vacuum dried, and sieved. The sieved powder is mixed with 4) (14 g) and 5) (1 g), and the mixture is punched by a tableting machine, whereby 1000 tablets containing 30 mg of the compound of Example 1 per tablet are obtained.

INDUSTRIAL APPLICABILITY

The compound of the present invention has a HDAC inhibitory activity, and may be useful for the treatment of

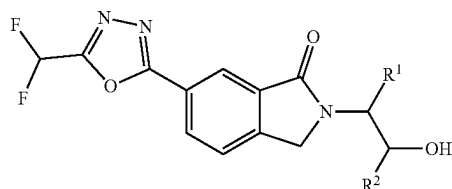
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central nervous system diseases including neurodegenerative diseases (Alzheimer's disease, progressive supranuclear palsy, etc.) and the like.

This application is based on patent applications No. 2019-177815 filed on Sep. 27, 2019 in Japan, the contents of which are encompassed in full herein.

The invention claimed is:

1. A compound represented by the formula (I):



wherein

R¹ is

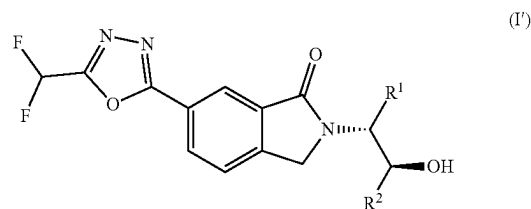
- (1) a C₃₋₁₀ cycloalkyl group optionally substituted by 1 to 3 halogen atoms,
- (2) a C₆₋₁₄ aryl group optionally substituted by 1 to 3 substituents selected from (i) a halogen atom, (ii) a cyano group, and (iii) a C₁₋₆ alkyl group optionally substituted by 1 to 3 halogen atoms,
- (3) a 5- or 6-membered monocyclic aromatic heterocyclic group optionally substituted by 1 to 3 substituents selected from (i) a halogen atom, (ii) a C₁₋₆ alkyl group optionally substituted by 1 to 3 halogen atoms, and (iii) a C₁₋₆ alkoxy group, or
- (4) a 3- to 8-membered monocyclic non-aromatic heterocyclic group, and

R² is

- (1) a C₃₋₁₀ cycloalkyl group optionally substituted by 1 to 3 halogen atoms,
 - (2) a C₆₋₁₄ aryl group optionally substituted by 1 to 3 substituents selected from (i) a halogen atom, and (ii) a C₁₋₆ alkyl group optionally substituted by 1 to 3 halogen atoms,
 - (3) a 5- or 6-membered monocyclic aromatic heterocyclic group optionally substituted by 1 to 3 substituents selected from (i) a halogen atom, (ii) a C₁₋₆ alkyl group optionally substituted by 1 to 3 halogen atoms, and (iii) a C₁₋₆ alkoxy group, or
 - (4) a 3- to 8-membered monocyclic non-aromatic heterocyclic group,
- or a salt thereof.

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2. The compound or salt according to claim 1, which is represented by the formula (I'):



wherein each symbol is as defined in claim 1.

3. The compound or salt according to claim 1, wherein R¹ is

- (1) a C₃₋₁₀ cycloalkyl group optionally substituted by 1 to 3 halogen atoms, or
- (2) a 5- or 6-membered monocyclic aromatic heterocyclic group optionally substituted by 1 to 3 substituents selected from (i) a halogen atom, (ii) a C₁₋₆ alkyl group optionally substituted by 1 to 3 halogen atoms, and (iii) a C₁₋₆ alkoxy group, and

R² is

- (1) a C₆₋₁₄ aryl group optionally substituted by 1 to 3 substituents selected from (i) a halogen atom, and (ii) a C₁₋₆ alkyl group optionally substituted by 1 to 3 halogen atoms, or
- (2) a 5- or 6-membered monocyclic aromatic heterocyclic group optionally substituted by 1 to 3 substituents selected from (i) a halogen atom, (ii) a C₁₋₆ alkyl group optionally substituted by 1 to 3 halogen atoms, and (iii) a C₁₋₆ alkoxy group.

4. The compound or salt according to claim 1, wherein R¹ is

- (1) a C₃₋₁₀ cycloalkyl group, or
- (2) a 6-membered monocyclic aromatic heterocyclic group optionally substituted by 1 to 3 halogen atoms, and

R² is

- (1) a C₆₋₁₄ aryl group optionally substituted by 1 to 3 halogen atoms, or
- (2) a 6-membered monocyclic aromatic heterocyclic group optionally substituted by 1 to 3 halogen atoms.

5. A pharmaceutical composition comprising the compound or salt according to claim 1 and a pharmaceutically acceptable carrier.

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